

generator configured to receive the first voltage from the first terminal and receive the first control signal from the charging controller; wherein the voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the detection signal indicating that the second voltage has decreased to the first predetermined threshold.

28 Claims, 17 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/465,678, filed on Sep. 2, 2021, now Pat. No. 11,671,021.

(51) Int. Cl.

H02M 1/36 (2007.01)

H02M 1/44 (2007.01)

(58) Field of Classification Search

CPC H02M 3/28; H02M 3/01; H02M 3/33569; H02M 3/33507; H02M 2007/4815; H02M 2007/4818; H02M 1/083; H02M 3/33538; H02M 3/33546; H02M 3/33515; H02M 3/33576; H02M 3/33592; H02M 3/33553; H02M 3/33523; H02M 3/33561; H02M 3/155; H02M 3/1582; H02M 1/4233; H02M 1/12; H02M 3/07; H02M 7/219; H02M 7/4815; H02M 1/0048; Y02B 70/1491

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,712,045 B2 *	7/2017	Mao	H02M 1/36
9,887,618 B2	2/2018	Cui et al.	
10,469,006 B2 *	11/2019	Pant	H02J 7/007182
10,644,607 B2	5/2020	Li et al.	
10,978,945 B2 *	4/2021	Matsuda	H02M 1/08
11,671,021 B2	6/2023	Yang et al.	
12,184,182 B2	12/2024	Yang et al.	
2012/0008343 A1	1/2012	Chen et al.	
2014/0098579 A1	4/2014	Kleinpenning	
2014/0362612 A1	12/2014	Shi et al.	
2014/0376277 A1	12/2014	Tsui	
2015/0003118 A1	1/2015	Shin et al.	
2015/0162841 A1	6/2015	Masuda et al.	
2019/0044449 A1	2/2019	Li et al.	
2022/0077786 A1	3/2022	Yang et al.	
2023/0396170 A1	12/2023	Yang et al.	

FOREIGN PATENT DOCUMENTS

CN	103023298 A	4/2013
CN	103023337 A	4/2013
CN	206135707 U	4/2017
CN	107231092 A	10/2017
CN	110518790 A	11/2019
JP	2016-146696 A	8/2016
TW	200934130	8/2009

OTHER PUBLICATIONS

Taiwan Intellectual Property Office, Office Action mailed Oct. 4, 2021, for Application No. 109135760.

* cited by examiner

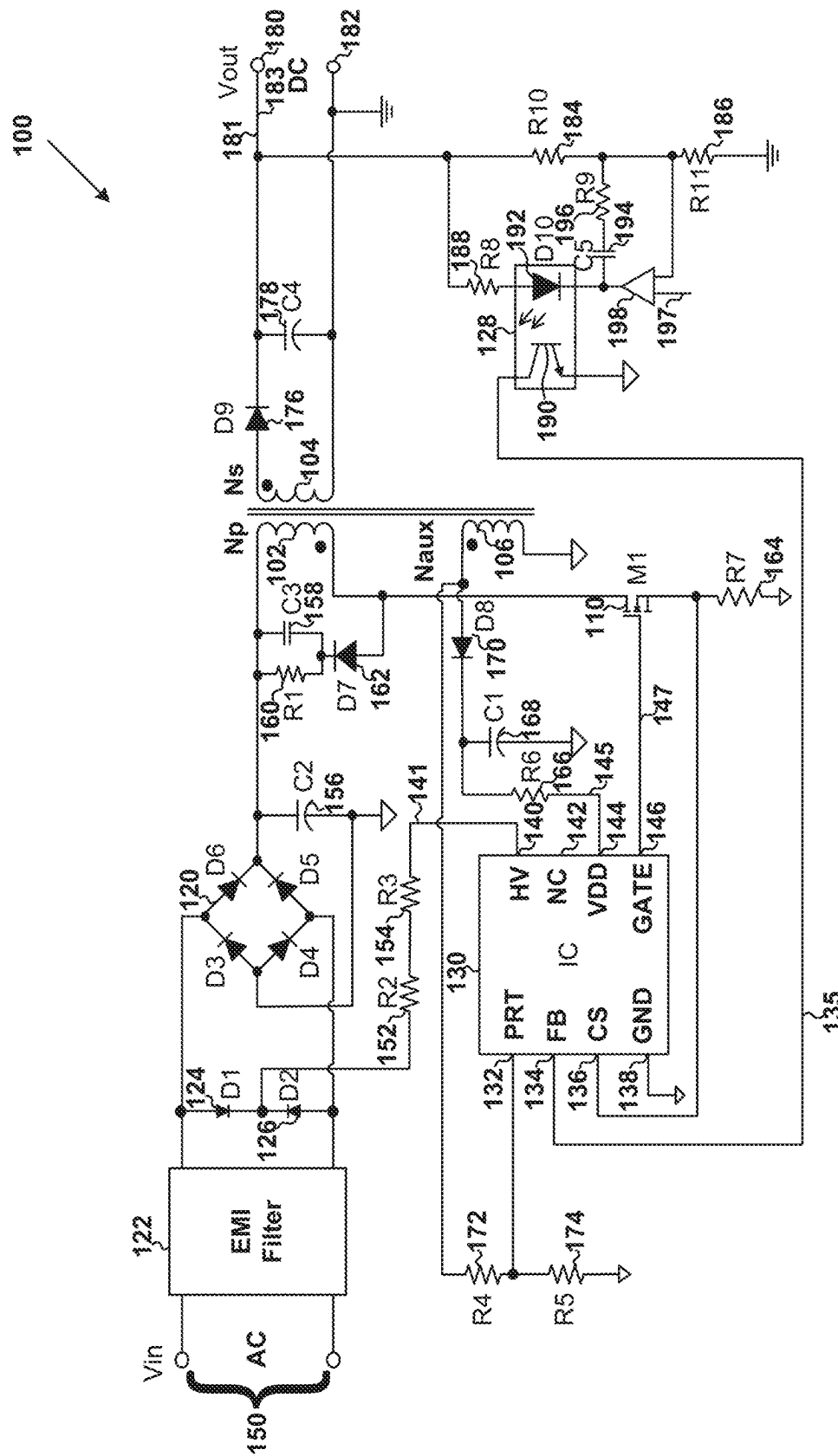


Fig. 1
Plot Art

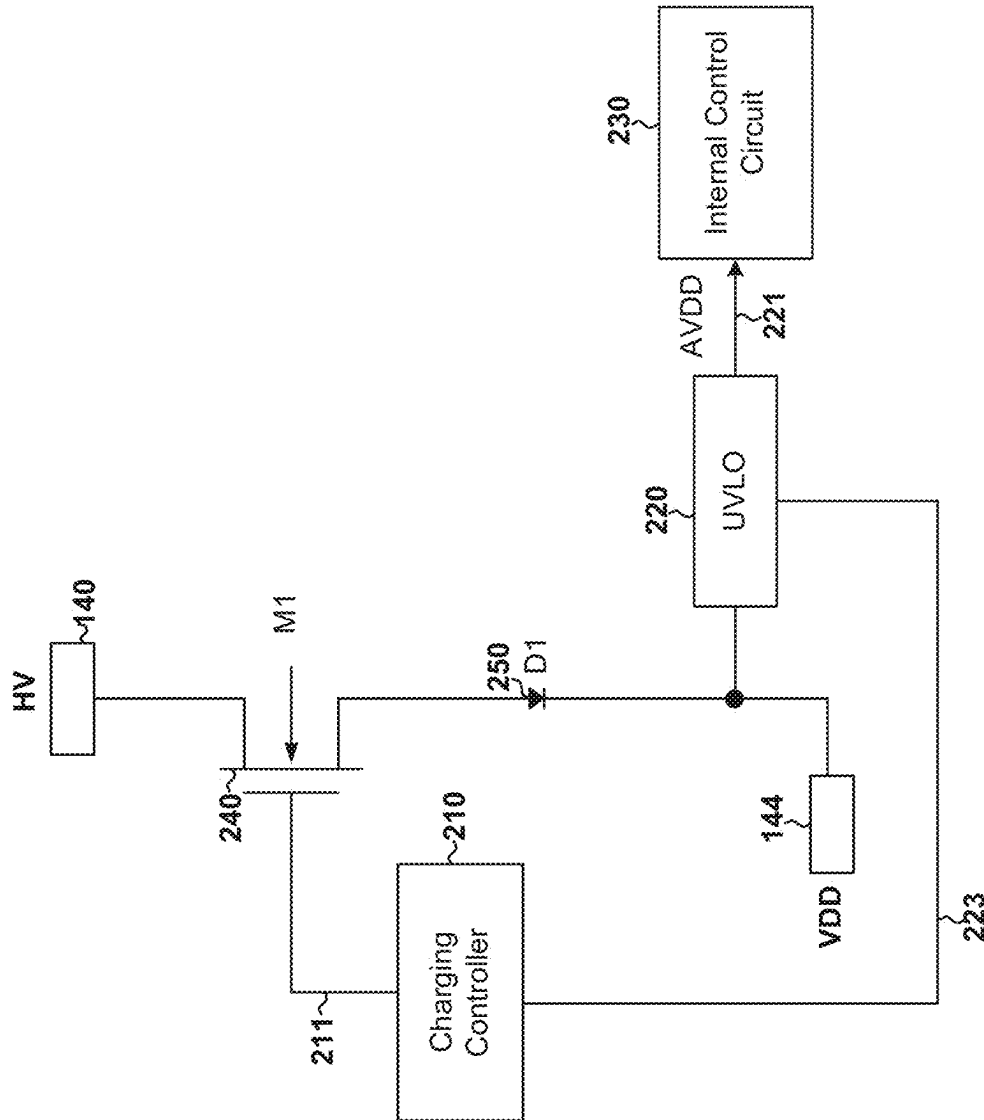


FIG. 2
Prior Art

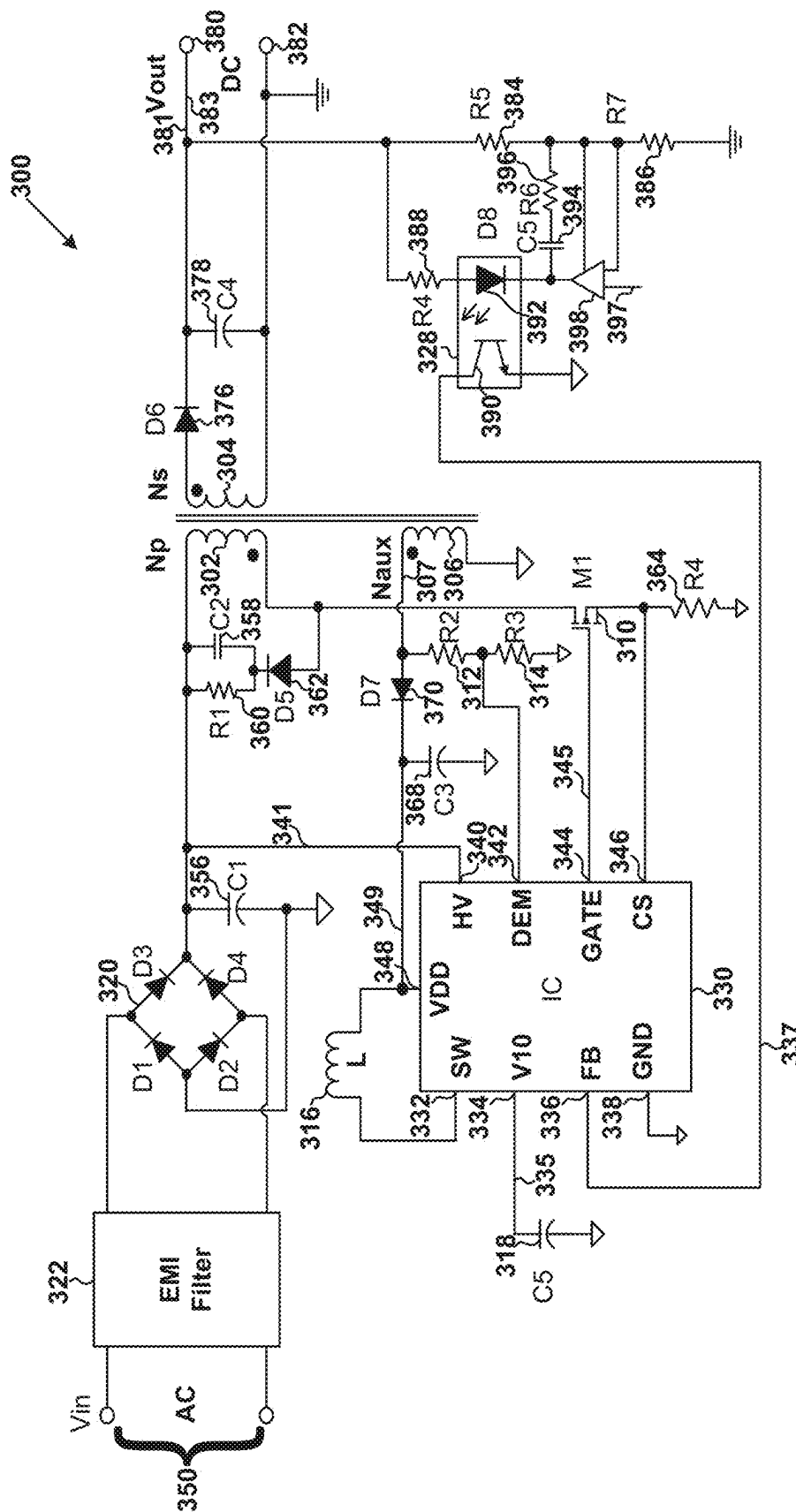


FIG. 3
Prior Art

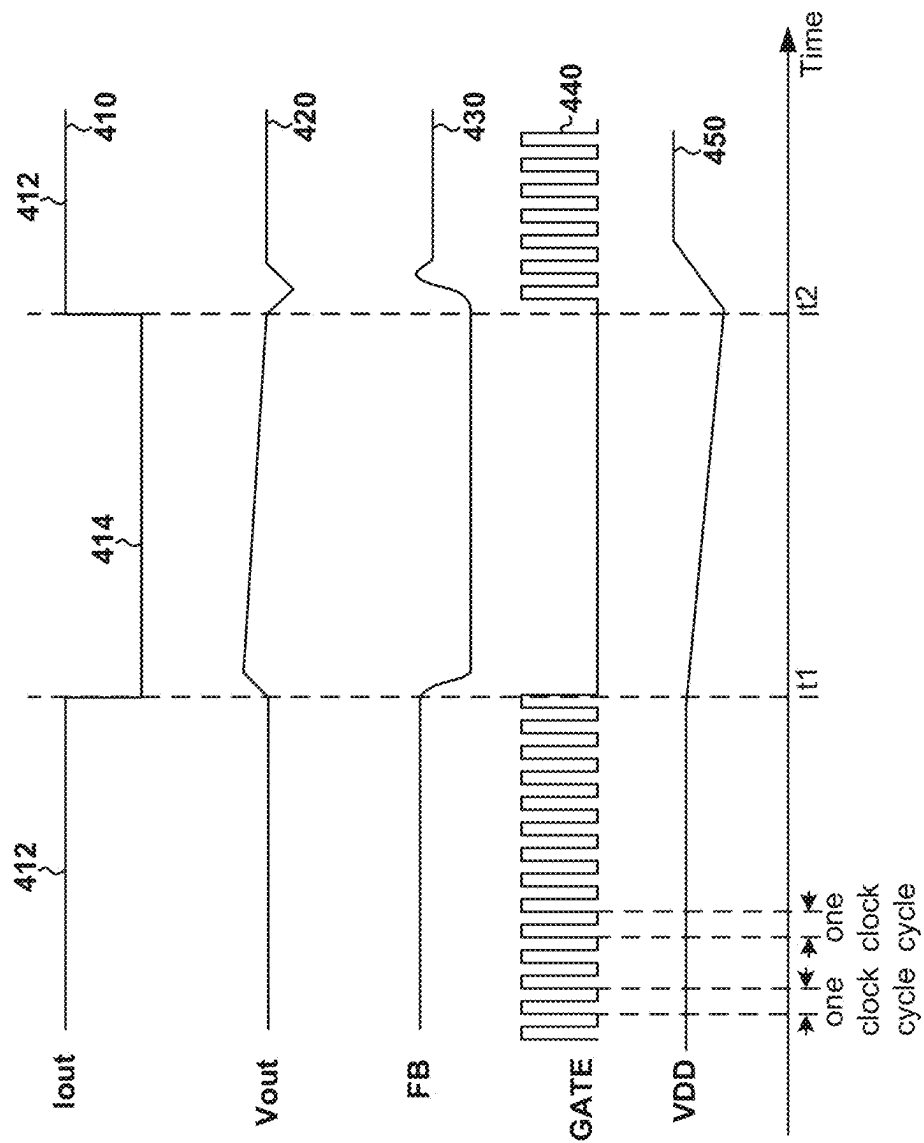


FIG. 4

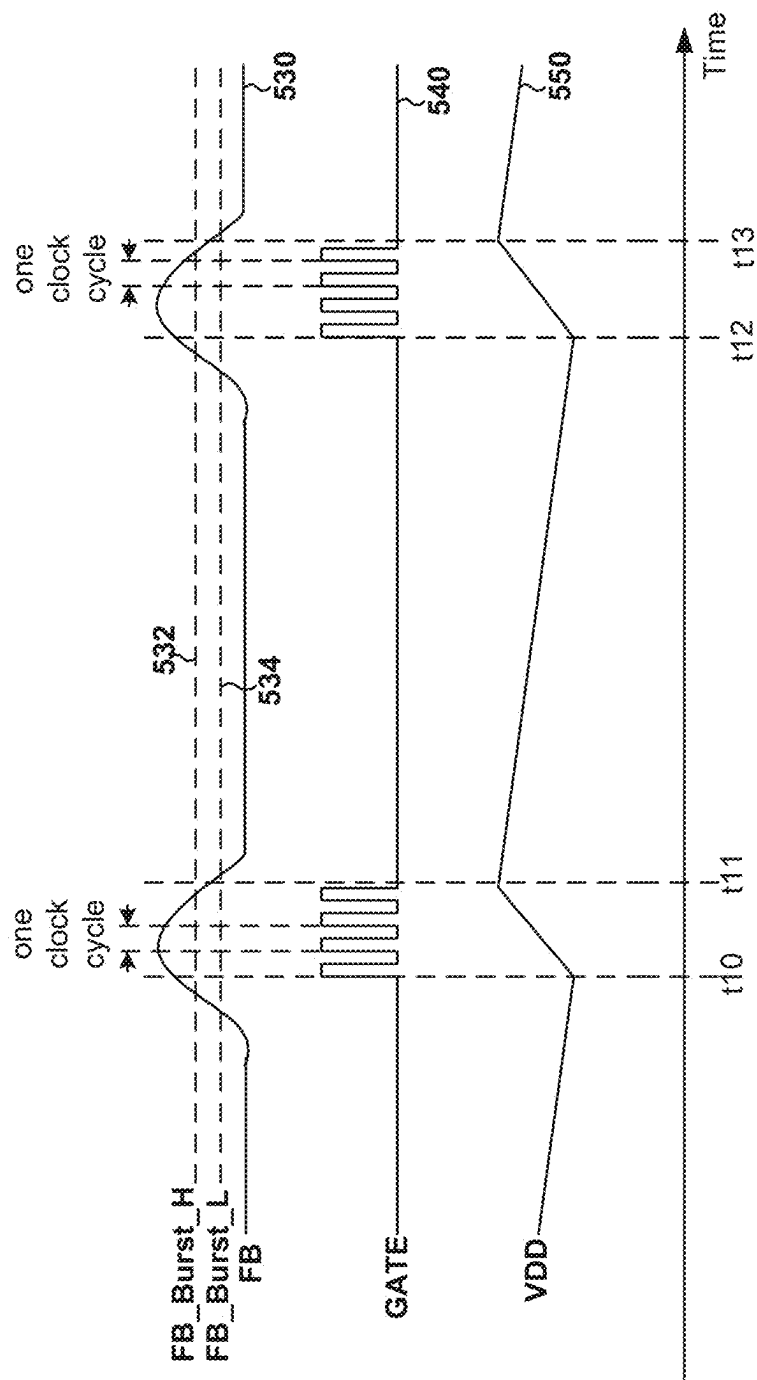


FIG. 5

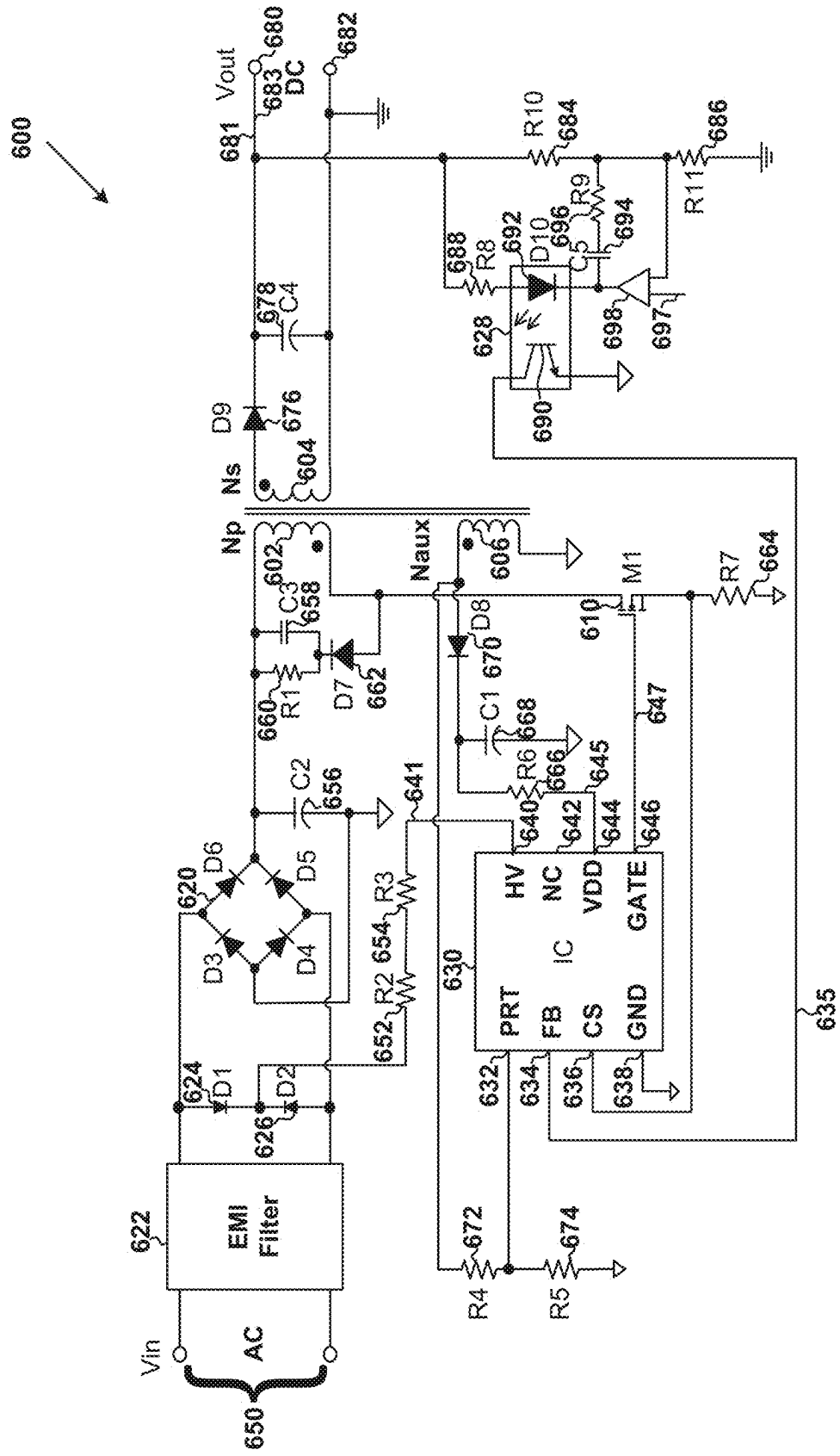


FIG. 6

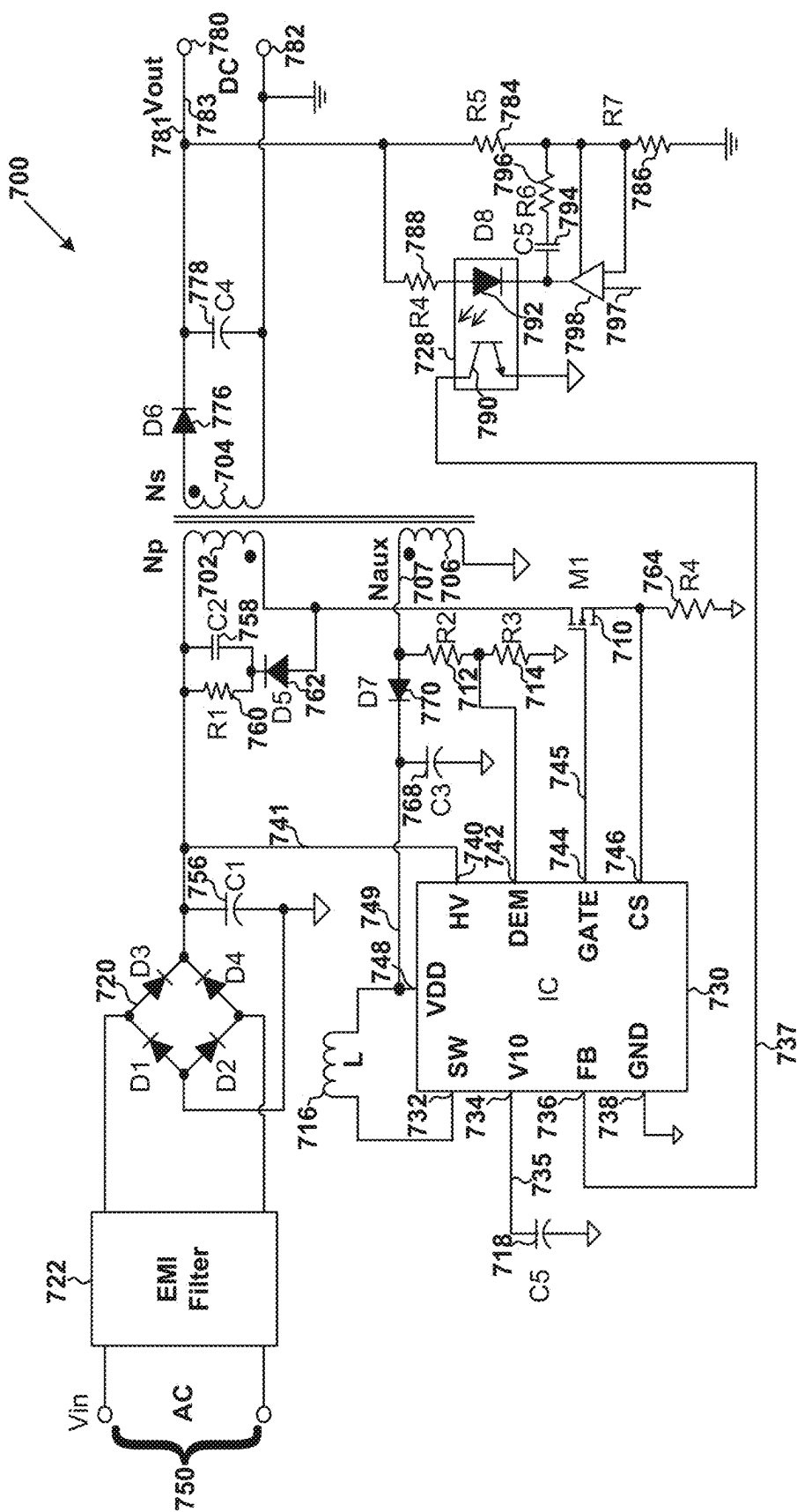


FIG. 7

800

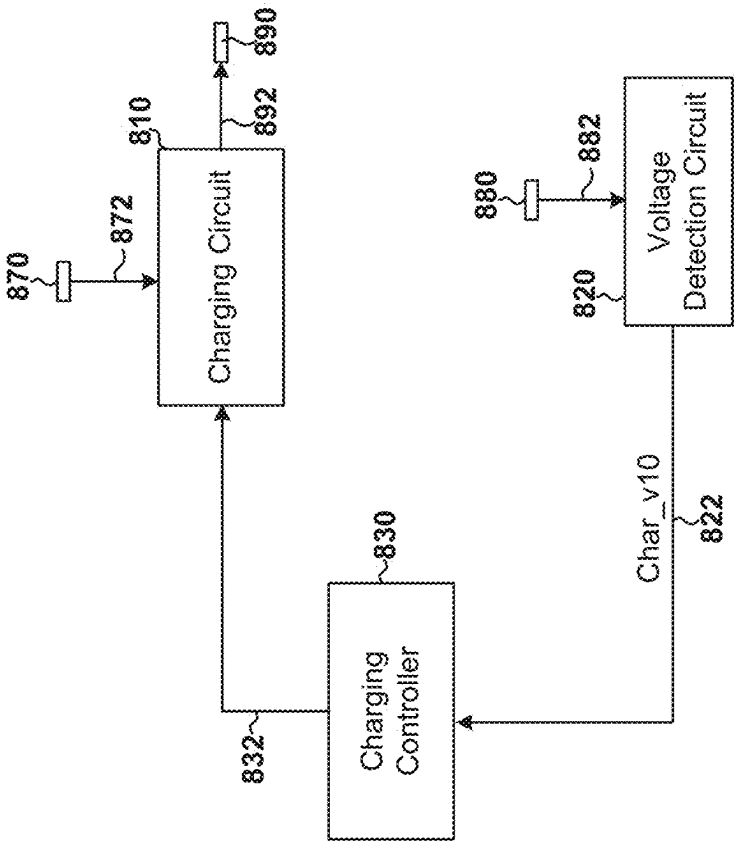


FIG. 8

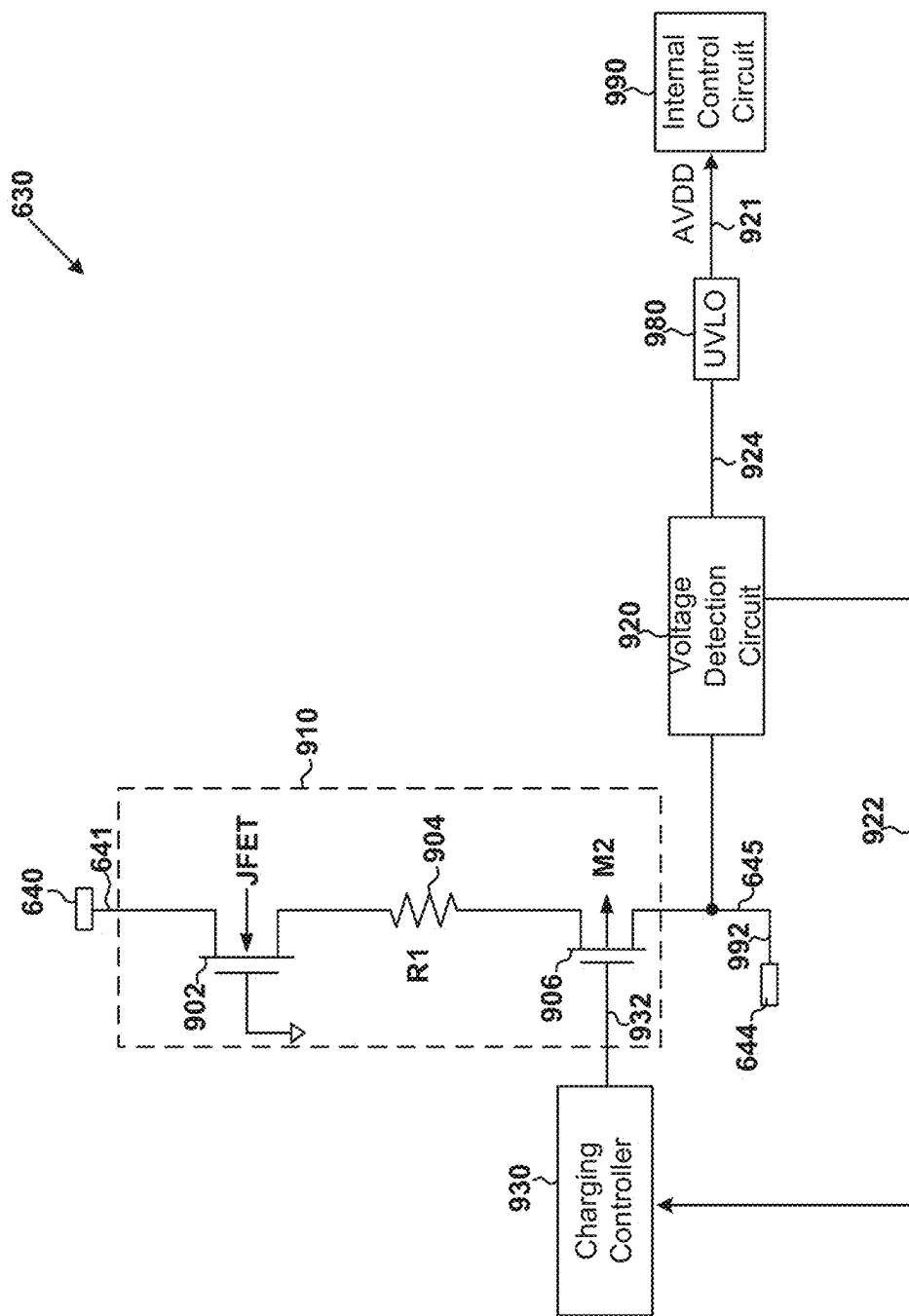


FIG. 9

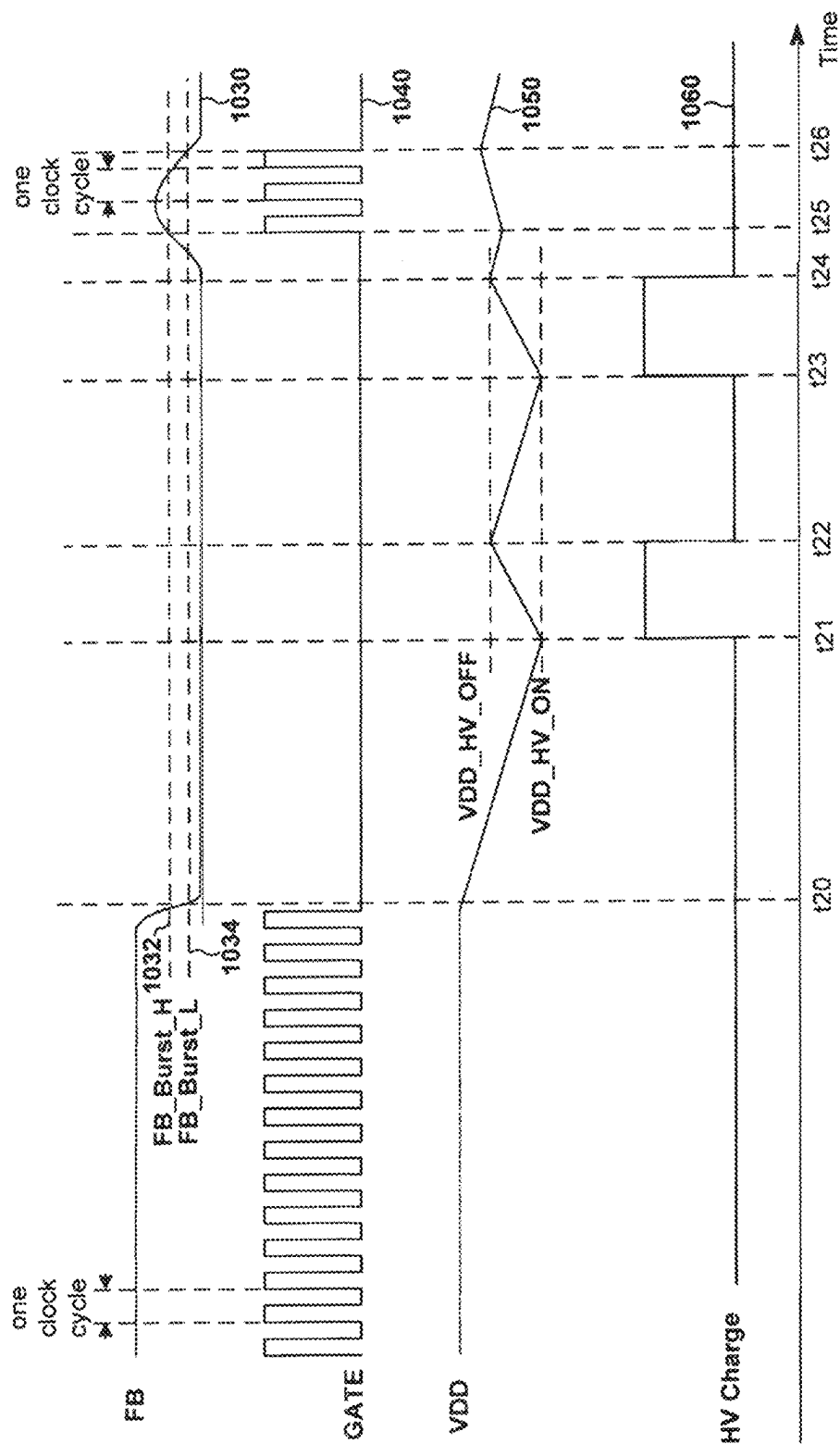


FIG. 10

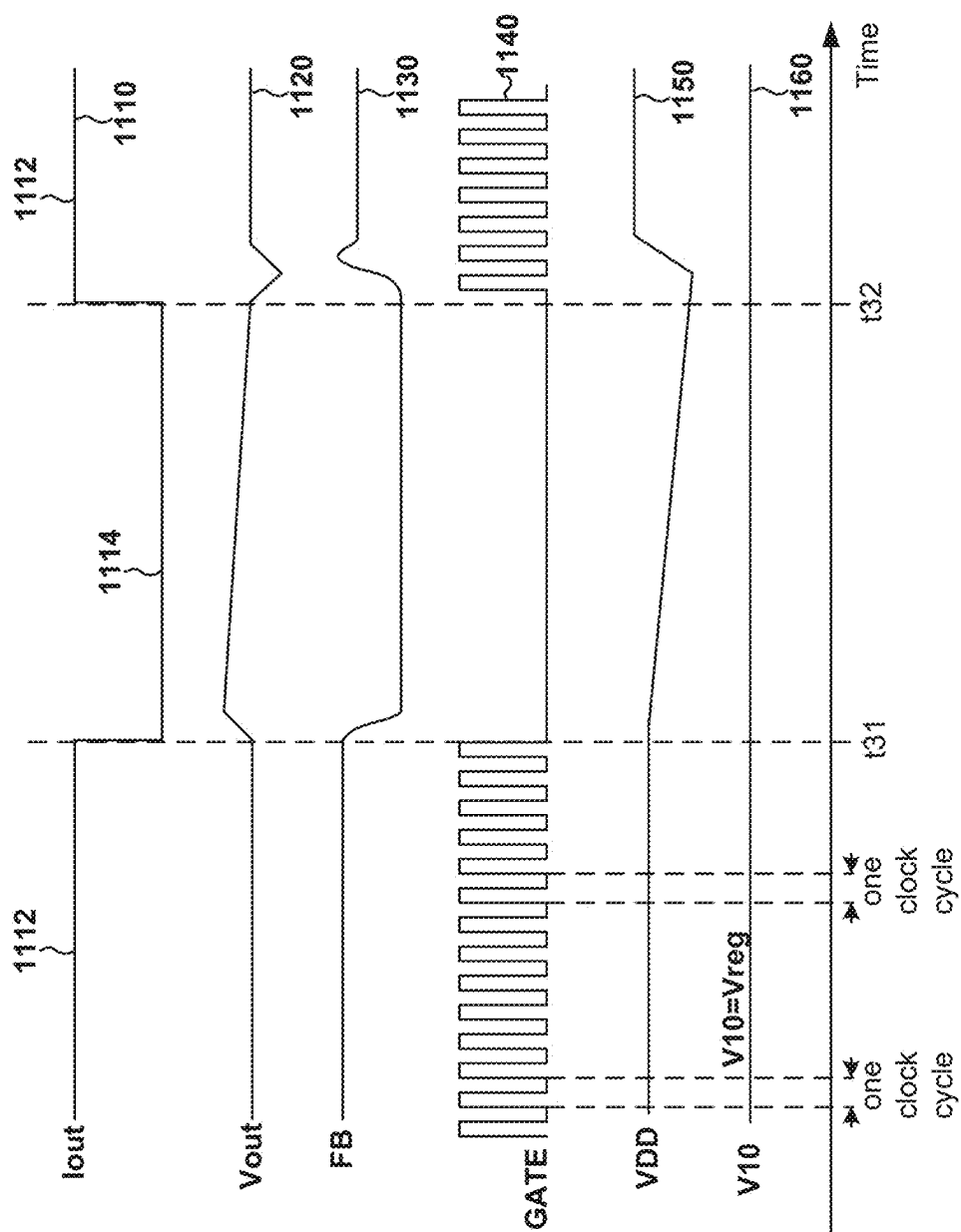


FIG. 11

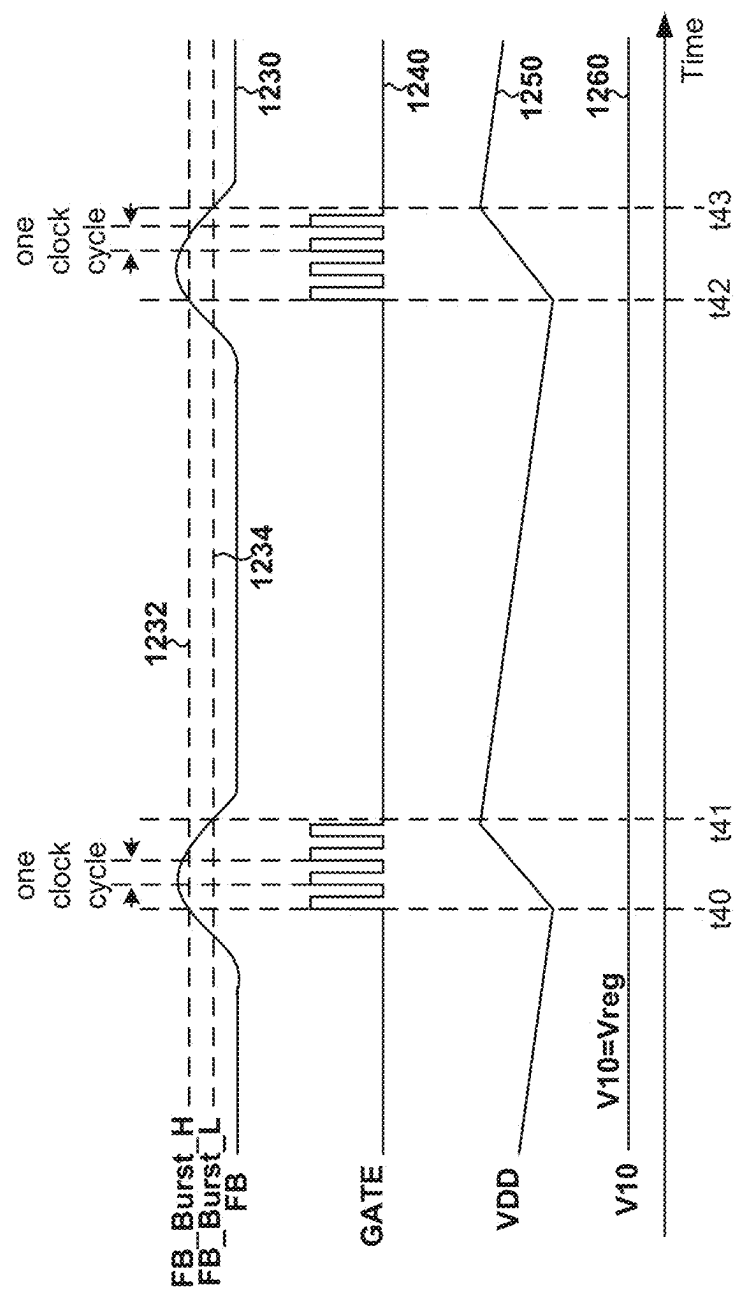


FIG. 12

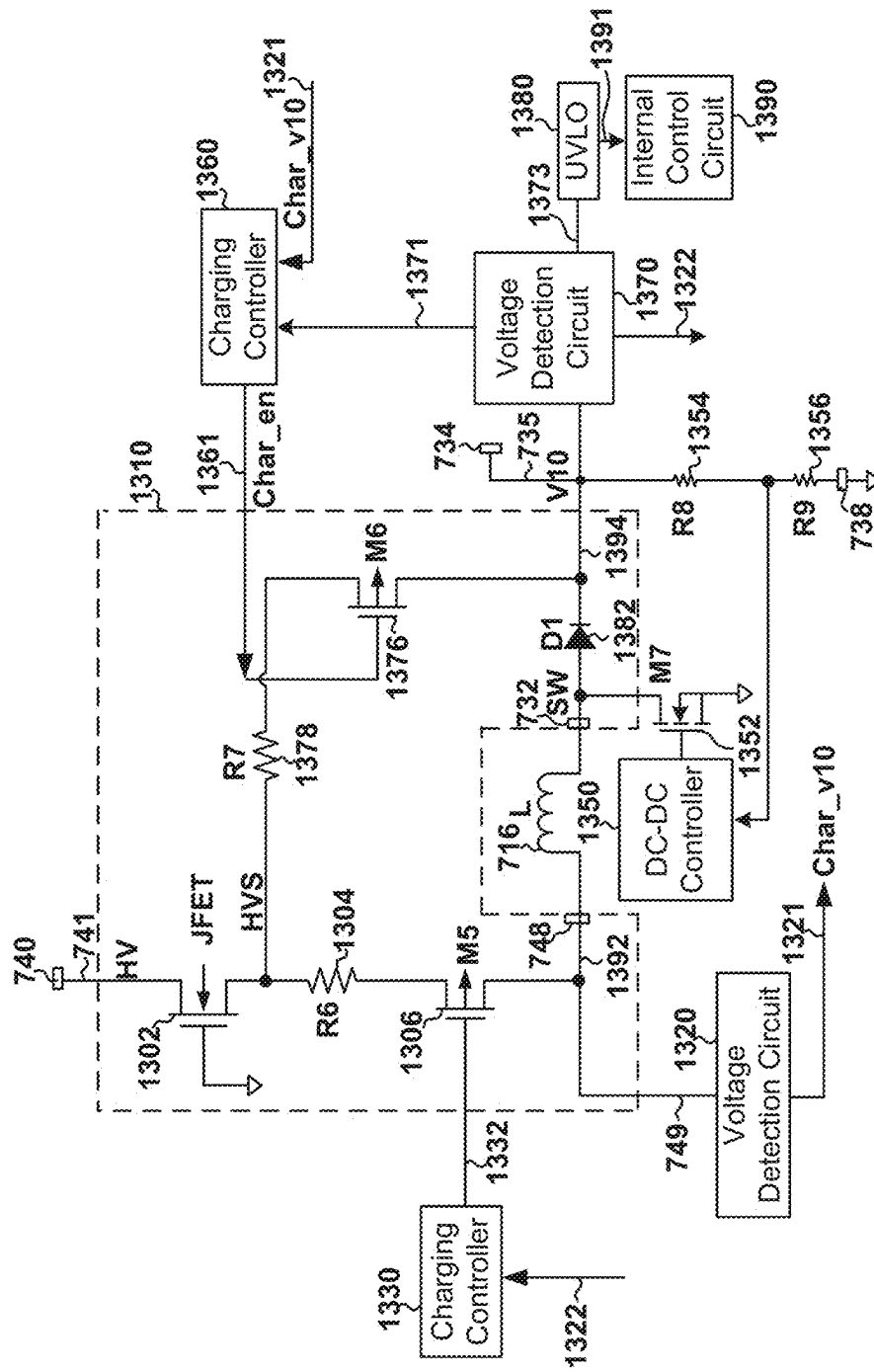


FIG. 13

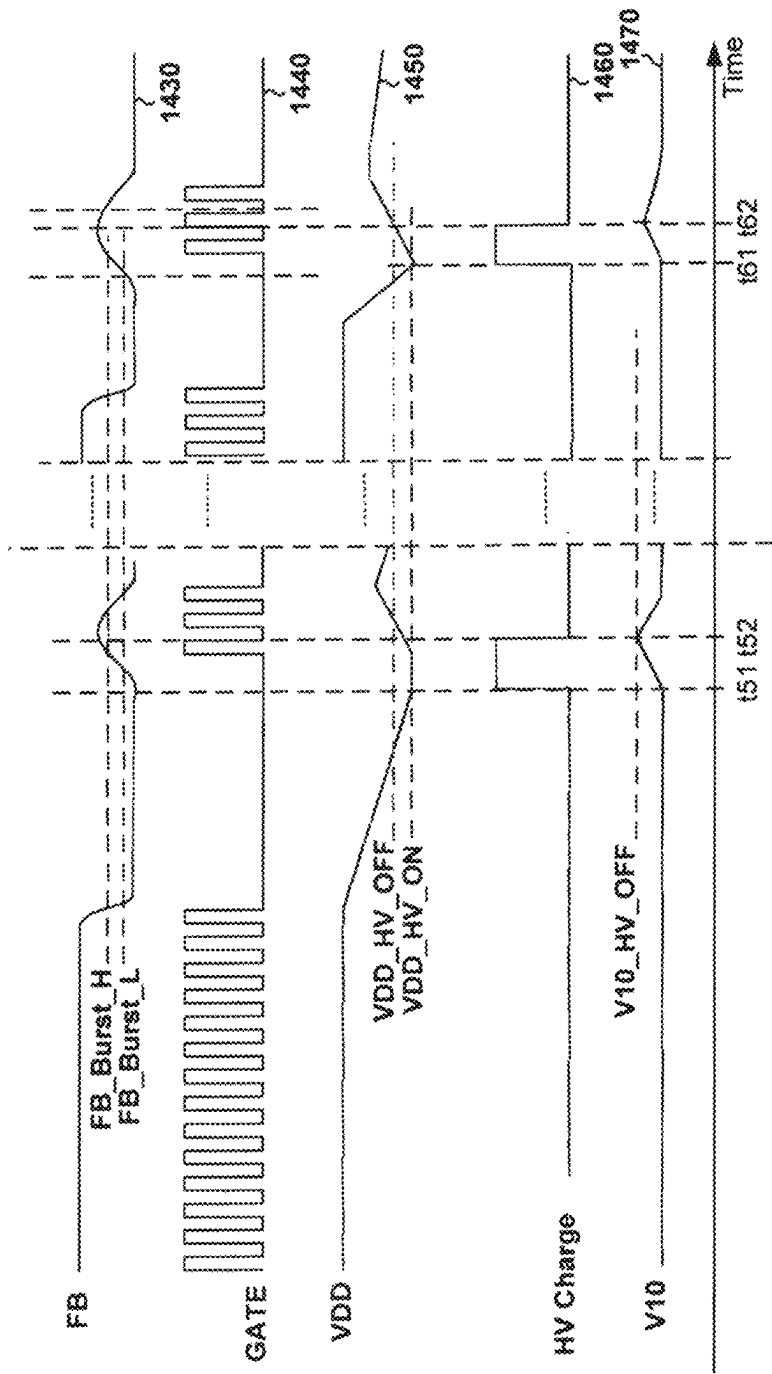
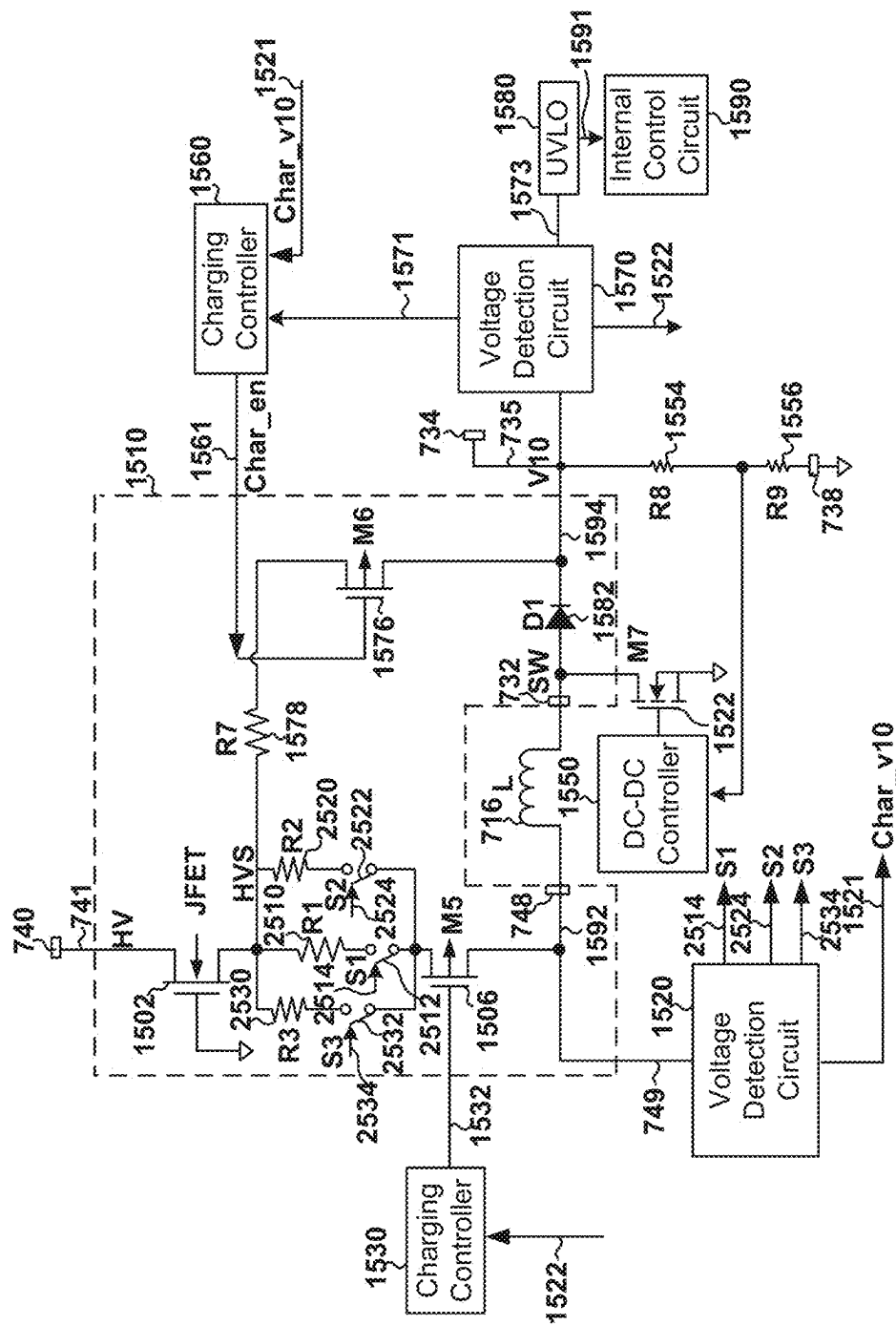


FIG. 14



15
6
E

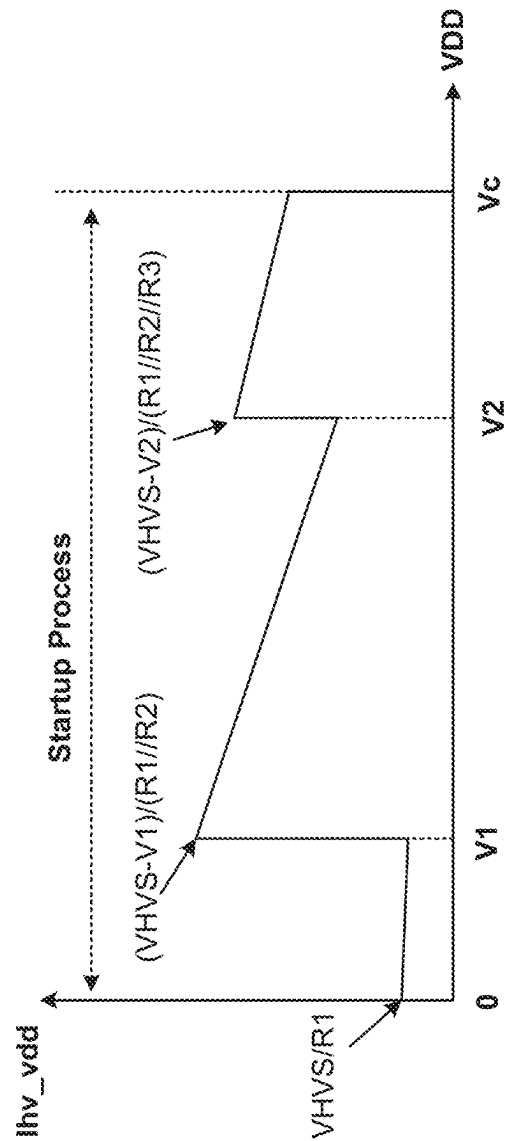


FIG. 16

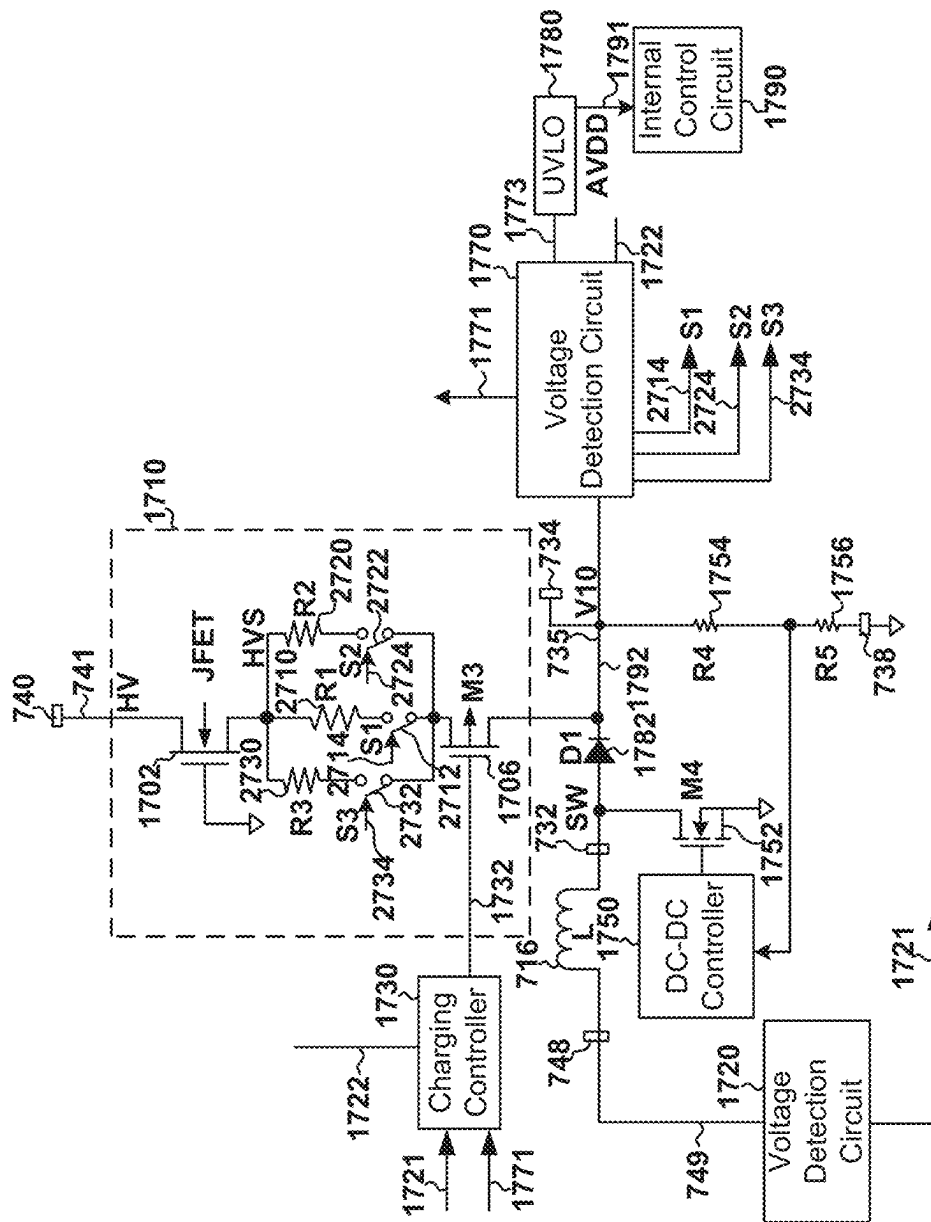


Fig. 17

SYSTEMS AND METHODS FOR PROVIDING POWER TO PULSE-WIDTH-MODULATION CONTROLLERS OF POWER CONVERTERS DURING NORMAL OPERATION

1. CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/131,788, filed Apr. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/465,678, filed Sep. 2, 2021, which claims priority to Chinese Patent Application No. 202010919275.0, filed Sep. 4, 2020, all of the above applications being incorporated by reference herein for all purposes.

2. BACKGROUND OF THE INVENTION

Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the invention provide systems and methods for providing power to pulse-width-modulation controllers of power converters. Merely by way of example, some embodiments of the invention have been applied to power converters during normal operation. But it would be recognized that the invention has a much broader range of applicability.

A conventional power converter often includes a semiconductor chip. For the semiconductor chip, a high-voltage pin (e.g., an HV pin) is generally connected to one or more AC terminals of the power converter through one or more components such as one or more diodes and/or is connected to one or more capacitors (e.g., a bulk capacitor) of the power converter. During the startup process of the semiconductor chip, the high-voltage pin (e.g., an HV pin) charges an external capacitor (e.g., a VDD capacitor) through an internal charging circuit of the chip.

After the voltage (e.g., a VDD voltage) provided by the external capacitor rises to a chip startup threshold, the internal charging circuit is disconnected and the startup process of the semiconductor chip ends. Upon completion of the startup process, the semiconductor chip starts normal operation for the power converter. As the output voltage of the power converter rises, an auxiliary winding of the power converter starts providing power to the semiconductor chip. Often, such start-up process that uses the high-voltage pin (e.g., an HV pin) can speed up the startup and also reduce power consumption of the semiconductor chip after the startup.

FIG. 1 is a simplified diagram showing a conventional power converter. The power converter **100** (e.g., an AC-DC power supply) includes a primary winding **102**, a secondary winding **104**, an auxiliary winding **106**, a transistor **110** (e.g., a transistor M1), a rectifier **120** (e.g., a bridge rectifier), a filter **122** for electromagnetic interference (EMI), diodes **124**, **126**, **162**, **170** and **176**, an optocoupler **128**, a pulse-width-modulation (PWM) controller **130** (e.g., an IC chip), resistors **152**, **154**, **160**, **164**, **166**, **172**, **174**, **184**, **186**, **188** and **196**, capacitors **156**, **158**, **168**, **178** and **194**, and an error amplifier **198** (e.g., TL431). The pulse-width-modulation (PWM) controller **130** (e.g., an IC chip) includes a terminal **132** (e.g., a PRT pin), a terminal **134** (e.g., an FB pin), a terminal **136** (e.g., a CS pin), a terminal **138** (e.g., a GND pin), a terminal **140** (e.g., a HV pin), a terminal **142** (e.g., an NC pin), a terminal **144** (e.g., a VDD pin), and a terminal **146** (e.g., a GATE pin). Additionally, the optocoupler **128** includes a light emitting diode **192** and a photosensitive detector **190**. Moreover, the terminal **134** (e.g., an FB pin) is

used to receive a feedback signal **135** that represents an output voltage **181** (e.g., V_{out}) of the power converter **100**, and the terminal **140** (e.g., a HV pin) is used to receive a voltage **141**.

As shown in FIG. 1, the EMI filter **122** receives an alternating current (AC) voltage **150**. One output terminal of the EMI filter **122** is connected to an anode of the diode **124** (e.g., a diode D1), and another output terminal of the EMI filter **122** is connected to an anode of the diode **126** (e.g., a diode D2). A cathode of the diode **124** (e.g., a diode D1) and a cathode of the diode **126** (e.g., a diode D2) both are connected to one terminal of the resistor **152** (e.g., a resistor R2), which includes another terminal that is connected to one terminal of the resistor **154** (e.g., a resistor R3). Another terminal of the resistor **154** (e.g., a resistor R3) is connected to the terminal **140** (e.g., a HV pin) of the PWM controller **130**.

The anode of the diode **124** (e.g., a diode D1) is connected to an input terminal of the rectifier **120** (e.g., a bridge rectifier), and the anode of the diode **126** (e.g., a diode D2) is connected to another input terminal of the rectifier **120** (e.g., a bridge rectifier). The rectifier **120** (e.g., a bridge rectifier) includes four diodes (e.g., diodes D3, D4, D5, and D6). One output terminal of the rectifier **120** (e.g., a bridge rectifier) is connected to one plate of the capacitor **156** (e.g., a capacitor C2) that is biased to a ground voltage on the primary side, and another output terminal of the rectifier **120** (e.g., a bridge rectifier) is connected to another plate of the capacitor **156** (e.g., a capacitor C2) that is connected to one terminal of the resistor **160** (e.g., a resistor R1), one terminal of the capacitor **158** (e.g., a capacitor C3), and one terminal of the primary winding **102**. Another terminal of the resistor **160** (e.g., a resistor R1) and another terminal of the capacitor **158** (e.g., a capacitor C3) both are connected to a cathode of the diode **162** (e.g., a diode D7). An anode of the diode **162** (e.g., a diode D7) is connected to a drain terminal of the transistor **110** (e.g., a transistor M1) and another terminal of the primary winding **102**.

A gate terminal of the transistor **110** (e.g., a transistor M1) receives a drive signal **147** from the terminal **146** (e.g., a GATE pin) of the PWM controller **130** (e.g., an IC chip), and a source terminal of the transistor **110** (e.g., a transistor M1) is connected to the terminal **136** (e.g., a CS pin) of the PWM controller **130** (e.g., an IC chip) and also connected to one terminal of the resistor **164** (e.g., a resistor R7). Another terminal of the resistor **164** (e.g., a resistor R7) is biased to the ground voltage on the primary side. The terminal **144** (e.g., a VDD pin) of the pulse-width-modulation (PWM) controller **130** (e.g., an IC chip) is connected to one terminal of the resistor **166** (e.g., a resistor R6), and another terminal of the resistor **166** (e.g., a resistor R6) is connected to one plate of the capacitor **168** (e.g., a capacitor C1) and a cathode of the diode **170** (e.g., a diode D8). Another plate of the capacitor **168** (e.g., a capacitor C1) is biased to the ground voltage on the primary side. An anode of the diode **170** (e.g., a diode D8) is connected to one terminal of the auxiliary winding **106**, and another terminal of the auxiliary winding **106** is biased to the ground voltage on the primary side.

The anode of the diode **170** (e.g., a diode D8) is connected to one terminal of the resistor **172** (e.g., a resistor R4), and another terminal of the resistor **172** (e.g., a resistor R4) is connected to the terminal **132** (e.g., a PRT pin) of the PWM controller **130** (e.g., an IC chip) and also connected to one terminal of the resistor **174** (e.g., a resistor R5). Another terminal of the resistor **174** (e.g., a resistor R5) is biased to the ground voltage on the primary side. One terminal of the secondary winding **104** is connected to an anode of the diode

176 (e.g., a diode D9), and a cathode of the diode 176 (e.g., a diode D9) is connected to an output terminal 180 and one terminal of the capacitor 178 (e.g., a capacitor C4). Another terminal of the secondary winding 104 is connected to an output terminal 182 and another terminal of the capacitor 178 (e.g., a capacitor C4). The output terminal 180 provides the output voltage 181 (e.g., V_{out}) and an output current 183 (e.g., I_{out}), and the output terminal 182 is biased to a ground voltage on the secondary side.

The output terminal 180 is connected to one terminal of the resistor 184 (e.g., a resistor R10), and another terminal of the resistor 184 (e.g., a resistor R10) is connected to one terminal of the resistor 196 (e.g., a resistor R9), one terminal of the resistor 186 (e.g., a resistor R11), and an input terminal of the error amplifier 198. Another terminal of the resistor 196 (e.g., a resistor R9) is connected to one terminal of the capacitor 194 (e.g., a capacitor C5), which includes another terminal connected to a cathode of the light emitting diode 192 (e.g., a light emitting diode D10) and connected to an output terminal of the error amplifier 198. Another terminal of the resistor 186 (e.g., R11) is biased to the ground voltage on the secondary side, and another input terminal of the error amplifier 198 is biased to a reference voltage 197. An anode of the light emitting diode 192 (e.g., a light emitting diode D10) is connected to one terminal of the resistor 188 (e.g., a resistor R8), and another terminal of the resistor 188 (e.g., a resistor R8) is connected to the output terminal 180. The terminal 138 (e.g., a GND pin) of the PWM controller 130 (e.g., an IC chip) is biased to the ground voltage on the primary side.

As shown in FIG. 1, after the EMI filter 122 receives the alternating current (AC) voltage 150, the voltage 141 (e.g., an AC voltage) is provided to the terminal 140 (e.g., a HV pin) of the PWM controller 130 (e.g., an IC chip) so that the capacitor 168 (e.g., a capacitor C1) is charged through the terminal 144 (e.g., a VDD pin). If a voltage 145 at the terminal 144 (e.g., a VDD pin) rises above a startup threshold, the PWM controller 130 (e.g., an IC chip) disconnects the charging circuit inside the controller 130 and starts normal operation for the power converter 100. Under the normal operation, the output voltage 181 (e.g., V_{out}) rises so that the auxiliary winding 106, together with the diode 170 (e.g., a diode D8), the capacitor 168 (e.g., a capacitor C1) and the resistor 166 (e.g., a resistor R6), provides power supply to the terminal 144 (e.g., a VDD pin). The voltage 145 at the terminal 144 (e.g., a VDD pin) of the PWM controller 130 (e.g., an IC chip) is determined as follows:

$$V_{DD} \approx \frac{N_{aux}}{N_s} \times V_{out} \quad (\text{Equation 1})$$

where V_{DD} represents the voltage 145 at the terminal 144 (e.g., a VDD pin), and V_{out} represents the output voltage 181 at the output terminal 180. Additionally, N_{aux} represents the number of turns of the auxiliary winding 106, and N_s represents the number of turns of the secondary winding 104.

FIG. 2 is a simplified diagram showing certain conventional components of the PWM controller 130 of the power converter 100 as shown in FIG. 1. The PWM controller 130 (e.g., an IC chip) includes a charging controller 210, a circuit 220 for undervoltage-lockout (UVLO), an internal control circuit 230, a transistor 240 (e.g., a transistor M1), and a diode 250 (e.g., a diode D1).

The circuit 220 for undervoltage-lockout (UVLO) is used to perform the power-up detection and/or the power-down detection for the PWM controller 130 (e.g., an IC chip). If the circuit 220 for undervoltage-lockout (UVLO) detects that the voltage 145 at the terminal 144 (e.g., a VDD pin) rises above the startup threshold, the circuit 220 for undervoltage-lockout (UVLO) outputs a signal 221 (e.g., AVDD) to the internal control circuit 230 in order to power on other components of the PWM controller 130 (e.g., an IC chip), where the signal 221 (e.g., AVDD) indicates the end of the startup process. If the circuit 220 for undervoltage-lockout (UVLO) detects that the voltage 145 at the terminal 144 (e.g., a VDD pin) falls below an undervoltage threshold, the circuit 220 for undervoltage-lockout (UVLO) outputs the signal 221 (e.g., AVDD) to the internal control circuit 230 in order to power off other components of the PWM controller 130 (e.g., an IC chip). As an example, the internal control circuit 230 includes various control circuits, and the internal control circuit 230 does not include the charging controller 210. For example, the internal control circuit 230 includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

A drain terminal of the transistor 240 (e.g., a transistor M1) is connected to the terminal 140 (e.g., a HV pin) of the PWM controller 130 (e.g., an IC chip), a gate terminal of the transistor 240 (e.g., a transistor M1) is connected to an output terminal of the charging controller 210, and a source terminal of the transistor 240 (e.g., a transistor M1) is connected to an anode of the diode 250. A cathode of the diode 250 is connected to the terminal 144 (e.g., a VDD pin) of the PWM controller 130 (e.g., an IC chip) and also connected to an input terminal of the circuit 220 for undervoltage-lockout (UVLO). At an output terminal of the circuit 220 for undervoltage-lockout (UVLO), the signal 221 (e.g., AVDD) is sent to an input terminal of the internal control circuit 230.

As shown in FIG. 1 and FIG. 2, after the EMI filter 122 receives the AC voltage 150, the PWM controller 130 (e.g., an IC chip) undergoes a startup process. During the startup process, if the circuit 220 for undervoltage-lockout (UVLO) detects that the voltage 145 at the terminal 144 (e.g., a VDD pin) has not reached the startup threshold, the circuit 220 for undervoltage-lockout (UVLO) outputs a signal 223 to an input terminal of the charging controller 210, and in response, the charging controller 210 generates a signal 211 at the output terminal to turn on the transistor 240 (e.g., a transistor M1) and to raise the voltage 145 at the terminal 144 (e.g., a VDD pin). If the circuit 220 for undervoltage-lockout (UVLO) detects that the voltage 145 at the terminal 144 (e.g., a VDD pin) has risen above the startup threshold, the circuit 220 for undervoltage-lockout (UVLO) outputs the signal 223 to the input terminal of the charging controller 210, and in response, the charging controller 210 generates the signal 211 at the output terminal to turn off the transistor 240 (e.g., a transistor M1) in order to disconnect the charging circuit inside the controller 130 and starts normal operation for the power converter 100.

FIG. 3 is a simplified diagram showing another conventional power converter. The power converter 300 (e.g., an AC-DC power supply) includes a primary winding 302, a secondary winding 304, an auxiliary winding 306, a transistor 310 (e.g., a transistor M1), a rectifier 320 (e.g., a bridge rectifier), a filter 322 for electromagnetic interference (EMI), an optocoupler 328, a pulse-width-modulation (PWM) controller 330 (e.g., an IC chip), resistors 312, 314,

360, 364, 384, 386, 388 and 396, capacitors 318, 356, 358, 368, 378 and 394, diodes 362, 370 and 376, an error amplifier 398 (e.g., TL431), and an inductor 316. The pulse-width-modulation (PWM) controller 330 (e.g., an IC chip) includes a terminal 332 (e.g., a SW pin), a terminal 334 (e.g., a V10 pin), a terminal 336 (e.g., an FB pin), a terminal 338 (e.g., a GND pin), a terminal 340 (e.g., a HV pin), a terminal 342 (e.g., a DEM pin), a terminal 344 (e.g., a GATE pin), a terminal 346 (e.g., a CS pin), and a terminal 348 (e.g., a VDD pin). Additionally, the optocoupler 328 includes a light emitting diode 392 and a photosensitive detector 390. Moreover, the terminal 336 (e.g., an FB pin) is used to receive a feedback signal 337 that represents an output voltage 381 (e.g., V_{out}) of the power converter 300, and the terminal 340 (e.g., a HV pin) is used to receive a voltage 341.

As shown in FIG. 3, the EMI filter 322 receives an alternating current (AC) voltage 350. Two output terminals of the EMI filter 322 are connected to two input terminals of the rectifier 320 (e.g., a bridge rectifier) respectively. The rectifier 320 (e.g., a bridge rectifier) includes four diodes (e.g., diodes D1, D2, D3, and D4). One output terminal of the rectifier 320 (e.g., a bridge rectifier) is connected to one plate of the capacitor 356 (e.g., a capacitor C1) that is biased to a ground voltage on the primary side, and another output terminal of the rectifier 320 (e.g., a bridge rectifier) is connected to another plate of the capacitor 356 (e.g., a capacitor C1) that is connected to one terminal of the resistor 360 (e.g., a resistor R1), one terminal of the capacitor 358 (e.g., a capacitor C2), and one terminal of the primary winding 302. Another terminal of the resistor 360 (e.g., a resistor R1) and another terminal of the capacitor 358 (e.g., a capacitor C2) both are connected to a cathode of the diode 362 (e.g., a diode D5). An anode of the diode 362 (e.g., a diode D5) is connected to a drain terminal of the transistor 310 (e.g., a transistor M1) and another terminal of the primary winding 302.

A gate terminal of the transistor 310 (e.g., a transistor M1) receives a drive signal 345 from the terminal 344 (e.g., a GATE pin) of the PWM controller 330 (e.g., an IC chip), and a source terminal of the transistor 310 (e.g., a transistor M1) is connected to the terminal 346 (e.g., a CS pin) of the PWM controller 330 (e.g., an IC chip) and also connected to one terminal of the resistor 364 (e.g., a resistor R4). Another terminal of the resistor 364 (e.g., a resistor R4) is biased to the ground voltage on the primary side. The terminal 348 (e.g., a VDD pin) of the pulse-width-modulation (PWM) controller 330 (e.g., an IC chip) is connected to one terminal of the inductor 316 (e.g., an inductor L), one plate of the capacitor 368 (e.g., a capacitor C3) and a cathode of the diode 370 (e.g., a diode D7). Another terminal of the inductor 316 (e.g., an inductor L) is connected to the terminal 332 (e.g., a SW pin) of the PWM controller 330 (e.g., an IC chip). Another plate of the capacitor 368 (e.g., a capacitor C3) is biased to the ground voltage on the primary side. An anode of the diode 370 (e.g., a diode D7) is connected to one terminal of the resistor 312 (e.g., a resistor R2) and one terminal of the auxiliary winding 306 that corresponds to a voltage 307. Another terminal of the auxiliary winding 306 is biased to the ground voltage on the primary side. Another terminal of the resistor 312 (e.g., a resistor R2) is connected to the terminal 342 (e.g., a DEM pin) of the PWM controller 330 (e.g., an IC chip) and one terminal of the resistor 314 (e.g., a resistor R3). Another terminal of the resistor 314 (e.g., a resistor R3) is biased to the ground voltage on the primary side. The terminal 334 (e.g., a V10 pin) of the PWM controller 330 (e.g., an IC

chip) is connected to one terminal of the capacitor 318 (e.g., a capacitor C5), and another terminal of the capacitor 318 (e.g., a capacitor C5) is biased to the ground voltage on the primary side.

One terminal of the secondary winding 304 is connected to an anode of the diode 376 (e.g., a diode D6), and a cathode of the diode 376 (e.g., a diode D6) is connected to an output terminal 380 and one terminal of the capacitor 378 (e.g., a capacitor C4). Another terminal of the secondary winding 304 is connected to an output terminal 382 and another terminal of the capacitor 378 (e.g., a capacitor C4). The output terminal 380 provides the output voltage 381 (e.g., V_{out}) and an output current 383 (e.g., I_{out}), and the output terminal 382 is biased to a ground voltage on the secondary side.

The output terminal 380 is connected to one terminal of the resistor 384 (e.g., a resistor R5), and another terminal of the resistor 384 (e.g., a resistor R5) is connected to one terminal of the resistor 396 (e.g., a resistor R6), one terminal of the resistor 386 (e.g., a resistor R7), and an input terminal of the error amplifier 398. Another terminal of the resistor 396 (e.g., a resistor R6) is connected to one terminal of the capacitor 394 (e.g., a capacitor C5), which includes another terminal connected to a cathode of the light emitting diode 392 (e.g., a light emitting diode D8) and connected to an output terminal of the error amplifier 398. Another terminal of the resistor 386 (e.g., R7) is biased to the ground voltage on the secondary side, and another input terminal of the error amplifier 398 is biased to a reference voltage 397. An anode of the light emitting diode 392 (e.g., a light emitting diode D8) is connected to one terminal of the resistor 388 (e.g., a resistor R4), and another terminal of the resistor 388 (e.g., a resistor R4) is connected to the output terminal 380. The terminal 338 (e.g., a GND pin) of the PWM controller 330 (e.g., an IC chip) is biased to the ground voltage on the primary side.

As shown in FIG. 3, after the EMI filter 322 receives the alternating current (AC) voltage 350, the voltage 341 is provided to the terminal 340 (e.g., a HV pin) of the PWM controller 330 (e.g., an IC chip) so that the capacitor 368 (e.g., a capacitor C3) is charged through the terminal 348 (e.g., a VDD pin). When an internal DC-DC converter of the PWM controller 330 (e.g., an IC chip) does not operate, the terminal 348 (e.g., a VDD pin) is used to raise a voltage 335 at the terminal 334 (e.g., a V10 pin) of the PWM controller 330 (e.g., an IC chip) by charging the capacitor 318 (e.g., a capacitor C5) through the external inductor 316 (e.g., an inductor L), the terminal 332 (e.g., a SW pin) and an internal Schottky diode of the PWM controller 330 (e.g., an IC chip). If the voltage 335 at the terminal 334 (e.g., a V10 pin) rises above a startup threshold, the PWM controller 330 (e.g., an IC chip) disconnects the charging circuit inside the controller 330 that is used to charge the capacitor 368 (e.g., a capacitor C3) by the voltage 341 at the terminal 340 (e.g., a HV pin), and the PWM controller 330 (e.g., an IC chip) also starts normal operation for the power converter 300.

Under the normal operation, the output voltage 381 (e.g., V_{out}) rises so that the voltage 307 at one terminal of the auxiliary winding 306 also rises. The auxiliary winding 306, together with the diode 370 (e.g., a diode D7) and the capacitor 368 (e.g., a capacitor C3), provides power supply, to the terminal 348 (e.g., a VDD pin). A voltage 349 at the terminal 348 (e.g., a VDD pin) of the PWM controller 330 (e.g., an IC chip) is determined as follows:

$$V_{DD} \approx \frac{N_{aux}}{N_s} \times V_{out} \quad (\text{Equation 2})$$

where V_{DD} represents the voltage **349** at the terminal **348** (e.g., a VDD pin), and V_{out} represents the output voltage **381** at the output terminal **380**. Additionally, N_{aux} represents the number of turns of the auxiliary winding **306**, and N_s represents the number of turns of the secondary winding **304**.

As shown by Equation 2, the higher the output voltage **381** (e.g., V_{out}), the higher the voltage **349** (e.g., V_{DD}), and the lower the output voltage **381** (e.g., V_{out}), the lower the voltage **349** (e.g., V_{DD}). When the output voltage **381** (e.g., V_{out}) is low, the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) uses the voltage **349** (e.g., V_{DD}) to generate the voltage **335** at the terminal **334** (e.g., a V10 pin), where the voltage **335** is stable and higher than the voltage **349** (e.g., V_{DD}). The voltage **335** at the terminal **334** (e.g., a V10 pin) is used to provide stable power supply to the PWM controller **330** (e.g., an IC chip), even though the output voltage **381** (e.g., V_{out}) varies widely.

If the voltage **349** (e.g., V_{DD}) is lower than the reference voltage (e.g., V_{reg}), the voltage **335** at the terminal **334** (e.g., a V10 pin) is generated by the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) using the voltage **349** (e.g., V_{DD}), and the generated voltage **335** is larger than the voltage **349** (e.g., V_{DD}) and is equal to the reference voltage (e.g., V_{reg}). If the voltage **349** (e.g., V_{DD}) is higher than the reference voltage (e.g., V_{reg}), the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) is idle, and the voltage **335** at the terminal **334** (e.g., a V10 pin) is approximately equal to the voltage **349** (e.g., V_{DD}).

Hence it is highly desirable to improve the techniques related to pulse-width-modulation (PWM) controllers.

3. BRIEF SUMMARY OF THE INVENTION

Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the invention provide systems and methods for providing power to pulse-width-modulation controllers of power converters. Merely by way of example, some embodiments of the invention have been applied to power converters during normal operation. But it would be recognized that the invention has a much broader range of applicability.

According to some embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a capacitor and biased to a second voltage; a voltage detector configured to receive the second voltage from the second terminal and generate a detection signal based at least in part on the second voltage; a charging controller configured to receive the detection signal and generate a first control signal based at least in part on the detection signal; and a charging current generator configured to receive the first voltage from the first terminal and receive the first control signal from the charging controller; wherein the voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated;

wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the second terminal to charge the capacitor associated with the second voltage.

According to certain embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a first capacitor and biased to a second voltage; a third terminal connected to a second capacitor and biased to a third voltage; a first voltage detector configured to receive the second voltage from the second terminal and generate a first detection signal based at least in part on the second voltage; a first charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a charging current generator configured to receive the first voltage from the first terminal, the charging current generator including a first current generator and a second current generator, the first current generator being configured to receive the first control signal from the first charging controller; wherein the first voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the first charging controller is further configured to: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a first charging current is to be generated; wherein the first current generator is further configured to: in response to the first control signal indicating that the first charging current is to be generated, generate the first charging current that flows through the third terminal to charge the second capacitor associated with the third voltage.

According to some embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a first capacitor and biased to a second voltage; a third terminal connected to a second capacitor and biased to a third voltage; a first voltage detector configured to receive the second voltage from the second terminal and generate a first detection signal based at least in part on the second voltage; a charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a charging current generator configured to receive the first voltage from the first terminal and configured to receive the first control signal from the charging controller; wherein the first voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated; wherein the current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the third terminal to charge the second capacitor associated with the third voltage.

According to certain embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a capacitor; generating a detection signal based at least in part on the second voltage;

receiving the detection signal; generating a first control signal based at least in part on the detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the detection signal includes: in response to the detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the charging current is to be generated, generating the charging current to charge the capacitor associated with the second voltage.

According to some embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a first capacitor; receiving a third voltage associated with a second capacitor; generating a first detection signal based at least in part on the second voltage; receiving the first detection signal; generating a first control signal based at least in part on the first detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a first detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the first detection signal includes: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a first charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the first charging current is to be generated, generating the first charging current to charge the second capacitor associated with the third voltage.

According to certain embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a first capacitor; receiving a third voltage associated with a second capacitor; generating a first detection signal based at least in part on the second voltage; receiving the first detection signal; generating a first control signal based at least in part on the first detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a first detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the first detection signal includes: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the charging current is to be generated, generating the charging current to charge the second capacitor associated with the third voltage.

Depending upon embodiment, one or more benefits may be achieved. These benefits and various additional objects,

features and advantages of the present invention can be fully appreciated with reference to the detailed description and accompanying drawings that follow.

4. BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified diagram showing a conventional power converter.

FIG. 2 is a simplified diagram showing certain conventional components of the PWM controller of the power converter as shown in FIG. 1.

FIG. 3 is a simplified diagram showing another conventional power converter.

FIG. 4 shows simplified timing diagrams under certain dynamic-load conditions for the power converter as shown in FIG. 1 according to certain embodiments.

FIG. 5 shows simplified timing diagrams under no-load condition for the power converter as shown in FIG. 1 according to some embodiments.

FIG. 6 is a simplified diagram showing a power converter according to certain embodiments of the present invention.

FIG. 7 is a simplified diagram showing a power converter according to some embodiments of the present invention.

FIG. 8 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 6 and/or certain components of the PWM controller of the power converter as shown in FIG. 7 according to some embodiments of the present invention.

FIG. 9 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 6 according to some embodiments of the present invention.

FIG. 10 shows simplified timing diagrams under certain dynamic-load and/or no-load conditions for the power converter that includes the PWM controller as shown in FIG. 6 and FIG. 9 according to some embodiments.

FIG. 11 shows simplified timing diagrams under certain dynamic-load conditions for the power converter as shown in FIG. 3 according to certain embodiments.

FIG. 12 shows simplified timing diagrams under no-load condition for the power converter as shown in FIG. 3 according to certain embodiments.

FIG. 13 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 7 according to some embodiments of the present invention.

FIG. 14 shows simplified timing diagrams under certain dynamic-load and/or no-load conditions for the power converter that includes the PWM controller as shown in FIG. 7 and FIG. 13 according to some embodiments of the present invention.

FIG. 15 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 7 according to some embodiments of the present invention.

FIG. 16 shows a simplified diagram showing certain charging current as a function of certain voltage for the power converter that includes the PWM controller as shown in FIG. 7 and FIG. 15 according to some embodiments of the present invention.

FIG. 17 is a simplified diagram showing certain components of the PWM controller of the power converter as shown in FIG. 7 according to some embodiments of the present invention.

5. DETAILED DESCRIPTION OF THE INVENTION

Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the

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invention provide systems and methods for providing power to pulse-width-modulation controllers of power converters. Merely by way of example, some embodiments of the invention have been applied to power converters during normal operation. But it would be recognized that the invention has a much broader range of applicability.

As shown in FIG. 1 and FIG. 2, after the power converter 100 starts the normal operation, the auxiliary winding 106 is used to supply power to the terminal 144 (e.g., a VDD pin) of the PWM controller 130 (e.g., an IC chip) according to some embodiments. For example, under certain dynamic-load and/or no-load conditions, the PWM controller 130 (e.g., an IC chip) keeps the gate signal 147 at a logic low level for multiple clock cycles due to the adjustment of the feedback signal 135. As an example, during these multiple clock cycles, the auxiliary winding 106 does not provide additional power, and the voltage 145 at the terminal 144 (e.g., a VDD pin) is supported by the capacitor 168 (e.g., a capacitor C1). According to certain embodiments, with the discharge of the capacitor 168 (e.g., a capacitor C1), the voltage 145 at the terminal 144 (e.g., a VDD pin) decreases with time. For example, if the voltage 145 at the terminal 144 (e.g., a VDD pin) drops below a predetermined threshold, the output voltage 181 (e.g., V_{out}) drifts up. As an example, under the low temperature and/or a harsh working condition, with the discharge of the capacitor 168 (e.g., a capacitor C1), the voltage 145 at the terminal 144 (e.g., a VDD pin) decreases more quickly due to the decrease in capacitance of the capacitor 168 (e.g., a capacitor C1), causing insufficient power supply for the PWM controller 130 (e.g., an IC chip) and also triggering undervoltage protection.

FIG. 4 shows simplified timing diagrams under certain dynamic-load conditions for the power converter 100 as shown in FIG. 1 according to certain embodiments. The waveform 410 represents the output current 183 (e.g., I_{out}) as a function of time, the waveform 420 represents the output voltage 181 (e.g., V_{out}) as a function of time, the waveform 430 represents the feedback signal 135 as a function of time, the waveform 440 represents the drive signal 147 as a function of time, and the waveform 450 represents the voltage 145 (e.g., V_{DD}) as a function of time.

In some embodiments, before time t_1 , the output current 183 (e.g., I_{out}) is at a current level 412, and within each clock cycle, the drive signal 147 changes between a logic high level and a logic low level. For example, at time t_1 , the output current 183 (e.g., I_{out}) decreases from the current level 412 to a current level 414. As an example, from time t_1 to time t_2 , the output current 183 (e.g., I_{out}) remains at the current level 414. For example, at time t_2 , the output current 183 (e.g., I_{out}) increases from the current level 414 to the current level 412. As an example, after time t_2 , the output current 183 (e.g., I_{out}) remains at the current level 412. In some examples, the current level 412 is higher than the current level 414. For example, if the output current 183 (e.g., I_{out}) is at the current level 412, the power converter 100 is under a high-load condition. As an example, if the output current 183 (e.g., I_{out}) is at the current level 414, the power converter 100 is under a light-load condition. In certain embodiments, from time t_1 to time t_2 , the power converter 100 is under the light-load condition. For example, under the light-load condition, the output voltage 181 (e.g., V_{out}) overshoots as shown by the waveform 420, the feedback signal 135 drops as shown by the waveform 430, the drive signal 147 remains at the logic low level during multiple clock cycles as shown by the waveform 440, and

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the voltage 145 (e.g., V_{DD}) decreases gradually from time t_1 to time t_2 as shown by the waveform 450.

FIG. 5 shows simplified timing diagrams under no-load condition for the power converter 100 as shown in FIG. 1 according to some embodiments. The waveform 530 represents the feedback signal 135 as a function of time, the waveform 540 represents the drive signal 147 as a function of time, and the waveform 550 represents the voltage 145 (e.g., V_{DD}) as a function of time.

According to certain embodiments, under no-load condition, the output current 183 (e.g., I_{out}) is equal to zero in magnitude. For example, at time t_{10} , the feedback signal 135 becomes larger than a signal threshold 532 (e.g., FB_Burst_H), and at time t_{11} , the feedback signal 135 becomes smaller than a signal threshold 534 (e.g., FB_Burst_L) as shown by the waveform 530. As an example, at time t_{12} , the feedback signal 135 becomes larger than the signal threshold 532 (e.g., FB_Burst_H), and at time t_{13} , the feedback signal 135 becomes smaller than the signal threshold 534 (e.g., FB_Burst_L) as shown by the waveform 530. For example, the signal threshold 532 (e.g., FB_Burst_H) is larger than the signal threshold 534 (e.g., FB_Burst_L) in magnitude.

In some embodiments, from time t_{10} to time t_{11} , the drive signal 147 changes between a logic high level and a logic low level within each clock cycle as shown by the waveform 540, and the voltage 145 (e.g., V_{DD}) increases as shown by the waveform 550. In certain embodiments, time t_{11} to time t_{12} , the drive signal 147 remains at the logic low level during multiple clock cycles as shown by the waveform 540, and the voltage 145 (e.g., V_{DD}) decreases gradually as shown by the waveform 550. In some embodiments, from time t_{12} to time t_{13} , the drive signal 147 changes between the logic high level and the logic low level within each clock cycle as shown by the waveform 540, and the voltage 145 (e.g., V_{DD}) increases as shown by the waveform 550.

As shown in FIG. 4 and FIG. 5, under certain dynamic-load and/or no-load conditions, when the drive signal 147 remains at the logic low level during multiple clock cycles, the voltage 145 (e.g., V_{DD}) decreases gradually according to certain embodiments. For example, if the voltage 145 (e.g., V_{DD}) becomes too small in magnitude, the undervoltage protection is triggered for the PWM controller 130 (e.g., an IC chip) because of insufficient power supply.

FIG. 6 is a simplified diagram showing a power converter according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The power converter 600 (e.g., an AC-DC power supply) includes a primary winding 602, a secondary winding 604, an auxiliary winding 606, a transistor 610 (e.g., a transistor M1), a rectifier 620 (e.g., a bridge rectifier), a filter 622 for electromagnetic interference (EMI), diodes 624, 626, 662, 670 and 676, an optocoupler 628, a pulse-width-modulation (PWM) controller 630 (e.g., an IC chip), resistors 652, 654, 660, 664, 666, 672, 674, 684, 686, 688 and 696, capacitors 656, 658, 668, 678 and 694, and an error amplifier 698, where the pulse-width-modulation (PWM) controller 630 (e.g., an IC chip) includes the PWM controller 800 as shown in FIG. 8 and/or the PWM controller 630 as shown in FIG. 9. For example, the pulse-width-modulation (PWM) controller 630 (e.g., an IC chip) includes a terminal 632 (e.g., a PRT pin), a terminal 634 (e.g., an FB pin), a terminal 636 (e.g., a CS pin), a terminal 638 (e.g., a GND pin), a terminal 640 (e.g., a HV pin), a terminal 642 (e.g., an NC pin), a terminal 644 (e.g., a VDD pin), and a

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terminal **646** (e.g., a GATE pin). As an example, the opto-coupler **628** includes a light emitting diode **692** and a photosensitive detector **690**. For example, the terminal **634** (e.g., an FB pin) is used to receive a feedback signal **635** that represents an output voltage **681** (e.g., V_{out}) of the power converter **600**, and the terminal **640** (e.g., a HV pin) is used to receive a voltage **641**. As an example, the error amplifier **698** includes TL431.

As shown in FIG. 6, the EMI filter **622** receives an alternating current (AC) voltage **650** according to some embodiments. For example, one output terminal of the EMI filter **622** is connected to an anode of the diode **624** (e.g., a diode D1), and another output terminal of the EMI filter **622** is connected to an anode of the diode **626** (e.g., a diode D2). As an example, a cathode of the diode **624** (e.g., a diode D1) and a cathode of the diode **626** (e.g., a diode D2) both are connected to one terminal of the resistor **652** (e.g., a resistor R2), which includes another terminal that is connected to one terminal of the resistor **654** (e.g., a resistor R3). For example, another terminal of the resistor **654** (e.g., a resistor R3) is connected to the terminal **640** (e.g., a HV pin) of the PWM controller **630**.

According to certain embodiments, the anode of the diode **624** (e.g., a diode D1) is connected to an input terminal of the rectifier **620** (e.g., a bridge rectifier), and the anode of the diode **626** (e.g., a diode D2) is connected to another input terminal of the rectifier **620** (e.g., a bridge rectifier). For example, the rectifier **620** (e.g., a bridge rectifier) includes four diodes (e.g., diodes D3, D4, D5, and D6). As an example, one output terminal of the rectifier **620** (e.g., a bridge rectifier) is connected to one plate of the capacitor **656** (e.g., a capacitor C2) that is biased to a ground voltage on the primary side, and another output terminal of the rectifier **620** (e.g., a bridge rectifier) is connected to another plate of the capacitor **656** (e.g., a capacitor C2) that is connected to one terminal of the resistor **660** (e.g., a resistor R1), one terminal of the capacitor **658** (e.g., a capacitor C3), and one terminal of the primary winding **602**. For example, another terminal of the resistor **660** (e.g., a resistor R1) and another terminal of the capacitor **658** (e.g., a capacitor C3) both are connected to a cathode of the diode **662** (e.g., a diode D7). As an example, an anode of the diode **662** (e.g., a diode D7) is connected to a drain terminal of the transistor **610** (e.g., a transistor M1) and another terminal of the primary winding **602**.

In some embodiments, a gate terminal of the transistor **610** (e.g., a transistor M1) receives a drive signal **647** from the terminal **646** (e.g., a GATE pin) of the PWM controller **630** (e.g., an IC chip), and a source terminal of the transistor **610** (e.g., a transistor M1) is connected to the terminal **636** (e.g., a CS pin) of the PWM controller **630** (e.g., an IC chip) and also connected to one terminal of the resistor **664** (e.g., a resistor R7). For example, another terminal of the resistor **664** (e.g., a resistor R7) is biased to the ground voltage on the primary side. As an example, the terminal **644** (e.g., a VDD pin) of the pulse-width-modulation (PWM) controller **630** (e.g., an IC chip) is connected to one terminal of the resistor **666** (e.g., a resistor R6), and another terminal of the resistor **666** (e.g., a resistor R6) is connected to one plate of the capacitor **668** (e.g., a capacitor C1) and a cathode of the diode **670** (e.g., a diode D8). For example, the terminal **644** (e.g., a VDD pin) is at a voltage **645**. As an example, another plate of the capacitor **668** (e.g., a capacitor C1) is biased to the ground voltage on the primary side. For example, an anode of the diode **670** (e.g., a diode D8) is connected to one

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terminal of the auxiliary winding **606**, and another terminal of the auxiliary winding **606** is biased to the ground voltage on the primary side.

In certain embodiments, the anode of the diode **670** (e.g., a diode D8) is connected to one terminal of the resistor **672** (e.g., a resistor R4), and another terminal of the resistor **672** (e.g., a resistor R4) is connected to the terminal **632** (e.g., a PRT pin) of the PWM controller **630** (e.g., an IC chip) and also connected to one terminal of the resistor **674** (e.g., a resistor R5). For example, another terminal of the resistor **674** (e.g., a resistor R5) is biased to the ground voltage on the primary side. As an example, one terminal of the secondary winding **604** is connected to an anode of the diode **676** (e.g., a diode D9), and a cathode of the diode **676** (e.g., a diode D9) is connected to an output terminal **680** and one terminal of the capacitor **678** (e.g., a capacitor C4). For example, another terminal of the secondary winding **604** is connected to an output terminal **682** and another terminal of the capacitor **678** (e.g., a capacitor C4). As an example, the output terminal **680** provides the output voltage **681** (e.g., V_{out}) and an output current **683** (e.g., I_{out}), and the output terminal **682** is biased to a ground voltage on the secondary side.

According to some embodiments, the output terminal **680** is connected to one terminal of the resistor **684** (e.g., a resistor R10), and another terminal of the resistor **684** (e.g., a resistor R10) is connected to one terminal of the resistor **696** (e.g., a resistor R9), one terminal of the resistor **686** (e.g., a resistor R11), and an input terminal of the error amplifier **698**. For example, another terminal of the resistor **696** (e.g., a resistor R9) is connected to one terminal of the capacitor **694** (e.g., a capacitor C5), which includes another terminal connected to a cathode of the light emitting diode **692** (e.g., a light emitting diode D10) and connected to an output terminal of the error amplifier **698**. As an example, another terminal of the resistor **686** (e.g., R11) is biased to the ground voltage on the secondary side, and another input terminal of the error amplifier **698** is biased to a reference voltage **697**. For example, an anode of the light emitting diode **692** (e.g., a light emitting diode D10) is connected to one terminal of the resistor **688** (e.g., a resistor R8), and another terminal of the resistor **688** (e.g., a resistor R8) is connected to the output terminal **680**. As an example, the terminal **638** (e.g., a GND pin) of the PWM controller **630** (e.g., an IC chip) is biased to the ground voltage on the primary side. According to certain embodiments, under the normal operation, when the output voltage **681** (e.g., V_{out}) rises, the auxiliary winding **606**, together with the diode **670** (e.g., a diode D8), the capacitor **668** (e.g., a capacitor C1) and the resistor **666** (e.g., a resistor R6), raise the voltage **645** at the terminal **644** (e.g., a VDD pin).

FIG. 7 is a simplified diagram showing a power converter according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The power converter **700** (e.g., an AC-DC power supply) includes a primary winding **702**, a secondary winding **704**, an auxiliary winding **706**, a transistor **710** (e.g., a transistor M1), a rectifier **720** (e.g., a bridge rectifier), a filter **722** for electromagnetic interference (EMI), an optocoupler **728**, a pulse-width-modulation (PWM) controller **730** (e.g., an IC chip), resistors **712**, **714**, **760**, **764**, **784**, **786**, **788** and **796**, capacitors **718**, **756**, **758**, **768**, **778** and **794**, diodes **762**, **770** and **776**, an error amplifier **798**, and an inductor **716**, where the pulse-width-modulation (PWM) controller **730** (e.g., an IC chip) includes the PWM control-

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ler 800 as shown in FIG. 8, the PWM controller 730 as shown in FIG. 13, the PWM controller 730 as shown in FIG. 15, and/or the PWM controller 730 as shown in FIG. 17. For example, the pulse-width-modulation (PWM) controller 730 (e.g., an IC chip) includes a terminal 732 (e.g., a SW pin), a terminal 734 (e.g., a V10 pin), a terminal 736 (e.g., an FB pin), a terminal 738 (e.g., a GND pin), a terminal 740 (e.g., a HV pin), a terminal 742 (e.g., a DEM pin), a terminal 744 (e.g., a GATE pin), a terminal 746 (e.g., a CS pin), and a terminal 748 (e.g., a VDD pin). As an example, the opto-coupler 728 includes a light emitting diode 792 and a photosensitive detector 790. For example, the terminal 736 (e.g., an FB pin) is used to receive a feedback signal 737 that represents an output voltage 781 (e.g., V_{out}) of the power converter 700, and the terminal 740 (e.g., a HV pin) is used to receive to receive a voltage 741. As an example, the error amplifier 798 includes TL431.

As shown in FIG. 7, the EMI filter 722 receives an alternating current (AC) voltage 750 according to some embodiments. For example, two output terminals of the EMI filter 722 are connected to two input terminals of the rectifier 720 (e.g., a bridge rectifier) respectively. As an example, the rectifier 720 (e.g., a bridge rectifier) includes four diodes (e.g., diodes D1, D2, D3, and D4). For example, one output terminal of the rectifier 720 (e.g., a bridge rectifier) is connected to one plate of the capacitor 756 (e.g., a capacitor C1) that is biased to a ground voltage on the primary side, and another output terminal of the rectifier 720 (e.g., a bridge rectifier) is connected to another plate of the capacitor 756 (e.g., a capacitor C1) that is connected to one terminal of the resistor 760 (e.g., a resistor R1), one terminal of the capacitor 758 (e.g., a capacitor C2), and one terminal of the primary winding 702. As an example, another terminal of the resistor 760 (e.g., a resistor R1) and another terminal of the capacitor 758 (e.g., a capacitor C2) both are connected to a cathode of the diode 762 (e.g., a diode D5). For example, an anode of the diode 762 (e.g., a diode D5) is connected to a drain terminal of the transistor 710 (e.g., a transistor M1) and another terminal of the primary winding 702.

According to certain embodiments, a gate terminal of the transistor 710 (e.g., a transistor M1) receives a drive signal 745 from the terminal 744 (e.g., a GATE pin) of the PWM controller 730 (e.g., an IC chip), and a source terminal of the transistor 710 (e.g., a transistor M1) is connected to the terminal 746 (e.g., a CS pin) of the PWM controller 730 (e.g., an IC chip) and also connected to one terminal of the resistor 764 (e.g., a resistor R4). For example, another terminal of the resistor 764 (e.g., a resistor R4) is biased to the ground voltage on the primary side. As an example, the terminal 748 (e.g., a VDD pin) of the pulse-width-modulation (PWM) controller 730 (e.g., an IC chip) is connected to one terminal of the inductor 716 (e.g., an inductor L), one plate of the capacitor 768 (e.g., a capacitor C3) and a cathode of the diode 770 (e.g., a diode D7). For example, the terminal 748 (e.g., a VDD pin) is at a voltage 749. As an example, another terminal of the inductor 716 (e.g., an inductor L) is connected to the terminal 732 (e.g., a SW pin) of the PWM controller 730 (e.g., an IC chip). For example, another plate of the capacitor 768 (e.g., a capacitor C3) is biased to the ground voltage on the primary side. As an example, an anode of the diode 770 (e.g., a diode D7) is connected to one terminal of the resistor 712 (e.g., a resistor R2) and one terminal of the auxiliary winding 706 that corresponds to a voltage 707. For example, another terminal of the auxiliary winding 706 is biased to the ground voltage on the primary side. As an example, another terminal of the

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resistor 712 (e.g., a resistor R2) is connected to the terminal 742 (e.g., a DEM pin) of the PWM controller 730 (e.g., an IC chip) and one terminal of the resistor 714 (e.g., a resistor R3). For example, another terminal of the resistor 714 (e.g., a resistor R3) is biased to the ground voltage on the primary side. As an example, the terminal 734 (e.g., a V10 pin) of the PWM controller 730 (e.g., an IC chip) is connected to one terminal of the capacitor 718 (e.g., a capacitor C5), and another terminal of the capacitor 718 (e.g., a capacitor C5) is biased to the ground voltage on the primary side. For example, the terminal 734 (e.g., a V10 pin) is at a voltage 735.

In some embodiments, one terminal of the secondary winding 704 is connected to an anode of the diode 776 (e.g., a diode D6), and a cathode of the diode 776 (e.g., a diode D6) is connected to an output terminal 780 and one terminal of the capacitor 778 (e.g., a capacitor C4). For example, another terminal of the secondary winding 704 is connected to an output terminal 782 and another terminal of the capacitor 778 (e.g., a capacitor C4). As an example, the output terminal 780 provides the output voltage 781 (e.g., V_{out}) and an output current 783 (e.g., I_{out}), and the output terminal 782 is biased to a ground voltage on the secondary side.

According to certain embodiments, the output terminal 780 is connected to one terminal of the resistor 784 (e.g., a resistor R5), and another terminal of the resistor 784 (e.g., a resistor R5) is connected to one terminal of the resistor 796 (e.g., a resistor R6), one terminal of the resistor 786 (e.g., a resistor R7), and an input terminal of the error amplifier 798. For example, another terminal of the resistor 796 (e.g., a resistor R6) is connected to one terminal of the capacitor 794 (e.g., a capacitor C5), which includes another terminal connected to a cathode of the light emitting diode 792 (e.g., a light emitting diode D8) and connected to an output terminal of the error amplifier 798. As an example, another terminal of the resistor 786 (e.g., R7) is biased to the ground voltage on the secondary side, and another input terminal of the error amplifier 798 is biased to a reference voltage 797. For example, an anode of the light emitting diode 792 (e.g., a light emitting diode D8) is connected to one terminal of the resistor 788 (e.g., a resistor R4), and another terminal of the resistor 788 (e.g., a resistor R4) is connected to the output terminal 780. As an example, the terminal 738 (e.g., a GND pin) of the PWM controller 730 (e.g., an IC chip) is biased to the ground voltage on the primary side. According to certain embodiments, under the normal operation, when the output voltage 781 (e.g., V_{out}) rises, the auxiliary winding 706, together with the diode 770 (e.g., a diode D7) and the capacitor 768 (e.g., a capacitor C3), raises the voltage 749 at the terminal 748 (e.g., a VDD pin).

FIG. 8 is a simplified diagram showing certain components of the PWM controller 630 of the power converter 600 as shown in FIG. 6 and/or certain components of the PWM controller 730 of the power converter 700 as shown in FIG. 7 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller 800 (e.g., the PWM controller 630 and/or the PWM controller 730) includes a charging circuit 810, a voltage detection circuit 820, and a charging controller 830. Additionally, the PWM controller 800 (e.g., an IC chip) includes a terminal 870 (e.g., a pin), a terminal 880 (e.g., a pin) and a terminal 890 (e.g., a pin). In some examples, the PWM controller 800 (e.g., an IC chip) is the PWM controller 630 (e.g., an IC chip), the terminal

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870 (e.g., a pin) is the terminal 640 (e.g., a HV pin), the terminal 880 (e.g., a pin) is the terminal 644 (e.g., a VDD pin), and the terminal 890 (e.g., a pin) is also the terminal 644 (e.g., a VDD pin), where the terminal 880 (e.g., a pin) and the terminal 890 (e.g., a pin) are the same terminal. In certain examples, the PWM controller 800 (e.g., an IC chip) is the PWM controller 730 (e.g., an IC chip), the terminal 870 (e.g., a pin) is the terminal 740 (e.g., a HV pin), the terminal 880 (e.g., a pin) is the terminal 748 (e.g., a VDD pin), and the terminal 890 (e.g., a pin) is the terminal 734 (e.g., a V10 pin).

According to certain embodiments, the voltage detection circuit 820 receives a signal 882 from the terminal 880 (e.g., terminal 644 and/or the terminal 748) and generates a signal 822 based at least in part on the signal 882. In some examples, the charging controller 830 receives the signal 822 and generates a signal 832 based in part on the signal 822. For example, the signal 832 is used to connect (e.g., turn on) the charging circuit 810 so that the charging circuit 810 is allowed to use a signal 872 from the terminal 870 (e.g., the terminal 640 and/or the terminal 740) to charge the terminal 890 (e.g., the terminal 644 and/or the terminal 734). As an example, the signal 832 is used to disconnect (e.g., turn off) the charging circuit 810 so that the charging circuit 810 is not allowed to use the signal 872 from the terminal 870 (e.g., the terminal 640 and/or the terminal 740) to charge the terminal 890 (e.g., the terminal 644 and/or the terminal 734). In certain examples, the charging circuit 810 receives the signal 832 from the charging controller 830 and the signal 872 from the terminal 870 (e.g., the terminal 640 and/or the terminal 740) and generates a signal 892 based at least in part on the signals 832 and 872. For example, the signal 892 is received by the terminal 890 (e.g., the terminal 644 and/or the terminal 734).

According to some embodiments, the voltage detection circuit 820 detects whether the signal 882 (e.g., the voltage 645 and/or the voltage 749) drops to a lower predetermined threshold (e.g., VDD_HV_ON) and whether the signal 882 (e.g., the voltage 645 and/or the voltage 749) rises to a higher predetermined threshold (e.g., VDD_HV_OFF). For example, the higher predetermined threshold (e.g., VDD_HV_OFF) is larger than the lower predetermined threshold (e.g., VDD_HV_ON). As an example, the voltage detection circuit 820 detects whether the signal 882 (e.g., the voltage 645 and/or the voltage 749) drops to the lower predetermined threshold (e.g., VDD_HV_ON) and whether the signal 882 (e.g., the voltage 645 and/or the voltage 749) rises to the higher predetermined threshold (e.g., VDD_HV_OFF) under certain dynamic-load and/or no-load conditions of the power converter 600 and/or the power converter 700.

In certain examples, if the voltage detection circuit 820 determines that the signal 882 (e.g., the voltage 645 and/or the voltage 749) drops to the lower predetermined threshold (e.g., VDD_HV_ON), the voltage detection circuit 820 generates the signal 822 (e.g., Char_v10), and in response to receiving the signal 822 (e.g., Char_v10), the charging controller 830 generates the signal 832 in order to connect (e.g., turn on) the charging circuit 810. As an example, if the charging circuit 810 is connected (e.g., turned on), the charging circuit 810 uses the signal 872 (e.g., the voltage 641 and/or the voltage 741) to generate the signal 892 and charge the terminal 890 (e.g., the terminal 644 and/or the terminal 734). For example, when the terminal 890 (e.g., the terminal 644 and/or the terminal 734) is being charged by the charging circuit 810, the voltage 645 and/or the voltage 735 at the terminal 890 (e.g., the terminal 644 and/or the

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terminal 734) is supported by the signal 892 from the charging circuit 810, in addition to being supported by charges on the capacitor 668 and/or the capacitor 718, in order to avoid triggering undervoltage protection in response to insufficient voltage 645 and/or the voltage 735 under certain dynamic-load and/or no-load conditions of the power converter 600 and/or the power converter 700 respectively.

In some examples, if the voltage detection circuit 820 determines that the signal 882 (e.g., the voltage 645 and/or the voltage 749) rises to the higher predetermined threshold (e.g., VDD_HV_OFF), the voltage detection circuit 820 generates the signal 822 (e.g., Char_v10), and in response to receiving the signal 822 (e.g., Char_v10), the charging controller 830 generates the signal 832 in order to disconnect (e.g., turn off) the charging circuit 810. As an example, if the charging circuit 810 is disconnected (e.g., turned off), the charging circuit 810 does not charge the terminal 890 (e.g., the terminal 644 and/or the terminal 734), in order to avoid extra power consumption by the charging circuit 810 and to improve power efficiency of the power converter 600 and/or the power converter 700 respectively.

FIG. 9 is a simplified diagram showing certain components of the PWM controller 630 of the power converter 600 as shown in FIG. 6 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller 630 includes a charging circuit 910, a voltage detection circuit 920, a charging controller 930, a circuit 980 for undervoltage-lockout (UVLO), and an internal control circuit 990. Additionally, the PWM controller 630 (e.g., an IC chip) includes the terminal 640 (e.g., a HV pin) and the terminal 644 (e.g., a VDD pin). In certain examples, the charging circuit 810 includes a transistor 902 (e.g., a junction field-effect transistor), a resistor 904 (e.g., a resistor R1), and a transistor 906 (e.g., a metal-oxide-semiconductor field-effect transistor). In some examples, the charging circuit 910 is an example of the charging circuit 810, the voltage detection circuit 920 is an example of the voltage detection circuit 820, and the charging controller 930 is an example of the charging controller 830.

In certain embodiments, a drain terminal of the transistor 902 (e.g., JFET) is connected to the terminal 640 (e.g., a HV pin), the gate terminal of the transistor 902 (e.g., JFET) is biased to the ground voltage on the primary side of the power converter 600, and a source terminal of the transistor 902 (e.g., JFET) is connected to one terminal of the resistor 904 (e.g., a resistor R1). For example, another terminal of the resistor 904 (e.g., a resistor R1) is connected to a source terminal of the transistor 906 (e.g., a transistor M2). As an example, a gate terminal of the transistor 906 (e.g., a transistor M2) receives a signal 932 (e.g., the signal 832) from the charging controller 930 (e.g., the charging controller 830). For example, a drain terminal of the transistor 906 (e.g., a transistor M2) is connected to the terminal 644 (e.g., a VDD pin).

In some embodiments, an input terminal of the voltage detection circuit 920 (e.g., the voltage detection circuit 820) receives the voltage 645 from the terminal 644 (e.g., a VDD pin). For example, the voltage detection circuit 920 (e.g., the voltage detection circuit 820) generates a signal 922 (e.g., the signal 822) and sends the signal 922 (e.g., the signal 822) to the charging controller 930 (e.g., the charging controller 830). As an example, the voltage detection circuit 920 (e.g., the voltage detection circuit 820) generates a signal 924 and sends the signal 924 to the circuit 980 for undervoltage-

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lockout (UVLO). In certain examples, an output terminal of the circuit 980 for undervoltage-lockout (UVLO) is connected to an input terminal of the internal control circuit 990. As an example, the internal control circuit 990 includes various control circuits, and the internal control circuit 990 does not include the charging controller 930. For example, the internal control circuit 990 includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

As shown in FIG. 6, after the EMI filter 622 receives the AC voltage 650, the PWM controller 630 (e.g., an IC chip) undergoes a startup process according to certain embodiments. In some examples, during the startup process, if the voltage detection circuit 920 (e.g., the voltage detection circuit 820) determines that the voltage 645 has not reached a startup threshold (e.g., VDD_UVLO), the voltage detection circuit 920 (e.g., the voltage detection circuit 820) generates the signal 922 (e.g., the signal 822), so that the charging controller 930 generates the signal 932 (e.g., the signal 832) to turn on the transistor 906 (e.g., a transistor M2) and the charging circuit 910 (e.g., the charging circuit 810) generates a charging current 992 (e.g., the signal 892) to charge the capacitor 668 (e.g., a capacitor C1) and raise the voltage 645, until the voltage 645 reaches the startup threshold (e.g., VDD_UVLO). In certain examples, before the voltage 645 reaches the startup threshold (e.g., VDD_UVLO), the voltage detection circuit 920 (e.g., the voltage detection circuit 820) generates the signal 924, so that the circuit 980 for undervoltage-lockout (UVLO) does not generate the signal 921 that indicates the end of the startup process and causes the internal control circuit 990 to power on other components of the PWM controller 630 (e.g., an IC chip). For example, during the startup process, other components of the PWM controller 630 (e.g., an IC chip) remain powered off.

According to some embodiments, if the voltage detection circuit 920 (e.g., the voltage detection circuit 820) determines that the voltage 645 has risen above the startup threshold (e.g., VDD_UVLO), the voltage detection circuit 920 (e.g., the voltage detection circuit 820) generates the signal 922 (e.g., the signal 822), so that the charging controller 930 generates the signal 932 (e.g., the signal 832) to turn off the transistor 906 (e.g., a transistor M2) and the charging circuit 910 (e.g., the charging circuit 810) does not charge the capacitor 668 (e.g., a capacitor C1). In some examples, after the voltage detection circuit 920 (e.g., the voltage detection circuit 820) determines that the voltage 645 has risen above the startup threshold (e.g., VDD_UVLO), the startup process of the PWM controller 630 (e.g., an IC chip) ends and the normal operation for the power converter 600 starts. For example, at the beginning of the normal operation, the voltage detection circuit 920 (e.g., the voltage detection circuit 820) generates the signal 924, so that the circuit 980 for undervoltage-lockout (UVLO) generates the signal 921 that indicates the end of the startup process and causes the internal control circuit 990 to power on other components of the PWM controller 630 (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller 630 (e.g., an IC chip) remain powered on.

According to certain embodiments, the voltage detection circuit 920 (e.g., the voltage detection circuit 820) also determines whether the voltage 645 drops to a lower predetermined threshold (e.g., VDD_HV_ON) and whether the voltage 645 rises to a higher predetermined threshold (e.g.,

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VDD_HV_OFF). For example, the higher predetermined threshold (e.g., VDD_HV_OFF) is larger than the lower predetermined threshold (e.g., VDD_HV_ON), and the lower predetermined threshold (e.g., VDD_HV_ON) is larger than the startup threshold (e.g., VDD_UVLO).

In certain embodiments, during the normal operation, if the voltage detection circuit 920 (e.g., the voltage detection circuit 820) determines that the voltage 645 drops to a lower predetermined threshold (e.g., VDD_HV_ON), the voltage detection circuit 920 (e.g., the voltage detection circuit 820) generates the signal 922 (e.g., the signal 822), so that the charging controller 930 generates the signal 932 (e.g., the signal 832) to turn on the transistor 906 (e.g., a transistor M2). For example, if the transistor 906 (e.g., a transistor M2) is turned on, the charging circuit 910 (e.g., the charging circuit 810) generates a charging current 992 (e.g., the signal 892) to flow through the terminal 644 (e.g., a VDD pin) and to charge the capacitor 668 (e.g., a capacitor C1) and raise the voltage 645. As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit 910 (e.g., the charging circuit 810) generates the charging current 992 (e.g., the signal 892) to flow through the terminal 644 (e.g., a VDD pin) and to charge the capacitor 668 (e.g., a capacitor C1) and raise the voltage 645, in order to prevent insufficient power supply to the PWM controller 630 (e.g., an IC chip).

In some embodiments, during the normal operation, if the voltage detection circuit 920 (e.g., the voltage detection circuit 820) determines that the voltage 645 rises to a higher predetermined threshold (e.g., VDD_HV_OFF), the voltage detection circuit 920 (e.g., the voltage detection circuit 820) generates the signal 922 (e.g., the signal 822), so that the charging controller 930 generates the signal 932 (e.g., the signal 832) to turn off the transistor 906 (e.g., a transistor M2). For example, if the transistor 906 (e.g., a transistor M2) is turned off, the charging circuit 910 (e.g., the charging circuit 810) does not charge the capacitor 668 (e.g., a capacitor C1). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit 910 (e.g., the charging circuit 810) does not charge the capacitor 668 (e.g., a capacitor C1) in order to reduce power consumption of the PWM controller 630 (e.g., an IC chip).

As discussed above and further emphasized here, FIG. 9 is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, the voltage detection circuit 920 is part of the circuit 980 for undervoltage-lockout (UVLO), and the signal 924 is internal to the circuit 980 for undervoltage-lockout (UVLO).

FIG. 10 shows simplified timing diagrams under certain dynamic-load and/or no-load conditions for the power converter 600 that includes the PWM controller 630 as shown in FIG. 6 and FIG. 9 according to some embodiments. These diagrams are merely examples, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The waveform 1030 represents the feedback signal 635 as a function of time, the waveform 1040 represents the drive signal 647 as a function of time, the waveform 1050 represents the voltage 645 (e.g., V_{DD}) as a function of time, and the waveform 1060 represents the signal 932 as a function of time.

According to some embodiments, the waveforms 1030, 1040, 1050, and 1060 are timing diagrams when the power converter 600 is under the normal operation after the voltage

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645 has risen above the startup threshold (e.g., VDD_UVLO). For example, at the beginning of the normal operation, the voltage detection circuit 920 generates the signal 924, so that the circuit 980 for undervoltage-lockout (UVLO) generates the signal 921 that indicates the end of the startup process and causes the internal control circuit 990 to power on other components of the PWM controller 630 (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller 630 (e.g., an IC chip) remains powered on. For example, the higher predetermined threshold (e.g., VDD_HV_OFF) is larger than the lower predetermined threshold (e.g., VDD_HV_ON), and the lower predetermined threshold (e.g., VDD_HV_ON) is larger than the startup threshold (e.g., VDD_UVLO).

As shown in FIG. 10, at time t_{20} , the feedback signal 635 becomes smaller than a signal threshold 1034 (e.g., FB_Burst_L), and at time t_{25} , the feedback signal 635 becomes larger than a signal threshold 1032 (e.g., FB_Burst_H) as shown by the waveform 1030 according to certain embodiments. For example, at time t_{26} , the feedback signal 635 again becomes smaller than the signal threshold 1034 (e.g., FB_Burst_L) as shown by the waveform 1030. As an example, the signal threshold 1032 (e.g., FB_Burst_H) is larger than the signal threshold 1034 (e.g., FB_Burst_L) in magnitude.

In some embodiments, before time t_{20} , the drive signal 647 changes between a logic high level and a logic low level within each clock cycle, and from time t_{25} to time t_{26} , the drive signal 647 changes between the logic high level and the logic low level within each clock cycle as shown by the waveform 1040. In certain embodiments, from time t_{20} to time t_{25} , the drive signal 647 remains at the logic low level during multiple clock cycles, and after time t_{26} , the drive signal 647 remains at the logic low level during multiple clock cycles as shown by the waveform 1040.

As shown by the waveform 1050, from time t_{20} to time t_{21} , the voltage 645 (e.g., VDD) decreases with time, and at time t_{21} , the voltage 645 (e.g., VDD) decreases to the lower predetermined threshold (e.g., VDD_HV_ON) according to some embodiments. For example, at time t_{21} , in response to the voltage 645 (e.g., VDD) decreasing to the lower predetermined threshold (e.g., VDD_HV_ON), the signal 932 changes from the logic low level to the logic high level in order to turn on the transistor 906 (e.g., a transistor M2) as shown by the waveform 1060.

According to certain embodiments, from time t_{21} to time t_{22} , the signal 932 remains at the logic high level as shown by the waveform 1060, the transistor 906 (e.g., a transistor M2) remains turned on, and the voltage 645 (e.g., VDD) increases with time as shown by the waveform 1050. For example, at time t_{22} , the voltage 645 (e.g., VDD) increases to the higher predetermined threshold (e.g., VDD_HV_OFF) as shown by the waveform 1050. As an example, at time t_{22} , in response to the voltage 645 (e.g., VDD) increasing to the higher predetermined threshold (e.g., VDD_HV_OFF), the signal 932 changes from the logic high level to the logic low level in order to turn off the transistor 906 (e.g., a transistor M2) as shown by the waveform 1060.

As shown by the waveform 1050, from time t_{22} to time t_{23} , the voltage 645 (e.g., VDD) decreases with time, and at time t_{23} , the voltage 645 (e.g., VDD) decreases to the lower predetermined threshold (e.g., VDD_HV_ON) according to some embodiments. For example, at time t_{23} , in response to the voltage 645 (e.g., VDD) decreasing to the lower predetermined threshold (e.g., VDD_HV_ON), the signal 932 changes from the logic low level to the logic high level in

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order to turn on the transistor 906 (e.g., a transistor M2) as shown by the waveform 1060.

According to certain embodiments, from time t_{23} to time t_{24} , the signal 932 remains at the logic high level as shown by the waveform 1060, the transistor 906 (e.g., a transistor M2) remains turned on, and the voltage 645 (e.g., VDD) increases with time as shown by the waveform 1050. For example, at time t_{24} , the voltage 645 (e.g., VDD) increases to the higher predetermined threshold (e.g., VDD_HV_OFF) as shown by the waveform 1050. As an example, at time t_{24} , in response to the voltage 645 (e.g., VDD) increasing to the higher predetermined threshold (e.g., VDD_HV_OFF), the signal 932 changes from the logic high level to the logic low level in order to turn off the transistor 906 (e.g., a transistor M2) as shown by the waveform 1060.

In some embodiments, as shown by the waveform 1060, the signal 932 remains at the logic low level before time t_{21} , change from the logic low level to the logic high level at time t_{21} , remains at the logic high level from time t_{21} to time t_{22} , changes from the logic high level to the logic low level at time t_{22} , remains at the logic low level from time t_{22} to time t_{23} , changes from the logic low level to the logic high level at time t_{23} , remains at the logic high level from time t_{23} to time t_{24} , changes from the logic high level to the logic low level at time t_{24} , and remains at the logic low level after time t_{24} .

FIG. 11 shows simplified timing diagrams under certain dynamic-load conditions for the power converter 300 as shown in FIG. 3 according to certain embodiments. The waveform 1110 represents the output current 383 (e.g., I_{out}) as a function of time, the waveform 1120 represents the output voltage 381 (e.g., V_{out}) as a function of time, the waveform 1130 represents the feedback signal 337 as a function of time, the waveform 1140 represents the drive signal 345 as a function of time, the waveform 1150 represents the voltage 349 (e.g., VDD) as a function of time, and the waveform 1160 represents the voltage 335 (e.g., V_{I0}) as a function of time.

In some embodiments, before time t_{31} , the output current 383 (e.g., I_{out}) is at a current level 1112, and within each clock cycle, the drive signal 345 changes between a logic high level and a logic low level. For example, at time t_{31} , the output current 383 (e.g., I_{out}) decreases from the current level 1112 to a current level 1114. As an example, from time t_{31} to time t_{32} , the output current 383 (e.g., I_{out}) remains at the current level 1114. For example, at time t_{32} , the output current 383 (e.g., I_{out}) increases from the current level 1114 to the current level 1112. As an example, after time t_{32} , the output current 383 (e.g., I_{out}) remains at the current level 1112. In some examples, the current level 1112 is higher than the current level 1114. For example, if the output current 383 (e.g., I_{out}) is at the current level 1112, the power converter 300 is under a high-load condition. As an example, if the output current 383 (e.g., I_{out}) is at the current level 1114, the power converter 300 is under a light-load condition. In certain embodiments, from time t_{31} to time t_{32} , the power converter 300 is under the light-load condition. For example, under the light-load condition, after time t_{31} but before time t_{32} , the output voltage 381 (e.g., V_{out}) overshoots as shown by the waveform 1120, the feedback signal 337 drops as shown by the waveform 1130, the drive signal 345 remains at the logic low level during multiple clock cycles as shown by the waveform 1140, and the voltage 349 (e.g., VDD) decreases gradually as shown by the waveform 1150. In certain embodiments, from time t_{31} to time t_{32} , even though the voltage 349 (e.g., VDD) decreases gradually as shown by the waveform 1150, the voltage 335 (e.g., V_{I0}) remains

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equal to the reference voltage (e.g., V_{reg}) as shown by the waveform **1160**. For example, the voltage **335** (e.g., V_{10}) is generated by the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) based on the voltage **349** (e.g., V_{DD}). As an example, the generated voltage **335** (e.g., V_{10}) remains constant as shown by the waveform **1160**.

According to some embodiments, under certain dynamic-load conditions, if the voltage **349** (e.g., V_{DD}) decreases too much, the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) cannot keep the voltage **335** (e.g., V_{10}) equal to the reference voltage (e.g., V_{reg}) and the generated voltage **335** (e.g., V_{10}) also decreases. For example, if the voltage **335** (e.g., V_{10}) becomes too small in magnitude, the undervoltage protection is triggered for the PWM controller **330** (e.g., an IC chip) because of insufficient power supply.

FIG. 12 shows simplified timing diagrams under no-load condition for the power converter **300** as shown in FIG. 3 according to certain embodiments. The waveform **1230** represents the feedback signal **337** as a function of time, the waveform **1240** represents the drive signal **345** as a function of time, the waveform **1250** represents the voltage **349** (e.g., VDD) as a function of time, and the waveform **1260** represents the voltage **335** (e.g., V_{10}) as a function of time.

According to some embodiments, under no-load condition, the output current **383** (e.g., I_{out}) is equal to zero in magnitude. For example, at time t_{40} , the feedback signal **337** becomes larger than a signal threshold **1232** (e.g., FB_Burst_H), and at time t_{41} , the feedback signal **337** becomes smaller than a signal threshold **1234** (e.g., FB_Burst_L) as shown by the waveform **1230**. As an example, at time t_{42} , the feedback signal **337** becomes larger than the signal threshold **1232** (e.g., FB_Burst_H), and at time t_{43} , the feedback signal **337** becomes smaller than the signal threshold **1234** (e.g., FB_Burst_L) as shown by the waveform **1230**. For example, the signal threshold **1232** (e.g., FB_Burst_H) is larger than the signal threshold **1234** (e.g., FB_Burst_L) in magnitude.

In some embodiments, from time t_{40} to time t_{41} , the drive signal **345** changes between a logic high level and a logic low level within each clock cycle as shown by the waveform **1240**, and the voltage **349** (e.g., V_{DD}) increases as shown by the waveform **1250**. In certain embodiments, time t_{41} to time t_{42} , the drive signal **345** remains at the logic low level during multiple clock cycles as shown by the waveform **1240**, and the voltage **349** (e.g., VDD) decreases gradually as shown by the waveform **1250**. In some embodiments, from time t_{42} to time t_{43} , the drive signal **345** changes between the logic high level and the logic low level within each clock cycle as shown by the waveform **1240**, and the voltage **349** (e.g., V_{DD}) increases as shown by the waveform **1250**.

In certain embodiments, from time t_{41} to time t_{42} , even though the voltage **349** (e.g., V_{DD}) decreases gradually as shown by the waveform **1250**, the voltage **335** (e.g., V_{10}) remains equal to the reference voltage (e.g., V_{reg}) as shown by the waveform **1260**. For example, the voltage **335** (e.g., V_{10}) is generated by the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) based on the voltage **349** (e.g., V_{DD}). As an example, the generated voltage **335** (e.g., V_{10}) remains constant as shown by the waveform **1260**.

According to some embodiments, under no-load condition, if the voltage **349** (e.g., VDD) decreases too much, the internal DC-DC converter of the PWM controller **330** (e.g., an IC chip) cannot keep the voltage **335** (e.g., V_{10}) equal to the reference voltage (e.g., V_{reg}) and the generated voltage **335** (e.g., V_{10}) also decreases. For example, if the voltage **335** (e.g., V_{10}) becomes too small in magnitude, the under-

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voltage protection is triggered for the PWM controller **330** (e.g., an IC chip) because of insufficient power supply.

FIG. 13 is a simplified diagram showing certain components of the PWM controller **730** of the power converter **700** as shown in FIG. 7 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller **730** includes a charging circuit **1310**, voltage detection circuits **1320** and **1370**, charging controllers **1330** and **1360**, a circuit **1380** for undervoltage-lockout (UVLO), an internal control circuit **1390**, a DC-DC controller **1350**, a transistor **1352**, resistors **1354** and **1356**. Additionally, the PWM controller **730** (e.g., an IC chip) includes the terminal **732** (e.g., a SW pin), the terminal **734** (e.g., a V10 pin), the terminal **738** (e.g., a GND pin), the terminal **740** (e.g., a HV pin), and the terminal **748** (e.g., a VDD pin). In certain examples, the charging circuit **1310** includes a transistor **1302** (e.g., a junction field-effect transistor), a resistor **1304** (e.g., a resistor R6), a resistor **1378** (e.g., a resistor R7), a transistor **1306** (e.g., a metal-oxide-semiconductor field-effect transistor), a transistor **1376** (e.g., a metal-oxide-semiconductor field-effect transistor), and a diode **1382** (e.g., a Schottky diode). In some examples, the charging circuit **1310** includes two current generators. For example, one current generator of the charging circuit **1310** includes the transistor **1302** (e.g., a junction field-effect transistor), the resistor **1304** (e.g., a resistor R6), and the transistor **1306** (e.g., a metal-oxide-semiconductor field-effect transistor). As an example, another current generator of the charging circuit **1310** includes the transistor **1302** (e.g., a junction field-effect transistor), the resistor **1378** (e.g., a resistor R7), and the transistor **1376** (e.g., a metal-oxide-semiconductor field-effect transistor). In certain examples, the inductor **716** of the power converter **700** (e.g., an AC-DC power supply) is not part of the PWM controller **730** (e.g., an IC chip). For example, a DC-DC converter includes the inductor **716**, the capacitor **768** (e.g., the capacitor C3), the capacitor **718** (e.g., a capacitor C5), the DC-DC controller **1350**, the transistor **1352** (e.g., a transistor M7), the resistor **1354** (e.g., a resistor R8), the resistor **1356** (e.g., a resistor R9), and the diode **1382**.

According to certain embodiments, a drain terminal of the transistor **1302** (e.g., JFET) is connected to the terminal **740** (e.g., a HV pin), the gate terminal of the transistor **1302** (e.g., JFET) is biased to the ground voltage on the primary side of the power converter **700**, and a source terminal of the transistor **1302** (e.g., JFET) is connected to one terminal of the resistor **1304** (e.g., a resistor R6). For example, another terminal of the resistor **1304** (e.g., a resistor R6) is connected to a source terminal of the transistor **1306** (e.g., a transistor M5). As an example, a gate terminal of the transistor **1306** (e.g., a transistor M5) receives a signal **1332** from the charging controller **1330**. For example, a drain terminal of the transistor **1306** (e.g., a transistor M5) is connected to the terminal **748** (e.g., a VDD pin) and is also connected to one terminal of the inductor **716** through the terminal **748** (e.g., a VDD pin). As an example, the transistor **1306** (e.g., a transistor M5) is a PMOS transistor (e.g., a high-voltage PMOS transistor).

In some examples, another terminal of the inductor **716** is connected to the terminal **732** (e.g., a SW pin) and is also connected to an anode of the diode **1382** (e.g., a diode D1) through the terminal **732** (e.g., a SW pin). For example, a cathode of the diode **1382** (e.g., a diode D1) is connected to the terminal **734** (e.g., a V10 pin), a drain terminal of the

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transistor **1376** (e.g., a transistor **M6**), and one terminal of the resistor **1354** (e.g., a resistor **R8**). As an example, another terminal of the resistor **1354** (e.g., a resistor **R8**) is connected to an input terminal of the DC-DC controller **1350** and one terminal of the resistor **1356** (e.g., a resistor **R9**). For example, another terminal of the resistor **1356** (e.g., a resistor **R9**) is connected to the terminal **738** (e.g., a GND pin), which is biased to the ground voltage on the primary side of the power converter **700**. As an example, the transistor **1376** (e.g., a transistor **M6**) is a PMOS transistor.

In certain examples, an output terminal of the DC-DC controller **1350** is connected to a gate terminal of the transistor **1352** (e.g., a transistor **M7**). For example, a source terminal of the transistor **1352** (e.g., a transistor **M7**) is biased to the ground voltage on the primary side of the power converter **700**. As an example, a drain terminal of the transistor **1352** (e.g., a transistor **M7**) is connected to the anode of the diode **1382** and the terminal **732** (e.g., a SW pin). For example, the transistor **1352** (e.g., a transistor **M7**) is an NMOS transistor.

According to some embodiments, the source terminal of the transistor **1302** (e.g., JFET) is connected to one terminal of the resistor **1378** (e.g., a resistor **R7**). For example, another terminal of the resistor **1378** (e.g., a resistor **R7**) is connected to a source terminal of the transistor **1376** (e.g., a transistor **M6**). As an example, the gate terminal of the transistor **1376** (e.g., a transistor **M6**) is connected to an output terminal of the charging controller **1360**. For example, a drain terminal of the transistor **1376** (e.g., a transistor **M6**) is connected to the cathode of the diode **1382** (e.g., a diode **D1**) and the terminal **734** (e.g., a **V10** pin). In certain examples, an input terminal of the charging controller **1360** receives a signal **1321** (e.g., **Char_v10**) from the voltage detection circuit **1320**, and another input terminal of the charging controller **1360** receives a signal **1371** from the voltage detection circuit **1370**.

According to certain embodiments, an input terminal of the voltage detection circuit **1320** is connected to the drain terminal of the transistor **1306** (e.g., a transistor **M5**) and the terminal **748** (e.g., a **VDD** pin). For example, the output terminal of the voltage detection circuit **1320** sends the signal **1321** to the charging controller **1360**. In some examples, an input terminal of the voltage detection circuit **1370** is connected to the terminal **734** (e.g., a **V10** pin). For example, the voltage detection circuit **1370** generates signals **1322**, **1371**, and **1373**. As an example, the voltage detection circuit **1370** sends the signal **1322** to the charging controller **1330**, sends the signal **1371** to the charging controller **1360**, and sends the signal **1373** to the circuit **1380** for undervoltage-lockout (UVLO). In certain examples, an output terminal of the circuit **1380** for undervoltage-lockout (UVLO) is connected to an input terminal of the internal control circuit **1390**. As an example, the internal control circuit **1390** includes various control circuits, and the internal control circuit **1390** does not include the charging controller **1330** and does not include the charging controller **1360**. For example, the internal control circuit **1390** includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

As shown in FIG. 7, after the EMI filter **722** receives the AC voltage **750**, the PWM controller **730** (e.g., an IC chip) undergoes a startup process according to certain embodiments. In some examples, during the startup process, if the voltage detection circuit **1370** determines that the voltage **735** has not reached a startup threshold (e.g., **V10_UVLO**),

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the voltage detection circuit **1370** generates the signal **1322**, so that the charging controller **1330** generates the signal **1332** to turn on the transistor **1306** (e.g., a transistor **M5**) and the charging circuit **1310** generates a charging current **1392**. For example, the charging current **1392** flows through the transistor **1302** (e.g., JFET), the resistor **1304** (e.g., a resistor **R6**), and the transistor **1306** (e.g., a transistor **M5**) to charge the capacitor **768** (e.g., the capacitor **C3**) and raise the voltage **749** at the terminal **748** (e.g., a **VDD** pin). As an example, the voltage **749** is used, through the inductor **716** and the diode **1382**, to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**, until the voltage **735** reaches the startup threshold (e.g., **V10_UVLO**). In certain examples, before the voltage **735** reaches the startup threshold (e.g., **V10_UVLO**), the voltage detection circuit **1370** generates the signal **1373**, so that the circuit **1380** for undervoltage-lockout (UVLO) does not generate the signal **1391** that indicates the end of the startup process and causes the internal control circuit **1390** to power on other components of the PWM controller **730** (e.g., an IC chip). For example, during the startup process, other components of the PWM controller **730** (e.g., an IC chip) remain powered off. As an example, during the startup process, the transistor **1376** (e.g., a transistor **M6**) remains turned off, and the DC-DC converter does not operate.

According to some embodiments, during the startup process, the charging current **1392** flows through the transistor **1302** (e.g., JFET), the resistor **1304** (e.g., a resistor **R6**), and the transistor **1306** (e.g., a transistor **M5**) to raise the voltage **749**, and the voltage **749** is used, through the inductor **716** and the diode **1382**, to raise the voltage **735**. As an example, the charging current **1392** is determined as follows:

$$I_{hv_vdd} = \frac{V_{HVS} - V_{DD}}{R_6} \quad (\text{Equation 3})$$

where I_{hv_vdd} represents the charging current **1392**. Additionally, V_{HVS} represents the voltage **741**, and V_{DD} represents the voltage **749**. Also, R_6 represents the resistance of the resistor **1304**. For example, according to Equation 3, the charging current **1392** (e.g., I_{hv_vdd}) decreases with the increasing voltage **749** (e.g., V_{DD}).

In certain embodiments, if the voltage detection circuit **1370** determines that the voltage **735** has risen above the startup threshold (e.g., **V10_UVLO**), the voltage detection circuit **1370** generates the signal **1322**, so that the charging controller **1330** generates the signal **1332** to turn off the transistor **1306** (e.g., a transistor **M5**) and the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor **C5**). In some examples, after the voltage detection circuit **1370** determines that the voltage **735** has risen above the startup threshold (e.g., **V10_UVLO**), the startup process of the PWM controller **730** (e.g., an IC chip) ends and the normal operation for the power converter **700** starts. For example, at the beginning of the normal operation, the voltage detection circuit **1370** generates the signal **1373**, so that the circuit **1380** for undervoltage-lockout (UVLO) generates the signal **1391** that indicates the end of the startup process and causes the internal control circuit **1390** to power on other components of the PWM controller **730** (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller **730** (e.g., an IC chip) remain powered on. In certain examples, under normal operation, the transistor **1306** (e.g., a transistor **M5**) remains turned off. In some examples, the voltage detection circuit

1370 determines whether the voltage **735** has risen to the startup threshold (e.g., **V10_UVLO**) during the startup process, and also determines whether the voltage **735** rises to a predetermined threshold (e.g., **V10_HV_OFF**) during the normal operation.

In some embodiments, during the normal operation, when the voltage **749** (e.g., **VDD**) is higher than a reference voltage (e.g., V_{reg}), the DC-DC converter is idle, and the auxiliary winding **706** charges the capacitor **768** (e.g., a capacitor **C3**) to support the voltage **749** (e.g., **VDD**), which is used to charge the capacitor **718** (e.g., a capacitor **C5**) and support the voltage **735** so that the voltage **735** at the terminal **734** (e.g., a **V10** pin) is equal to the voltage **749** (e.g., **VDD**) minus the forward bias voltage of the diode **1382**. In some examples, during the normal operation, when the voltage **749** (e.g., **VDD**) is smaller than the reference voltage (e.g., V_{reg}) but is not smaller than a lower predetermined threshold (e.g., **VDD_HV_ON**), the voltage **735** at the terminal **734** (e.g., a **V10** pin) is generated by the DC-DC converter using the voltage **749** (e.g., **VDD**), and the generated voltage **735** is larger than the voltage **749** (e.g., **VDD**) and is equal to the reference voltage (e.g., V_{reg}).

According to certain embodiments, during the normal operation, if the voltage detection circuit **1320** determines that the voltage **749** drops to the lower predetermined threshold (e.g., **VDD_HV_ON**), the voltage detection circuit **1320** generates the signal **1321** (e.g., **Char_v10**), so that the charging controller **1360** generates a signal **1361** (e.g., **Char_en**) to turn on the transistor **1376** (e.g., a transistor **M6**). In some examples, if the transistor **1376** (e.g., a transistor **M6**) is turned on, the charging circuit **1310** generates a charging current **1394** to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**. For example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1310** generates the charging current **1394** to flow through the terminal **734** (e.g., a **V10** pin) and to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**, in order to prevent insufficient power supply to the PWM controller **730** (e.g., an IC chip). In certain examples, the charging current **1394** is determined as follows:

$$I_{hv_v10} = \frac{V_{HVS} - V_{10}}{R_7} \quad (\text{Equation 4})$$

where I_{hv_v10} represents the charging current **1394**. Additionally, V_{HVS} represents the voltage **741**, and V_{10} represents the voltage **735**. Also, R_7 represents the resistance of the resistor **1378**.

In some embodiments, during the normal operation, if the voltage detection circuit **1320** determines that the voltage **749** rises to a higher predetermined threshold (e.g., **VDD_HV_OFF**), the voltage detection circuit **1320** generates the signal **1321**, so that the charging controller **1360** generates the signal **1361** (e.g., **Char_en**) to turn off the transistor **1376** (e.g., a transistor **M6**). For example, if the transistor **1376** (e.g., a transistor **M6**) is turned off, the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor **C5**). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor **C5**) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

In certain embodiments, during the normal operation, the voltage detection circuit **1370** determines whether the volt-

age **735** rises to the predetermined threshold (e.g., **V10_HV_OFF**) and generates the signal **1371**. For example, if the voltage **735** has not risen to the predetermined threshold (e.g., **V10_HV_OFF**), the signal **1371** is at a first logic level (e.g., a logic low level). As an example, if the voltage **735** has risen to the predetermined threshold (e.g., **V10_HV_OFF**), the signal **1371** is at a second logic level (e.g., a logic high level). In certain examples, if the signal **1371** indicates that the voltage **735** has risen to the predetermined threshold (e.g., **V10_HV_OFF**), the charging controller **1360** generates the signal **1361** (e.g., **Char_en**) to turn off the transistor **1376** (e.g., a transistor **M6**). For example, if the transistor **1376** (e.g., a transistor **M6**) is turned off, the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor **C5**). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1310** does not charge the capacitor **718** (e.g., a capacitor **C5**) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

As shown in FIG. 13, the charging controller **1360** receives the signal **1321** and the signal **1371** and generates the signal **1361** (e.g., **Char_en**) to turn on or turn off the transistor **1376** (e.g., a transistor **M6**) according to some embodiments. For example, if the signal **1321** indicates that the voltage **749** drops to the lower predetermined threshold (e.g., **VDD_HV_ON**), the charging controller **1360** generates the signal **1361** (e.g., **Char_en**) to turn on the transistor **1376** (e.g., a transistor **M6**) for generating the charging current **1394** to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**. As an example, if the signal **1321** indicates that the voltage **749** rises to the higher predetermined threshold (e.g., **VDD_HV_OFF**), the charging controller **1360** generates the signal **1361** (e.g., **Char_en**) to turn off the transistor **1376** (e.g., a transistor **M6**). For example, if the signal **1371** indicates that the voltage **735** has risen to the predetermined threshold (e.g., **V10_HV_OFF**), the charging controller **1360** generates the signal **1361** (e.g., **Char_en**) to turn off the transistor **1376** (e.g., a transistor **M6**).

As discussed above and further emphasized here, FIG. 13 is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In some examples, the PWM controller **730** also includes a reference-current and reference-voltage generator that receives a signal from the voltage detection circuit **1370** and is connected to the internal control circuit **1390**. In certain examples, the voltage detection circuit **1370** outputs a signal for overvoltage protection. In certain examples, the diode **1382** (e.g., a Schottky diode) is replaced by a parasitic body diode of a PMOS transistor and/or by a PMOS switch. In some examples, the PWM controller **730** also includes a timer that, after being triggered, during a predetermined duration, prevents the signal **1371** from causing the charging controller **1360** to turn off the transistor **1376** (e.g., a transistor **M6**).

FIG. 14 shows simplified timing diagrams under certain dynamic-load and/or no-load conditions for the power converter **700** that includes the PWM controller **730** as shown in FIG. 7 and FIG. 13 according to some embodiments of the present invention. These diagrams are merely examples, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The waveform **1430** represents the feedback signal **737** as a function of time, the waveform **1440** represents the drive signal **745** as a function of time, the waveform **1450** represents the voltage **749** (e.g.,

V_{DD}) as a function of time, the waveform **1460** represents the signal **1361** (e.g., Char_en) as a function of time, and the waveform **1470** represents the voltage **735** (e.g., V_{10}) as a function of time.

According to some embodiments, the waveforms **1430**, **1440**, **1450**, **1460** and **1470** are timing diagrams when the power converter **700** is under the normal operation after the voltage **735** has risen above the startup threshold (e.g., V_{10_UVLO}). For example, at the beginning of the normal operation, the voltage detection circuit **1370** generates the signal **1373**, so that the circuit **1380** for undervoltage-lockout (UVLO) generates the signal **1391** that indicates the end of the startup process and causes the internal control circuit **1390** to power on other components of the PWM controller **730** (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller **730** (e.g., an IC chip) remains powered on.

As shown in FIG. 10, at time t_{s1} , the voltage **749** (e.g., V_{DD}) drops to the lower predetermined threshold (e.g., $V_{DD_HV_ON}$) as shown by the waveform **1450**, and in response, the transistor **1306** (e.g., the transistor **M8**) is turned on in order to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735** (e.g., V_{10}), according to certain embodiments. In some examples, from time t_{s1} to time t_{s2} , the voltage **735** (e.g., V_{10}) increases with time as shown by the waveform **1470**. In certain examples, at time t_{s2} , the voltage **735** (e.g., V_{10}) rises to the predetermined threshold (e.g., $V_{10_HV_OFF}$) and in response, the transistor **1306** (e.g., the transistor **M8**) is turned off.

According to some embodiments, at time t_{61} , the voltage **749** (e.g., V_{DD}) drops to the lower predetermined threshold (e.g., $V_{DD_HV_ON}$) as shown by the waveform **1450**, and in response, the transistor **1306** (e.g., the transistor **M8**) is turned on in order to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735** (e.g., V_{10}). In certain examples, from time t_{61} to time t_{62} , the voltage **735** (e.g., V_{10}) increases with time as shown by the waveform **1470**. In some examples, at time t_{62} , the voltage **749** (e.g., V_{DD}) rises to the higher predetermined threshold (e.g., $V_{DD_HV_OFF}$) as shown by the waveform **1450** and in response, the transistor **1306** (e.g., the transistor **M8**) is turned off.

FIG. 15 is a simplified diagram showing certain components of the PWM controller **730** of the power converter **700** as shown in FIG. 7 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller **730** includes a charging circuit **1510**, voltage detection circuits **1520** and **1570**, charging controllers **1530** and **1560**, a circuit **1580** for undervoltage-lockout (UVLO), an internal control circuit **1590**, a DC-DC controller **1550**, a transistor **1552**, resistors **1554** and **1556**. Additionally, the PWM controller **730** (e.g., an IC chip) includes the terminal **732** (e.g., a SW pin), the terminal **734** (e.g., a V_{10} pin), the terminal **738** (e.g., a GND pin), the terminal **740** (e.g., a HV pin), and the terminal **748** (e.g., a VDD pin). In certain examples, the charging circuit **1510** includes a transistor **1502** (e.g., a junction field-effect transistor), a resistor **1578** (e.g., a resistor **R7**), a transistor **1506** (e.g., a metal-oxide-semiconductor field-effect transistor), a transistor **1576** (e.g., a metal-oxide-semiconductor field-effect transistor), a diode **1582** (e.g., a Schottky diode), a resistor **2510** (e.g., a resistor **R1**), a resistor **2520** (e.g., a resistor **R2**), a resistor **2530** (e.g., a resistor **R3**), a switch **2512** (e.g., a switch **S1**), a switch **2522** (e.g., a switch **S2**), and a switch **2532** (e.g., a switch **S3**). In

some examples, the charging circuit **1510** includes two current generators. For example, one current generator of the charging circuit **1510** includes the transistor **1502** (e.g., a junction field-effect transistor), the resistor **2510** (e.g., a resistor **R1**), the resistor **2520** (e.g., a resistor **R2**), the resistor **2530** (e.g., a resistor **R3**), the switch **2512** (e.g., a switch **S1**), the switch **2522** (e.g., a switch **S2**), the switch **2532** (e.g., a switch **S3**), and the transistor **1506** (e.g., a metal-oxide-semiconductor field-effect transistor). As an example, another current generator of the charging circuit **1510** includes the transistor **1502** (e.g., a junction field-effect transistor), the resistor **1578** (e.g., a resistor **R7**), and the transistor **1576** (e.g., a metal-oxide-semiconductor field-effect transistor). In certain examples, the inductor **716** of the power converter **700** (e.g., an AC-DC power supply) is not part of the PWM controller **730** (e.g., an IC chip). For example, a DC-DC converter includes the inductor **716**, the capacitor **768** (e.g., the capacitor **C3**), the capacitor **718** (e.g., a capacitor **C5**), the DC-DC controller **1550**, the transistor **1552** (e.g., a transistor **M7**), the resistor **1554** (e.g., a resistor **R8**), the resistor **1556** (e.g., a resistor **R9**), and the diode **1582**.

According to certain embodiments, a drain terminal of the transistor **1502** (e.g., JFET) is connected to the terminal **740** (e.g., a HV pin), the gate terminal of the transistor **1502** (e.g., JFET) is biased to the ground voltage on the primary side of the power converter **700**, and a source terminal of the transistor **1502** (e.g., JFET) is connected to one terminal of the resistor **2510** (e.g., a resistor **R1**), one terminal of the resistor **2520** (e.g., a resistor **R2**), and one terminal of the resistor **2530** (e.g., a resistor **R3**). For example, another terminal of the resistor **2510** (e.g., a resistor **R1**) is connected to one terminal of the switch **2512** (e.g., a switch **S1**), another terminal of the resistor **2520** (e.g., a resistor **R2**) is connected to one terminal of the switch **2522** (e.g., a switch **S2**), and another terminal of the resistor **2530** (e.g., a resistor **R3**) is connected to one terminal of the switch **2532** (e.g., a switch **S3**). As an example, another terminal of the switch **2512** (e.g., a switch **S1**), another terminal of the switch **2522** (e.g., a switch **S2**), and another terminal of the switch **2532** (e.g., a switch **S3**) are all connected to a source terminal of the transistor **1506** (e.g., a transistor **M5**). For example, the transistor **1506** (e.g., a transistor **M5**) is a PMOS transistor (e.g., a high-voltage PMOS transistor).

In some examples, the switch **2512** (e.g., a switch **S1**) receives a signal **2514** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2512** (e.g., a switch **S1**), the switch **2522** (e.g., a switch **S2**) receives a signal **2524** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2522** (e.g., a switch **S2**), and the switch **2532** (e.g., a switch **S3**) receives a signal **2534** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2532** (e.g., a switch **S3**). In certain examples, a gate terminal of the transistor **1506** (e.g., a transistor **M5**) receives a signal **1532** from the charging controller **1530**. For example, a drain terminal of the transistor **1506** (e.g., a transistor **M5**) is connected to the terminal **748** (e.g., a VDD pin) and is also connected to one terminal of the inductor **716** through the terminal **748** (e.g., a VDD pin).

In some embodiments, another terminal of the inductor **716** is connected to the terminal **732** (e.g., a SW pin) and is also connected to an anode of the diode **1582** (e.g., a diode **D1**) through the terminal **732** (e.g., a SW pin). For example, a cathode of the diode **1582** (e.g., a diode **D1**) is connected to the terminal **734** (e.g., a V_{10} pin), a drain terminal of the transistor **1576** (e.g., a transistor **M6**), and one terminal of the resistor **1554** (e.g., a resistor **R8**). As an example, another

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terminal of the resistor **1554** (e.g., a resistor **R8**) is connected to an input terminal of the DC-DC controller **1550** and one terminal of the resistor **1556** (e.g., a resistor **R9**). For example, another terminal of the resistor **1556** (e.g., a resistor **R9**) is connected to the terminal **738** (e.g., a GND pin), which is biased to the ground voltage on the primary side of the power converter **700**. As an example, the transistor **1576** (e.g., a transistor **M6**) is a PMOS transistor.

In certain examples, an output terminal of the DC-DC controller **1550** is connected to a gate terminal of the transistor **1552** (e.g., a transistor **M7**). For example, a source terminal of the transistor **1552** (e.g., a transistor **M7**) is biased to the ground voltage on the primary side of the power converter **700**. As an example, a drain terminal of the transistor **1552** (e.g., a transistor **M7**) is connected to the anode of the diode **1582** and the terminal **732** (e.g., a SW pin). For example, the transistor **1552** (e.g., a transistor **M7**) is an NMOS transistor.

According to certain embodiments, the source terminal of the transistor **1502** (e.g., JFET) is connected to one terminal of the resistor **1578** (e.g., a resistor **R7**). For example, another terminal of the resistor **1578** (e.g., a resistor **R7**) is connected to a source terminal of the transistor **1576** (e.g., a transistor **M6**). As an example, a gate terminal of the transistor **1576** (e.g., a transistor **M6**) is connected to an output terminal of the charging controller **1560**. For example, a drain terminal of the transistor **1576** (e.g., a transistor **M6**) is connected to the cathode of the diode **1582** (e.g., a diode **D1**) and the terminal **734** (e.g., a **V10** pin). In certain examples, an input terminal of the charging controller **1560** receives a signal **1521** (e.g., **Char_v10**) from the voltage detection circuit **1520**, and another input terminal of the charging controller **1560** receives a signal **1571** from the voltage detection circuit **1570**.

According to some embodiments, an input terminal of the voltage detection circuit **1520** is connected to the drain terminal of the transistor **1506** (e.g., a transistor **M5**) and the terminal **748** (e.g., a **VDD** pin). For example, the output terminal of the voltage detection circuit **1520** sends the signal **1521** to the charging controller **1560**. In some examples, an input terminal of the voltage detection circuit **1570** is connected to the terminal **734** (e.g., a **V10** pin). For example, the voltage detection circuit **1570** generates signals **1522**, **1571**, and **1573**. As an example, the voltage detection circuit **1570** sends the signal **1522** to the charging controller **1530**, sends the signal **1571** to the charging controller **1560**, and sends the signal **1573** to the circuit **1580** for undervoltage-lockout (UVLO). In certain examples, an output terminal of the circuit **1580** for undervoltage-lockout (UVLO) is connected to an input terminal of the internal control circuit **1590**. As an example, the internal control circuit **1590** includes various control circuits, and the internal control circuit **1590** does not include the charging controller **1530** and does not include the charging controller **1560**. For example, the internal control circuit **1590** includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

As shown in FIG. 7, after the EMI filter **722** receives the AC voltage **750**, the PWM controller **730** (e.g., an IC chip) undergoes a startup process according to certain embodiments. In some examples, during the startup process, if the voltage detection circuit **1570** determines that the voltage **735** has not reached a startup threshold (e.g., **V10_UVLO**), the voltage detection circuit **1570** generates the signal **1522**, so that the charging controller **1530** generates the signal

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1532 to turn on the transistor **1506** (e.g., a transistor **M5**). For example, if the voltage detection circuit **1520** determines that the voltage **749** is smaller than a first reference voltage (e.g., **V1**), the voltage detection circuit **1520** generates the signal **2514** to close (e.g., turn on) the switch **2512**, generates the signal **2524** to open (e.g., turn off) the switch **2522**, and generates the signal **2534** to open (e.g., turn off) the switch **2532**. As an example, if the voltage detection circuit **1520** determines that the voltage **749** is larger than the first reference voltage (e.g., **V1**) but smaller than a second reference voltage (e.g., **V2**), the voltage detection circuit **1520** generates the signal **2514** to close (e.g., turn on) the switch **2512**, generates the signal **2524** to close (e.g., turn on) the switch **2522**, and generates the signal **2534** to open (e.g., turn off) the switch **2532**, where the second reference voltage (e.g., **V2**) is larger than the first reference voltage (e.g., **V1**). For example, if the voltage detection circuit **1520** determines that the voltage **749** is larger than both the first reference voltage (e.g., **V1**) and the second reference voltage (e.g., **V2**), the voltage detection circuit **1520** generates the signal **2514** to close (e.g., turn on) the switch **2512**, generates the signal **2524** to close (e.g., turn on) the switch **2522**, and generates the signal **2534** to close (e.g., turn on) the switch **2532**.

In certain examples, during the startup process, if the voltage **735** has not reached the startup threshold (e.g., **V10_UVLO**), the charging circuit **1510** generates a charging current **1592** to charge the capacitor **768** (e.g., the capacitor **C3**) and raise the voltage **749** at the terminal **748** (e.g., a **VDD** pin). For example, the charging current **1592** is determined as follows:

$$I_{hv_vdd} = \begin{cases} \frac{V_{HVS} - V_{DD}}{R_1} & V_{DD} < V_1 \\ \frac{V_{HVS} - V_{DD}}{R_1 // R_2} & V_1 < V_{DD} < V_2 \\ \frac{V_{HVS} - V_{DD}}{R_1 // R_2 // R_3} & V_{DD} > V_2 \end{cases} \quad (\text{Equation 5})$$

where I_{hv_vdd} represents the charging current **1592**. Additionally, V_{HVS} represents the voltage **741**, and **VDD** represents the voltage **749**. Moreover, R_1 represents the resistance of the resistor **2510**, R_2 represents the resistance of the resistor **2520**, and R_3 represents the resistance of the resistor **2530**. Also, R_1/R_2 represents the resistance of a combination of the resistors **2510** and **2520**, which are connected in parallel, and $R_1/R_2/R_3$ represents the resistance of a combination of the resistors **2510**, **2520** and **2530**, which are connected in parallel. Additionally, V_1 represents the first reference voltage, and V_2 represents the second reference voltage.

In some examples, if the voltage **735** is smaller than the startup threshold (e.g., **V10_UVLO**) and the voltage **749** is smaller than the first reference voltage (e.g., **V1**), the charging current **1592** flows through the transistor **1502** (e.g., FET), the resistor **2510** (e.g., a resistor **R1**), and the transistor **1506** (e.g., a transistor **M5**). For example, if the voltage **735** is smaller than the startup threshold (e.g., **V10_UVLO**) and the voltage **749** is larger than the first reference voltage (e.g., **V1**) but smaller than the second reference voltage (e.g., **V2**), the charging current **1592** flows through the transistor **1502** (e.g., FET), the resistor **2510** (e.g., a resistor **R1**), the resistor **2520** (e.g., a resistor **R2**), and the transistor **1506** (e.g., a transistor **M5**). As an example, if the voltage **735** is smaller than the startup

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threshold (e.g., V10_UVLO) and the voltage 749 is larger than both the first reference voltage (e.g., V1) and the second reference voltage (e.g., V2), the charging current 1592 flows through the transistor 1502 (e.g., FET), the resistor 2510 (e.g., a resistor R1), the resistor 2520 (e.g., a resistor R2), the resistor 2530 (e.g., a resistor R3), and the transistor 1506 (e.g., a transistor M5).

According to some embodiments, before the voltage 735 reaches the startup threshold (e.g., V10_UVLO), the voltage detection circuit 1570 generates the signal 1573, so that the circuit 1580 for undervoltage-lockout (UVLO) does not generate the signal 1591 that indicates the end of the startup process and causes the internal control circuit 1590 to power on other components of the PWM controller 730 (e.g., an IC chip). For example, during the startup process, other components of the PWM controller 730 (e.g., an IC chip) remain powered off. As an example, during the startup process, the transistor 1576 (e.g., a transistor M6) remains turned off, and the DC-DC converter does not operate.

In certain embodiments, if the voltage detection circuit 1570 determines that the voltage 735 has risen above the startup threshold (e.g., V10_UVLO), the voltage detection circuit 1570 generates the signal 1522, so that the charging controller 1530 generates the signal 1532 to turn off the transistor 1506 (e.g., a transistor M5) and the charging circuit 1510 does not charge the capacitor 718 (e.g., a capacitor C5). In some examples, after the voltage detection circuit 1570 determines that the voltage 735 has risen above the startup threshold (e.g., V10_UVLO), the startup process of the PWM controller 730 (e.g., an IC chip) ends and the normal operation for the power converter 700 starts. For example, at the beginning of the normal operation, the voltage detection circuit 1570 generates the signal 1573, so that the circuit 1580 for undervoltage-lockout (UVLO) generates the signal 1591 that indicates the end of the startup process and causes the internal control circuit 1590 to power on other components of the PWM controller 730 (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller 730 (e.g., an IC chip) remain powered on. In certain examples, under normal operation, the transistor 1506 (e.g., a transistor M5) remains turned off. In some examples, the voltage detection circuit 1570 determines whether the voltage 735 has risen to the startup threshold (e.g., V10_UVLO) during the startup process, and also determines whether the voltage 735 rises to a predetermined threshold (e.g., V10_HV_OFF) during the normal operation.

According to certain embodiments, during the normal operation, when the voltage 749 (e.g., V_{DD}) is higher than a reference voltage (e.g., V_{ref}), the DC-DC converter is idle, and the auxiliary winding 706 charges the capacitor 768 (e.g., a capacitor C3) to support the voltage 749 (e.g., V_{DD}), which is used to charge the capacitor 718 (e.g., a capacitor C5) and support the voltage 735 so that the voltage 735 at the terminal 734 (e.g., a V10 pin) is equal to the voltage 749 (e.g., V_{DD}) minus the forward bias voltage of the diode 1582. In some examples, during the normal operation, when the voltage 749 (e.g., V_{DD}) is smaller than the reference voltage (e.g., V_{ref}) but is not smaller than a lower predetermined threshold (e.g., VDD_HV_ON), the voltage 735 at the terminal 734 (e.g., a V10 pin) is generated by the DC-DC converter using the voltage 749 (e.g., V_{DD}), and the generated voltage 735 is larger than the voltage 749 (e.g., V_{DD}) and is equal to the reference voltage (e.g., V_{ref}).

According to some embodiments, during the normal operation, if the voltage detection circuit 1520 determines that the voltage 749 drops to the lower predetermined

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threshold (e.g., VDD_HV_ON), the voltage detection circuit 1520 generates the signal 1521 (e.g., Char_v10), so that the charging controller 1560 generates a signal 1561 (e.g., Char_en) to turn on the transistor 1576 (e.g., a transistor M6). In some examples, if the transistor 1576 (e.g., a transistor M6) is turned on, the charging circuit 1510 generates a charging current 1594 to charge the capacitor 718 (e.g., a capacitor C5) and raise the voltage 735. For example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit 1510 generates the charging current 1594 to flow through the terminal 734 (e.g., a V10 pin) and to charge the capacitor 718 (e.g., a capacitor C5) and raise the voltage 735, in order to prevent insufficient power supply to the PWM controller 730 (e.g., an IC chip). In certain examples, the charging current 1594 is determined as follows:

$$I_{hv_v10} = \frac{V_{HVS} - V_{10}}{R_7} \quad (\text{Equation 6})$$

where I_{hv_v10} represents the charging current 1594. Additionally, V_{HVS} represents the voltage 741, and V_{10} represents the voltage 735. Also, R_7 represents the resistance of the resistor 1578.

In some embodiments, during the normal operation, if the voltage detection circuit 1520 determines that the voltage 749 rises to a higher predetermined threshold (e.g., VDD_HV_OFF), the voltage detection circuit 1520 generates the signal 1521, so that the charging controller 1560 generates the signal 1561 (e.g., Char_en) to turn off the transistor 1576 (e.g., a transistor M6). For example, if the transistor 1576 (e.g., a transistor M6) is turned off, the charging circuit 1510 does not charge the capacitor 718 (e.g., a capacitor C5). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit 1510 does not charge the capacitor 718 (e.g., a capacitor C5) in order to reduce power consumption of the PWM controller 730 (e.g., an IC chip).

In certain embodiments, during the normal operation, the voltage detection circuit 1570 determines whether the voltage 735 rises to the predetermined threshold (e.g., V10_HV_OFF) and generates the signal 1571. For example, if the voltage 735 has not risen to the predetermined threshold (e.g., V10_HV_OFF), the signal 1571 is at a first logic level (e.g., a logic low level). As an example, if the voltage 735 has risen to the predetermined threshold (e.g., V10_HV_OFF), the signal 1571 is at a second logic level (e.g., a logic high level). In certain examples, if the signal 1571 (e.g., Char_en) indicates that the voltage 735 has risen to the predetermined threshold (e.g., V10_HV_OFF), the charging controller 1560 generates the signal 1561 (e.g., Char_en) to turn off the transistor 1576 (e.g., a transistor M6). For example, if the transistor 1576 (e.g., a transistor M6) is turned off, the charging circuit 1510 does not charge the capacitor 718 (e.g., a capacitor C5). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit 1510 does not charge the capacitor 718 (e.g., a capacitor C5) in order to reduce power consumption of the PWM controller 730 (e.g., an IC chip).

As shown in FIG. 15, the charging controller 1560 receives the signals 1521 and 1571 and generates the signal 1561 (e.g., Char_en) to turn on or turn off the transistor 1576 (e.g., a transistor M6) according to some embodiments. For example, if the signal 1521 indicates that the voltage 749

drops to the lower predetermined threshold (e.g., VDD_HV_ON), the charging controller 1560 generates the signal 1561 (e.g., Char_en) to turn on the transistor 1576 (e.g., a transistor M6) for generating the charging current 1594 to charge the capacitor 718 (e.g., a capacitor C5) and raise the voltage 735. As an example, if the signal 1521 indicates that the voltage 749 rises to the higher predetermined threshold (e.g., VDD_HV_OFF), the charging controller 1560 generates the signal 1561 (e.g., Char_en) to turn off the transistor 1576 (e.g., a transistor M6). For example, if the signal 1571 indicates that the voltage 735 has risen to the predetermined threshold (e.g., V10_HV_OFF), the charging controller 1560 generates the signal 1561 (e.g., Char_en) to turn off the transistor 1576 (e.g., a transistor M6).

As discussed above and further emphasized here, FIG. 15 is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In some examples, the PWM controller 730 also includes a reference-current and reference-voltage generator that receives a signal from the voltage detection circuit 1570 and is connected to the internal control circuit 1590. In certain examples, the voltage detection circuit 1570 outputs a signal for overvoltage protection. In certain examples, the diode 1582 (e.g., a Schottky diode) is replaced by a parasitic body diode of a PMOS transistor and/or by a PMOS switch. In some examples, the PWM controller 730 also includes a timer that, after being triggered, during a predetermined duration, prevents the signal 1571 from causing the charging controller 1560 to turn off the transistor 1576 (e.g., a transistor M6).

FIG. 16 shows a simplified diagram showing the charging current 1592 as a function of the voltage 749 for the power converter 700 that includes the PWM controller 730 as shown in FIG. 7 and FIG. 15 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown in FIG. 16, I_{hv_vdd} represents the charging current 1592, and V_{DD} represents the voltage 749 according to certain embodiments. In some examples, V_1 represents the first reference voltage, and V_2 represents the second reference voltage. In certain examples, V_c represents a voltage magnitude of the voltage 749, at which the startup process of the PWM controller 730 (e.g., an IC chip) ends. For example, V_c represents the voltage magnitude of the voltage 749, at which the voltage detection circuit 1570 determines that the voltage 735 has risen above the startup threshold (e.g., V10_UVLO). As an example, V_{HVS} represents the voltage 741, R_1 represents the resistance of the resistor 2510, R_2 represents the resistance of the resistor 2520, and R_3 represents the resistance of the resistor 2530.

In certain embodiments, according to Equation 5, if V_{DD} is equal to zero, I_{hv_vdd} is equal to

$$\frac{V_{HVS}}{R_1},$$

and when V_{DD} increases from zero to V_1 , I_{hv_vdd} decreases with the increasing V_{DD} . In some embodiments, according to Equation 5, if V_{DD} reaches V_1 , I_{hv_vdd} rises to

$$\frac{V_{HVS} - V_1}{R_1 // R_2},$$

and when V_{DD} increases from V_1 to V_2 , I_{hv_vdd} decreases with the increasing V_{DD} . In certain embodiments, according to Equation 5, if V_{DD} reaches V_2 , I_{hv_vdd} rises to

$$\frac{V_{HVS} - V_2}{R_1 // R_2 // R_3},$$

and when V_{DD} increases from V_2 to V_c , I_{hv_vdd} decreases with the increasing V_{DD} . In some embodiments, if V_{DD} reaches V_c , the startup process of the PWM controller 730 (e.g., an IC chip) ends and I_{hv_vdd} drops to zero.

FIG. 17 is a simplified diagram showing certain components of the PWM controller 730 of the power converter 700 as shown in FIG. 7 according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The PWM controller 730 includes a charging circuit 1710, voltage detection circuits 1720 and 1770, a charging controller 1730, a circuit 1780 for undervoltage-lockout (UVLO), an internal control circuit 1790, a DC-DC controller 1750, a transistor 1752, resistors 1754 and 1756. Additionally, the PWM controller 730 (e.g., an IC chip) includes the terminal 732 (e.g., a SW pin), the terminal 734 (e.g., a V10 pin), the terminal 738 (e.g., a GND pin), the terminal 740 (e.g., a HV pin), and the terminal 748 (e.g., a VDD pin). In certain examples, the charging circuit 1710 includes a transistor 1702 (e.g., a junction field-effect transistor), a transistor 1706 (e.g., a metal-oxide-semiconductor field-effect transistor), a diode 1782 (e.g., a Schottky diode), a resistor 2710 (e.g., a resistor R1), a resistor 2720 (e.g., a resistor R2), a resistor 2730 (e.g., a resistor R3), a switch 2712 (e.g., a switch S1), a switch 2722 (e.g., a switch S2), and a switch 2732 (e.g., a switch S3). As an example, the inductor 716 of the power converter 700 (e.g., an AC-DC power supply) is not part of the PWM controller 730 (e.g., an IC chip). For example, a DC-DC converter includes the inductor 716, the capacitor 768 (e.g., the capacitor C3), the capacitor 718 (e.g., a capacitor C5), the DC-DC controller 1750, the transistor 1752 (e.g., a transistor M4), the resistor 1754 (e.g., a resistor R4), the resistor 1756 (e.g., a resistor R5), and the diode 1782.

According to certain embodiments, a drain terminal of the transistor 1702 (e.g., JFET) is connected to the terminal 740 (e.g., a HV pin), the gate terminal of the transistor 1702 (e.g., JFET) is biased to the ground voltage on the primary side of the power converter 700, and a source terminal of the transistor 1702 (e.g., JFET) is connected to one terminal of the resistor 2710 (e.g., a resistor R1), one terminal of the resistor 2720 (e.g., a resistor R2), and one terminal of the resistor 2730 (e.g., a resistor R3). For example, another terminal of the resistor 2710 (e.g., a resistor R1) is connected to one terminal of the switch 2712 (e.g., a switch S1), another terminal of the resistor 2720 (e.g., a resistor R2) is connected to one terminal of the switch 2722 (e.g., a switch S2), and another terminal of the resistor 2730 (e.g., a resistor R3) is connected to one terminal of the switch 2732 (e.g., a switch S3). As an example, another terminal of the switch 2712 (e.g., a switch S1), another terminal of the switch 2722 (e.g., a switch S2), and another terminal of the switch 2732 (e.g., a switch S3) are all connected to a source terminal of

the transistor **1706** (e.g., a transistor **M3**). For example, the transistor **1706** (e.g., a transistor **M3**) is a PMOS transistor (e.g., a high-voltage PMOS transistor).

In some examples, the switch **2712** (e.g., a switch **S1**) receives a signal **2714** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2712** (e.g., a switch **S1**), the switch **2722** (e.g., a switch **S2**) receives a signal **2724** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2722** (e.g., a switch **S1**), and the switch **2732** (e.g., a switch **S3**) receives a signal **2734** that is used to close (e.g., turn on) or open (e.g., turn off) the switch **2732** (e.g., a switch **S1**). In certain examples, a gate terminal of the transistor **1706** (e.g., a transistor **M3**) receives a signal **1732** from the charging controller **1730**. For example, a drain terminal of the transistor **1706** (e.g., a transistor **M3**) is connected to the terminal **748** (e.g., a VDD pin) and is also connected to one terminal of the inductor **716** through the terminal **748** (e.g., a VDD pin).

In some embodiments, another terminal of the inductor **716** is connected to the terminal **732** (e.g., a SW pin) and is also connected to an anode of the diode **1782** (e.g., a diode **D1**) through the terminal **732** (e.g., a SW pin). For example, a cathode of the diode **1782** (e.g., a diode **D1**) is connected to the terminal **734** (e.g., a V10 pin) and one terminal of the resistor **1754** (e.g., a resistor **R4**). As an example, another terminal of the resistor **1754** (e.g., a resistor **R4**) is connected to an input terminal of the DC-DC controller **1750** and one terminal of the resistor **1756** (e.g., a resistor **R5**). For example, another terminal of the resistor **1756** (e.g., a resistor **R5**) is connected to the terminal **738** (e.g., a GND pin), which is biased to the ground voltage on the primary side of the power converter **700**. In certain examples, an output terminal of the DC-DC controller **1750** is connected to a gate terminal of the transistor **1752** (e.g., a transistor **M4**). For example, a source terminal of the transistor **1752** (e.g., a transistor **M4**) is biased to the ground voltage on the primary side of the power converter **700**. As an example, a drain terminal of the transistor **1752** (e.g., a transistor **M4**) is connected to the anode of the diode **1782** and the terminal **732** (e.g., a SW pin). For example, the transistor **1752** (e.g., a transistor **M4**) is an NMOS transistor.

According to some embodiments, an input terminal of the voltage detection circuit **1720** is connected to the terminal **748** (e.g., a VDD pin). For example, the output terminal of the voltage detection circuit **1720** sends a signal **1721** to the charging controller **1730**. In some examples, an input terminal of the voltage detection circuit **1770** is connected to the terminal **734** (e.g., a V10 pin). For example, the voltage detection circuit **1770** generates signals **1722**, **1771**, **1773**, **2714**, **2724**, and **2734**. As an example, the voltage detection circuit **1770** sends the signals **1722** and **1771** to the charging controller **1730** and sends the signal **1773** to the circuit **1780** for undervoltage-lockout (UVLO). In certain examples, an output terminal of the circuit **1780** for undervoltage-lockout (UVLO) is connected to an input terminal of the internal control circuit **1790**. As an example, the internal control circuit **1790** includes various control circuits, and the internal control circuit **1790** does not include the charging controller **1730**. For example, the internal control circuit **1790** includes one or more control circuits for one or more oscillators, one or more control circuits for one or more timers, one or more drivers, one or more logic control circuits, and/or one or more protection circuits.

As shown in FIG. 7, after the EMI filter **722** receives the AC voltage **750**, the PWM controller **730** (e.g., an IC chip) undergoes a startup process according to certain embodiments. According to some embodiments, during the startup

process, if the voltage detection circuit **1770** determines that the voltage **735** has not reached a startup threshold (e.g., V10_UVLO), the voltage detection circuit **1770** generates the signal **1722**, so that the charging controller **1730** generates the signal **1732** to turn on the transistor **1706** (e.g., a transistor **M3**). In certain examples, if the voltage detection circuit **1770** determines that the voltage **735** is smaller than a first reference voltage (e.g., V_x), the voltage detection circuit **1770** generates the signal **2714** to close (e.g., turn on) the switch **2712**, generates the signal **2724** to open (e.g., turn off) the switch **2722**, and generates the signal **2734** to open (e.g., turn off) the switch **2732**. In some examples, if the voltage detection circuit **1770** determines that the voltage **735** is larger than the first reference voltage (e.g., V_x) but smaller than a second reference voltage (e.g., V_y), the voltage detection circuit **1770** generates the signal **2714** to close (e.g., turn on) the switch **2712**, generates the signal **2724** to close (e.g., turn on) the switch **2722**, and generates the signal **2734** to open (e.g., turn off) the switch **2732**. For example, the first reference voltage (e.g., V_x) is smaller than the second reference voltage (e.g., V_y), and the second reference voltage (e.g., V_y) is smaller than the startup threshold (e.g., V10_UVLO). In certain examples, if the voltage detection circuit **1720** determines that the voltage **735** is larger than both the first reference voltage (e.g., V_x) and the second reference voltage (e.g., V_y), the voltage detection circuit **1720** generates the signal **2714** to close (e.g., turn on) the switch **2712**, generates the signal **2724** to close (e.g., turn on) the switch **2722**, and generates the signal **2734** to close (e.g., turn on) the switch **2732**.

In certain embodiments, during the startup process, if the voltage **735** has not reached the startup threshold (e.g., V10_UVLO), the charging circuit **1710** generates a charging current **1792** to charge the capacitor **718** (e.g., the capacitor **C5**) and raise the voltage **735** at the terminal **734** (e.g., a V10 pin). For example, the charging current **1792** is determined as follows:

$$I_{hv_v10} = \begin{cases} \frac{V_{HVS} - V_{10}}{R_1} & V_{10} < V_x \\ \frac{V_{HVS} - V_{10}}{R_1 // R_2} & V_x < V_{10} < V_y \\ \frac{V_{HVS} - V_{10}}{R_1 // R_2 // R_3} & V_y < V_{10} < V_{10_UVLO} \end{cases} \quad (\text{Equation 7})$$

where I_{hv_v10} represents the charging current **1792**. Additionally, V_{HVS} represents the voltage **741**, and V_{10} represents the voltage **735**. Moreover, R_1 represents the resistance of the resistor **2710**, R_2 represents the resistance of the resistor **2720**, and R_3 represents the resistance of the resistor **2730**. Also, R_1/R_2 represents the resistance of a combination of the resistors **2710** and **2720**, which are connected in parallel, and $R_1/R_2/R_3$ represents the resistance of a combination of the resistors **2710**, **2720** and **2730**, which are connected in parallel. Additionally, V_x represents the first reference voltage, V_y represents the second reference voltage, and V10_UVLO represents the startup threshold.

In some embodiments, if the voltage **735** is smaller than the first reference voltage (e.g., V_x), the charging current **1792** flows through the transistor **1702** (e.g., FET), the resistor **2710** (e.g., a resistor **R1**), and the transistor **1706** (e.g., a transistor **M3**). For example, if the voltage **735** is larger than the first reference voltage (e.g., V_x) but smaller than the second reference voltage (e.g., V_y), the charging current **1792** flows through the transistor **1702** (e.g., FET),

the resistor **2710** (e.g., a resistor **R1**), the resistor **2720** (e.g., a resistor **R2**), and the transistor **1706** (e.g., a transistor **M3**). As an example, if the voltage **735** is larger than both the first reference voltage (e.g., V_x) and the second reference voltage (e.g., V_y) but smaller than the startup threshold (e.g., V_{10_UVLO}), the charging current **1792** flows through the transistor **1702** (e.g., FET), the resistor **2710** (e.g., a resistor **R1**), the resistor **2720** (e.g., a resistor **R2**), the resistor **2730** (e.g., a resistor **R3**), and the transistor **1706** (e.g., a transistor **M5**).

According to certain embodiments, before the voltage **735** reaches the startup threshold (e.g., V_{10_UVLO}), the voltage detection circuit **1770** generates the signal **1773**, so that the circuit **1780** for undervoltage-lockout (UVLO) does not generate the signal **1791** that indicates the end of the startup process and causes the internal control circuit **1790** to power on other components of the PWM controller **730** (e.g., an IC chip). For example, during the startup process, other components of the PWM controller **730** (e.g., an IC chip) remain powered off. As an example, during the startup process, the DC-DC converter does not operate.

According to some embodiments, if the voltage detection circuit **1770** determines that the voltage **735** has risen above the startup threshold (e.g., V_{10_UVLO}), the voltage detection circuit **1770** generates the signal **1722**, so that the charging controller **1730** generates the signal **1732** to turn off the transistor **1706** (e.g., a transistor **M3**) and the charging circuit **1710** does not charge the capacitor **718** (e.g., a capacitor **C5**). In some examples, after the voltage detection circuit **1770** determines that the voltage **735** has risen above the startup threshold (e.g., V_{10_UVLO}), the startup process of the PWM controller **730** (e.g., an IC chip) ends and the normal operation for the power converter **700** starts. For example, at the beginning of the normal operation, the voltage detection circuit **1770** generates the signal **1773**, so that the circuit **1780** for undervoltage-lockout (UVLO) generates the signal **1791** that indicates the end of the startup process and causes the internal control circuit **1790** to power on other components of the PWM controller **730** (e.g., an IC chip). As an example, during the normal operation, other components of the PWM controller **730** (e.g., an IC chip) remain powered on. In certain examples, during the startup process, the voltage detection circuit **1770** determines whether the voltage **735** is smaller than the first reference voltage (e.g., V_x), whether the voltage **735** is larger than the first reference voltage (e.g., V_x) but smaller than the second reference voltage (e.g., V_y), and/or whether the voltage **735** is larger than both the first reference voltage (e.g., V_x) and the second reference voltage (e.g., V_y) but smaller than the startup threshold (e.g., V_{10_UVLO}). In some examples, during the normal operation, the voltage detection circuit **1770** determines whether the voltage **735** is smaller than the first reference voltage (e.g., V_x), whether the voltage **735** is larger than the first reference voltage (e.g., V_x) but smaller than the second reference voltage (e.g., V_y), and/or whether the voltage **735** is larger than both the first reference voltage (e.g., V_x) and the second reference voltage (e.g., V_y), and also determines whether the voltage **735** rises to a predetermined threshold (e.g., $V_{10_HV_OFF}$).

In certain embodiments, during the normal operation, when the voltage **749** (e.g., V_{DD}) is higher than a reference voltage (e.g., V_{reg}), the DC-DC converter is idle, and the auxiliary winding **706** charges the capacitor **768** (e.g., a capacitor **C3**) to support the voltage **749** (e.g., V_{DD}), which is used to charge the capacitor **718** (e.g., a capacitor **C5**) and support the voltage **735** so that the voltage **735** at the

terminal **734** (e.g., a **V10** pin) is equal to the voltage **749** (e.g., V_{DD}) minus the forward bias voltage of the diode **1582**. In some examples, during the normal operation, when the voltage **749** (e.g., V_{DD}) is smaller than the reference voltage (e.g., V_{reg}) but is not smaller than a lower predetermined threshold (e.g., $V_{DD_HV_ON}$), the voltage **735** at the terminal **734** (e.g., a **V10** pin) is generated by the DC-DC converter using the voltage **749** (e.g., V_{DD}), and the generated voltage **735** is larger than the voltage **749** (e.g., V_{DD}) and is equal to the reference voltage (e.g., V_{reg}).

According to some embodiments, during the normal operation, if the voltage detection circuit **1720** determines that the voltage **749** drops to the lower predetermined threshold (e.g., $V_{DD_HV_ON}$), the voltage detection circuit **1720** generates the signal **1721**, so that the charging controller **1730** generates the signal **1732** to turn on the transistor **1706** (e.g., a transistor **M3**). In some examples, if the transistor **1706** (e.g., a transistor **M3**) is turned on, the charging circuit **1710** generates the charging current **1792** to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**. For example, during the normal operation, when the transistor **1706** (e.g., a transistor **M3**) is turned on, if the voltage detection circuit **1770** determines the voltage **735** is smaller than the first reference voltage (e.g., V_x), the switch **2512** is closed but the switches **2522** and **2532** are open. As an example, during the normal operation, when the transistor **1706** (e.g., a transistor **M3**) is turned on, if the voltage **735** is larger than the first reference voltage (e.g., V_x) but smaller than the second reference voltage (e.g., V_y), the switches **2512** and **2522** are closed but the switch **2532** is open. For example, during the normal operation, when the transistor **1706** (e.g., a transistor **M3**) is turned on, if the voltage **735** is larger than both the first reference voltage (e.g., V_x) and the second reference voltage (e.g., V_y), the switches **2512**, **2522**, and **2532** are closed. In certain examples, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1710** generates the charging current **1792** to flow through the terminal **734** (e.g., a **V10** pin) and to charge the capacitor **718** (e.g., a capacitor **C5**) and raise the voltage **735**, in order to prevent insufficient power supply to the PWM controller **730** (e.g., an IC chip).

In certain embodiments, during the normal operation, if the voltage detection circuit **1720** determines that the voltage **749** rises to a higher predetermined threshold (e.g., $V_{DD_HV_OFF}$), the voltage detection circuit **1720** generates the signal **1721**, so that the charging controller **1730** generates the signal **1732** to turn off the transistor **1706** (e.g., a transistor **M3**). For example, if the transistor **1706** (e.g., a transistor **M3**) is turned off, the charging circuit **1710** does not charge the capacitor **718** (e.g., a capacitor **C5**). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit **1710** does not charge the capacitor **718** (e.g., a capacitor **C5**) in order to reduce power consumption of the PWM controller **730** (e.g., an IC chip).

In some embodiments, during the normal operation, the voltage detection circuit **1770** determines whether the voltage **735** rises to the predetermined threshold (e.g., $V_{10_HV_OFF}$) and generates the signal **1771**. For example, if the voltage **735** has not risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the signal **1771** is at a first logic level (e.g., a logic low level). As an example, if the voltage **735** has risen to the predetermined threshold (e.g., $V_{10_HV_OFF}$), the signal **1771** is at a second logic level (e.g., a logic high level). In certain examples, if the signal **1771** indicates that the voltage **735** has risen to the prede-

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terminated threshold (e.g., V10_HV_OFF), the charging controller 1730 generates the signal 1732 to turn off the transistor 1706 (e.g., a transistor M3). For example, if the transistor 1706 (e.g., a transistor M3) is turned off, the charging circuit 1710 does not charge the capacitor 718 (e.g., a capacitor C5). As an example, during the normal operation, under certain dynamic-load and/or no-load conditions, the charging circuit 1710 does not charge the capacitor 718 (e.g., a capacitor C5) in order to reduce power consumption of the PWM controller 730 (e.g., an IC chip).

As shown in FIG. 17, during the normal operation, the charging controller 1730 receives the signals 1721 and 1771 and generates the signal 1732 to turn on or turn off the transistor 1706 according to some embodiments. For example, if the signal 1721 indicates that the voltage 749 drops to the lower predetermined threshold (e.g., VDD_HV_ON), the charging controller 1730 generates the signal 1732 to turn on the transistor 1706 (e.g., a transistor M3) for generating the charging current 1792 to charge the capacitor 718 (e.g., a capacitor C5) and raise the voltage 735. As an example, if the signal 1721 indicates that the voltage 749 rises to the higher predetermined threshold (e.g., VDD_HV_OFF), the charging controller 1730 generates the signal 1732 to turn off the transistor 1706 (e.g., a transistor M3). For example, if the signal 1771 indicates that the voltage 735 has risen to the predetermined threshold (e.g., V10_HV_OFF), the charging controller 1730 generates the signal 1732 to turn off the transistor 1706 (e.g., a transistor M3).

As discussed above and further emphasized here, FIG. 17 is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In some examples, the PWM controller 730 also includes a reference-current and reference-voltage generator that receives a signal from the voltage detection circuit 1770 and is connected to the internal control circuit 1790. In certain examples, the voltage detection circuit 1770 outputs a signal for overvoltage protection. In certain examples, the diode 1782 (e.g., a Schottky diode) is replaced by a parasitic body diode of a PMOS transistor and/or by a PMOS switch. In some examples, the PWM controller 730 also includes a timer that, after being triggered, during a predetermined duration, prevents the signal 1771 from causing the charging controller 1730 to turn off the transistor 1706 (e.g., a transistor M3).

According to some embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a capacitor and biased to a second voltage; a voltage detector configured to receive the second voltage from the second terminal and generate a detection signal based at least in part on the second voltage; a charging controller configured to receive the detection signal and generate a first control signal based at least in part on the detection signal; and a charging current generator configured to receive the first voltage from the first terminal and receive the first control signal from the charging controller; wherein the voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated; wherein the charging current generator is further configured

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to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the second terminal to charge the capacitor associated with the second voltage. For example, the controller is implemented according to at least FIG. 9.

As an example, the voltage detector is further configured to: detect that the second voltage has increased to a second predetermined threshold; and generate the detection signal indicating that the second voltage has increased to the second predetermined threshold; wherein the second predetermined threshold is larger than the first predetermined threshold. For example, the charging controller is further configured to: in response to the detection signal indicating that the second voltage has increased to the second predetermined threshold, generate the first control signal indicating that the charging current is not to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is not to be generated, not generate the charging current that flows through the second terminal.

As an example, the second voltage decreases to the first predetermined threshold during normal operation after a startup process; and the second voltage increases to the second predetermined threshold during the normal operation. For example, during the startup process: the voltage detector is further configured to: detect that the second voltage has not risen above a third predetermined threshold; and generate the detection signal indicating that the second voltage has not risen above the third predetermined threshold; the charging controller is further configured to: in response to the detection signal indicating that the second voltage has not risen above the third predetermined threshold, generate the first control signal indicating that the charging current is to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the second terminal to charge the capacitor and raise the second voltage. As an example, during the startup process: the voltage detector is further configured to: detect that the second voltage has risen above the third predetermined threshold; and generate the detection signal indicating that the second voltage has risen above the third predetermined threshold; the charging controller is further configured to: in response to the detection signal indicating that the second voltage has risen above the third predetermined threshold, generate the first control signal indicating that the charging current is not to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is not to be generated, not generate the charging current that flows through the second terminal to charge the capacitor and raise the second voltage. For examples, the controller further includes: a circuit for undervoltage-lockout configured to: in response to the second voltage has risen above the third predetermined threshold, generating a second control signal indicating the startup process has ended; and an internal control circuit configured to: in response to the second control signal indicating the startup process has ended, power on one or more components of the controller to start the normal operation.

According to certain embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a first capacitor and biased to a second voltage; a third terminal connected to a second capacitor and biased to a third voltage; a first voltage detector configured to receive the

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second voltage from the second terminal and generate a first detection signal based at least in part on the second voltage; a first charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a charging current generator configured to receive the first voltage from the first terminal, the charging current generator including a first current generator and a second current generator, the first current generator being configured to receive the first control signal from the first charging controller; wherein the first voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the first charging controller is further configured to: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a first charging current is to be generated; wherein the first current generator is further configured to: in response to the first control signal indicating that the first charging current is to be generated, generate the first charging current that flows through the third terminal to charge the second capacitor associated with the third voltage. For example, the controller is implemented according to at least FIG. 13 and/or FIG. 15.

As an example, the first voltage detector is further configured to: detect that the second voltage has increased to a second predetermined threshold; and generate the first detection signal indicating that the second voltage has increased to the second predetermined threshold; wherein the second predetermined threshold is larger than the first predetermined threshold. For example, the controller further includes: a second voltage detector configured to receive the third voltage from the third terminal and generate a second detection signal based at least in part on the third voltage; wherein the second voltage detector is further configured to: detect that the third voltage has increased to a third predetermined threshold; and generate the second detection signal indicating that the third voltage has increased to the third predetermined threshold. As an example, the first charging controller is further configured to: in response to the first detection signal indicating that the second voltage has increased to the second predetermined threshold, generate the first control signal indicating that the first charging current is not to be generated; and in response to the second detection signal indicating that the third voltage has increased to the third predetermined threshold, generate the first control signal indicating that the first charging current is not to be generated; wherein the first current generator is further configured to: in response to the first control signal indicating that the first charging current is not to be generated, not generate the first charging current that flows through the third terminal.

For example, the second voltage decreases to the first predetermined threshold during normal operation after a startup process; the second voltage increases to the second predetermined threshold during the normal operation; and the third voltage increases to the third predetermined threshold during the normal operation. As an example, during the startup process: the second voltage detector is further configured to: detect that the third voltage has not risen above a fourth predetermined threshold; and generate a third detection signal indicating that the third voltage has not risen above the fourth predetermined threshold. For example, the controller further includes a second charging controller configured to receive the third detection signal and generate

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a second control signal based at least in part on the third detection signal. As an example, the second charging controller is further configured to: in response to the third detection signal indicating that the third voltage has not risen above the fourth predetermined threshold, generate the second control signal indicating that a second charging current is to be generated; wherein the second current generator is further configured to: receive the second control signal from the second charging controller; and in response to the second control signal indicating that the second charging current is to be generated, generate the second charging current that flows through the second terminal.

For example, the first voltage detector is further configured to: determine whether the second voltage is smaller than a first reference voltage, whether the second voltage is larger than the first reference voltage but smaller than a second reference voltage, and whether the second voltage is larger than the second reference voltage; generate a fourth detection signal indicating that the second voltage is smaller than the first reference voltage; generate a fifth detection signal indicating that the second voltage is larger than the first reference voltage but smaller than the second reference voltage; and generate a sixth detection signal indicating that the second voltage is larger than the second reference voltage. As an example, the second current generator is further configured to, in response to the second control signal indicating that the second charging current is to be generated: generate the second charging current that flows through a first resistor but not a second resistor and not a third resistor; if the second current generator receives the fourth detection signal indicating that the second voltage is smaller than the first reference voltage; generate the second charging current that flows through the first resistor and the second resistor but not the third resistor, if the second current generator receives the fifth detection signal indicating that the second voltage is larger than the first reference voltage but smaller than the second reference voltage; and generate the second charging current that flows through the first resistor, the second resistor and the third resistor, if the second current generator receives the sixth detection signal indicating that the second voltage is larger than the second reference voltage. For example, the controller further includes: a circuit for undervoltage-lockout configured to: in response to the third voltage has risen above the fourth predetermined threshold, generating a third control signal indicating the startup process has ended; and an internal control circuit configured to: in response to the third control signal indicating the startup process has ended, power on one or more components of the controller to start the normal operation.

According to some embodiments, a controller for a power converter includes: a first terminal configured to receive a first voltage; a second terminal connected to a first capacitor and biased to a second voltage; a third terminal connected to a second capacitor and biased to a third voltage; a first voltage detector configured to receive the second voltage from the second terminal and generate a first detection signal based at least in part on the second voltage; a charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and a charging current generator configured to receive the first voltage from the first terminal and configured to receive the first control signal from the charging controller; wherein the first voltage detector is further configured to: detect that the second voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the second voltage has

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decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated; wherein the current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the third terminal to charge the second capacitor associated with the third voltage. For example, the controller is implemented according to at least FIG. 17.

As an example, the first voltage detector is further configured to: detect that the second voltage has increased to a second predetermined threshold; and generate the first detection signal indicating that the second voltage has increased to the second predetermined threshold; wherein the second predetermined threshold is larger than the first predetermined threshold. For example, the controller further includes: a second voltage detector configured to receive the third voltage from the third terminal and generate a second detection signal based at least in part on the third voltage; wherein the second voltage detector is further configured to: detect that the third voltage has increased to a third predetermined threshold; and generate the second detection signal indicating that the third voltage has increased to the third predetermined threshold. As an example, the charging controller is further configured to: in response to the first detection signal indicating that the second voltage has increased to the second predetermined threshold, generate the first control signal indicating that the charging current is not to be generated; and in response to the second detection signal indicating that the third voltage has increased to the third predetermined threshold, generate the first control signal indicating that the charging current is not to be generated; wherein the charging current generator is further configured to: in response to the first control signal indicating that the charging current is not to be generated, not generate the charging current that flows through the third terminal.

For example, the second voltage decreases to the first predetermined threshold during normal operation after a startup process; the second voltage increases to the second predetermined threshold during the normal operation; and the third voltage increases to the third predetermined threshold during the normal operation. As an example, during the startup process: the second voltage detector is further configured to: detect that the third voltage has not risen above a fourth predetermined threshold; and generate a third detection signal indicating that the third voltage has not risen above the fourth predetermined threshold. For example, the charging controller is further configured to: in response to the third detection signal indicating that the third voltage has not risen above the fourth predetermined threshold, generate the first control signal indicating that the charging current is to be generated; wherein the current generator is further configured to: in response to the first control signal indicating that the charging current is to be generated, generate the charging current that flows through the third terminal to charge the second capacitor associated with the third voltage.

As an example, the second voltage detector is further configured to: determine whether the third voltage is smaller than a first reference voltage, whether the third voltage is larger than the first reference voltage but smaller than a second reference voltage, and whether the third voltage is larger than the second reference voltage; generate a fourth

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detection signal indicating that the third voltage is smaller than the first reference voltage; generate a fifth detection signal indicating that the third voltage is larger than the first reference voltage but smaller than the second reference voltage; and generate a sixth detection signal indicating that the third voltage is larger than the second reference voltage. For example, wherein the current generator is further configured to, in response to the first control signal indicating that the charging current is to be generated: generate the charging current that flows through a first resistor but not a second resistor and not a third resistor, if the current generator receives the fourth detection signal indicating that the third voltage is smaller than the first reference voltage; generate the charging current that flows through the first resistor and the second resistor but not the third resistor, if the current generator receives the fifth detection signal indicating that the third voltage is larger than the first reference voltage but smaller than the second reference voltage; and generate the charging current that flows through the first resistor, the second resistor and the third resistor, if the current generator receives the sixth detection signal indicating that the third voltage is larger than the second reference voltage. As an example, the controller further includes: a circuit for undervoltage-lockout configured to: in response to the third voltage has risen above the fourth predetermined threshold, generating a second control signal indicating the startup process has ended; and an internal control circuit configured to: in response to the second control signal indicating the startup process has ended, power on one or more components of the controller to start the normal operation.

According to certain embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a capacitor; generating a detection signal based at least in part on the second voltage; receiving the detection signal; generating a first control signal based at least in part on the detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the detection signal includes: in response to the detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the charging current is to be generated, generating the charging current to charge the capacitor associated with the second voltage. For example, the method is implemented according to at least FIG. 9.

According to some embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a first capacitor; receiving a third voltage associated with a second capacitor; generating a first detection signal based at least in part on the second voltage; receiving the first detection signal; generating a first control signal based at least in part on the first detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a first detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the first detection signal indicating that the second voltage has decreased to the

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first predetermined threshold; wherein the generating a first control signal based at least in part on the first detection signal includes: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal 5 indicating that a first charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the first charging current is to be generated, generating the first charging current to charge the second capacitor associated with the third voltage. For example, the method is implemented according to at least FIG. 13 and/or FIG. 15.

According to certain embodiments, a method for a power converter includes: receiving a first voltage; receiving a second voltage associated with a first capacitor; receiving a third voltage associated with a second capacitor; generating a first detection signal based at least in part on the second voltage; receiving the first detection signal; generating a first control signal based at least in part on the first detection signal; receiving the first control signal; and processing the first control signal; wherein the generating a first detection signal based at least in part on the second voltage includes: detecting that the second voltage has decreased to a first predetermined threshold; and generating the first detection signal indicating that the second voltage has decreased to the first predetermined threshold; wherein the generating a first control signal based at least in part on the first detection signal includes: in response to the first detection signal indicating that the second voltage has decreased to the first predetermined threshold, generating the first control signal indicating that a charging current is to be generated; wherein the processing the first control signal includes: in response to the first control signal indicating that the charging current is to be generated, generating the charging current to charge the second capacitor associated with the third voltage. For example, the method is implemented according to at least FIG. 17.

For example, some or all components of various embodiments of the present invention each are, individually and/or in combination with at least another component, implemented using one or more software components, one or more hardware components, and/or one or more combinations of software and hardware components. As an example, some or all components of various embodiments of the present invention each are, individually and/or in combination with at least another component, implemented in one or more circuits, such as one or more analog circuits and/or one or more digital circuits. For example, various embodiments and/or examples of the present invention can be combined.

Although specific embodiments of the present invention have been described, it will be understood by those of skill in the art that there are other embodiments that are equivalent to the described embodiments. Accordingly, it is to be understood that the invention is not to be limited by the specific illustrated embodiments.

What is claimed is:

1. A converter controller for a power converter, the converter controller comprising:

a voltage detector configured to receive a voltage and generate a detection signal based at least in part on the voltage; and

a charging controller configured to receive the detection signal and generate a first control signal based at least in part on the detection signal;

wherein the voltage detector is further configured to:

detect that the voltage has decreased to a first predetermined threshold; and

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generate the detection signal indicating that the voltage has decreased to the first predetermined threshold; wherein the charging controller is further configured to: in response to the detection signal indicating that the voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a charging current is to be generated for raising the voltage;

wherein the voltage detector is further configured to:

detect that the voltage has increased to a second predetermined threshold; and

generate the detection signal indicating that the voltage has increased to the second predetermined threshold;

wherein the second predetermined threshold is larger than the first predetermined threshold.

2. The converter controller of claim 1 wherein:

the charging controller is further configured to:

in response to the detection signal indicating that the voltage has increased to the second predetermined threshold, generate the first control signal indicating that the charging current is not to be generated.

3. The converter controller of claim 1 wherein:

the voltage decreases to the first predetermined threshold during normal operation after a startup process; and the voltage increases to the second predetermined threshold during the normal operation.

4. The converter controller of claim 3 wherein, during the startup process:

the voltage detector is further configured to:

detect that the voltage has not reached a third predetermined threshold; and

generate the detection signal indicating that the voltage has not reached the third predetermined threshold;

the charging controller is further configured to:

in response to the detection signal indicating that the voltage has not risen above the third predetermined threshold, generate the first control signal indicating that the charging current is to be generated for raising the voltage.

5. The converter controller of claim 4 wherein, during the startup process:

the voltage detector is further configured to:

detect that the voltage has risen above the third predetermined threshold; and

generate the detection signal indicating that the voltage has risen above the third predetermined threshold;

the charging controller is further configured to:

in response to the detection signal indicating that the voltage has risen above the third predetermined threshold, generate the first control signal indicating that the charging current is not to be generated.

6. The converter controller of claim 5, and further comprising:

a circuit for undervoltage-lockout configured to:

in response to the voltage has risen above the third predetermined threshold, generating a second control signal indicating the startup process has ended; and

an internal control circuit configured to:

in response to the second control signal indicating the startup process has ended, power on one or more components of the converter controller to start the normal operation.

7. A converter controller for a power converter, the converter controller comprising:

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a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage;

a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage; and

a first charging controller configured to receive the first detection signal and the second detection signal and generate a first control signal based at least in part on the first detection signal and the second detection signal;

wherein the first voltage detector is further configured to: detect that the first voltage has decreased to a first predetermined threshold; and generate the first detection signal indicating that the first voltage has decreased to the first predetermined threshold;

wherein the first charging controller is further configured to:

in response to the first detection signal indicating that the first voltage has decreased to the first predetermined threshold, generate the first control signal indicating that a first charging current is to be generated;

wherein the second voltage detector is further configured to:

detect that the second voltage has increased to a second predetermined threshold; and

generate the second detection signal indicating that the second voltage has increased to the second predetermined threshold;

wherein the first voltage and the second voltage are two different voltages.

8. The converter controller of claim 7 wherein the first charging controller is further configured to:

in response to the second detection signal indicating that the second voltage has increased to the second predetermined threshold, generate the first control signal indicating that the first charging current is not to be generated.

9. The converter controller of claim 8, and further comprising:

a first current generator configured to receive the first control signal from the first charging controller;

wherein the first current generator is further configured to:

in response to the first control signal indicating that the first charging current is to be generated, generate the first charging current for raising the second voltage; and

in response to the first control signal indicating that the first charging current is not to be generated, not generate the first charging current.

10. The converter controller of claim 7 wherein the first voltage detector is further configured to:

detect that the first voltage has increased to a third predetermined threshold; and

generate the first detection signal indicating that the first voltage has increased to the third predetermined threshold;

wherein the third predetermined threshold is larger than the first predetermined threshold.

11. The converter controller of claim 10 wherein the first charging controller is further configured to:

in response to the first detection signal indicating that the first voltage has increased to the third predetermined threshold, generate the first control signal indicating that the first charging current is not to be generated.

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12. The converter controller of claim 10 wherein the first voltage decreases to the first predetermined threshold during normal operation after a startup process.

13. The converter controller of claim 12 wherein the second voltage increases to the second predetermined threshold during the normal operation.

14. The converter controller of claim 12 wherein the first voltage increases to the third predetermined threshold during the normal operation.

15. A converter controller for a power converter, the converter controller comprising:

a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage;

a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage;

a first charging controller configured to receive the first detection signal and generate a first control signal based at least in part on the first detection signal; and

a second charging controller configured to receive the second detection signal and generate a second control signal based at least in part on the second detection signal;

wherein, during a startup process before normal operation,

the second voltage detector is further configured to:

detect that the second voltage has not reached a first predetermined threshold; and

generate the second detection signal indicating that the second voltage has not reached the first predetermined threshold;

the second charging controller is further configured to:

in response to the second detection signal indicating that the second voltage has not reached the first predetermined threshold, generate the second control signal indicating that a first charging current is to be generated to raise the first voltage;

wherein, during the normal operation after the startup process,

the first voltage detector is further configured to:

detect that the first voltage has decreased to a second predetermined threshold; and

generate the first detection signal indicating that the first voltage has decreased to the second predetermined threshold;

the first charging controller is further configured to:

in response to the first detection signal indicating that the first voltage has decreased to the second predetermined threshold, generate the first control signal indicating that a second charging current is to be generated to raise the second voltage.

16. The converter controller of claim 15, and further comprising:

a first current generator, the first current generator being configured to:

receive the first control signal from the first charging controller; and

in response to the first control signal indicating that the second charging current is to be generated, generate the second charging current to raise the second voltage.

17. The converter controller of claim 16, and further comprising:

a second current generator, the second current generator being configured to:

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receive the second control signal from the second charging controller; and
 in response to the second control signal indicating that the first charging current is to be generated, generate the first charging current to raise the first voltage.

18. The converter controller of claim 17 wherein, during the startup process:

the second voltage detector is further configured to:

detect that the second voltage has risen above the first predetermined threshold; and

generate the second detection signal indicating that the second voltage has risen above the first predetermined threshold.

19. The converter controller of claim 18 wherein the second charging controller is further configured to:

in response to the second detection signal indicating that the second voltage has risen above the first predetermined threshold, generate the second control signal indicating that the first charging current is not to be generated.

20. The converter controller of claim 19 wherein the second current generator is further configured to, in response to the second control signal indicating that the first charging current is not to be generated, not generate the first charging current.

21. The converter controller of claim 18, and further comprising:

a circuit for undervoltage-lockout configured to:

in response to the second voltage has risen above the first predetermined threshold, generating a third control signal indicating that the startup process has ended; and

an internal control circuit configured to:

in response to the third control signal indicating that the startup process has ended, power on one or more components of the converter controller to start the normal operation.

22. A converter controller for a power converter, the converter controller comprising:

a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage;

a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage;

a charging controller configured to receive the first detection signal and the second detection signal and generate a control signal based at least in part on the second detection signal; and

a current generator configured to receive the control signal from the charging controller;

wherein, during a startup process before normal operation,

the second voltage detector is further configured to: detect that the second voltage has not reached a predetermined threshold; and

generate the second detection signal indicating that the second voltage has not reached the predetermined threshold;

the charging controller is further configured to:

in response to the second detection signal indicating that the second voltage has not reached the predetermined threshold, generate the control signal indicating that a charging current is to be generated to raise the first voltage;

the current generator is further configured to:

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in response to the control signal indicating that the charging current is to be generated, generate the charging current to raise the first voltage;

wherein, during the startup process before the normal operation,

the first voltage detector is further configured to:

determine that the first voltage is smaller than a first reference voltage, that the first voltage is larger than the first reference voltage but smaller than a second reference voltage, or that the first voltage is larger than the second reference voltage;

generate a third detection signal indicating that the first voltage is smaller than the first reference voltage, that the first voltage is larger than the first reference voltage but smaller than the second reference voltage, or that the first voltage is larger than the second reference voltage;

wherein the second reference voltage is larger than the first reference voltage.

23. The converter controller of claim 22 wherein the current generator is further configured to receive the third detection signal indicating that the first voltage is smaller than the first reference voltage, that the first voltage is larger than the first reference voltage but smaller than the second reference voltage, or that the first voltage is larger than the second reference voltage.

24. The converter controller of claim 23 wherein the current generator is further configured to:

in response to the third detection signal indicating that the first voltage is smaller than the first reference voltage, generate the charging current that flows through a first resistor but not a second resistor and a third resistor;

in response to the third detection signal indicating that the first voltage is larger than the first reference voltage but smaller than the second reference voltage, generate the charging current that flows through the first resistor and the second resistor but not the third resistor; and

in response to the third detection signal indicating that the first voltage is larger than the second reference voltage, generate the charging current that flows through the first resistor, the second resistor and the third resistor.

25. A converter controller for a power converter, the converter controller comprising:

a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage;

a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage; and

a charging controller configured to receive the first detection signal and the second detection signal and generate a control signal based at least in part on the first detection signal and the second detection signal;

wherein, during a startup process before normal operation,

the second voltage detector is further configured to:

detect that the second voltage has not reached a first predetermined threshold; and

generate the second detection signal indicating that the second voltage has not reached the first predetermined threshold;

the charging controller is further configured to:

in response to the second detection signal indicating that the second voltage has not reached the first predetermined threshold, generate the control signal indicating that a charging current is to be generated to raise the second voltage;

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wherein, during the normal operation after the startup process,

the first voltage detector is further configured to:

detect that the first voltage has decreased to a second predetermined threshold; and

generate the first detection signal indicating that the first voltage has decreased to the second predetermined threshold;

the charging controller is further configured to:

in response to the first detection signal indicating that the first voltage has decreased to the second predetermined threshold, generate the control signal indicating that the charging current is to be generated to raise the second voltage.

26. A converter controller for a power converter, the converter controller comprising:

a first voltage detector configured to receive a first voltage and generate a first detection signal based at least in part on the first voltage;

a second voltage detector configured to receive a second voltage and generate a second detection signal based at least in part on the second voltage;

a charging controller configured to receive the first detection signal and the second detection signal and generate a control signal based at least in part on the second detection signal; and

a current generator configured to receive the control signal from the charging controller;

wherein, during a startup process before normal operation,

the second voltage detector is further configured to:

detect that the second voltage has not reached a predetermined threshold; and

generate the second detection signal indicating that the second voltage has not reached the predetermined threshold;

the charging controller is further configured to:

in response to the second detection signal indicating that the second voltage has not reached the predetermined threshold, generate the control signal indicating that a charging current is to be generated to raise the second voltage;

the current generator is further configured to:

in response to the control signal indicating that the charging current is to be generated, generate the charging current to raise the second voltage;

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wherein, during the startup process before the normal operation,

the second voltage detector is further configured to:

determine that the second voltage is smaller than a first reference voltage, that the second voltage is larger than the first reference voltage but smaller than a second reference voltage, or that the second voltage is larger than the second reference voltage but smaller than the predetermined threshold; and generate a third detection signal indicating that the second voltage is smaller than the first reference voltage, that the second voltage is larger than the first reference voltage but smaller than the second reference voltage, or that the second voltage is larger than the second reference voltage but smaller than the predetermined threshold;

wherein:

the second reference voltage is larger than the first reference voltage; and

the predetermined threshold is larger than the second reference voltage.

27. The converter controller of claim 26 wherein the current generator is further configured to receive the third detection signal indicating that the second voltage is smaller than the first reference voltage, that the second voltage is larger than the first reference voltage but smaller than the second reference voltage, or that the second voltage is larger than the second reference voltage but smaller than the predetermined threshold.

28. The converter controller of claim 27 wherein the current generator is further configured to:

in response to the third detection signal indicating that the second voltage is smaller than the first reference voltage, generate the charging current that flows through a first resistor but not a second resistor and a third resistor;

in response to the third detection signal indicating that the second voltage is larger than the first reference voltage but smaller than the second reference voltage, generate the charging current that flows through the first resistor and the second resistor but not the third resistor; and in response to the third detection signal indicating that the second voltage is larger than the second reference voltage but smaller than the predetermined threshold, generate the charging current that flows through the first resistor, the second resistor and the third resistor.

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